

RENEET 2026



MAHA MARATHON

Complete

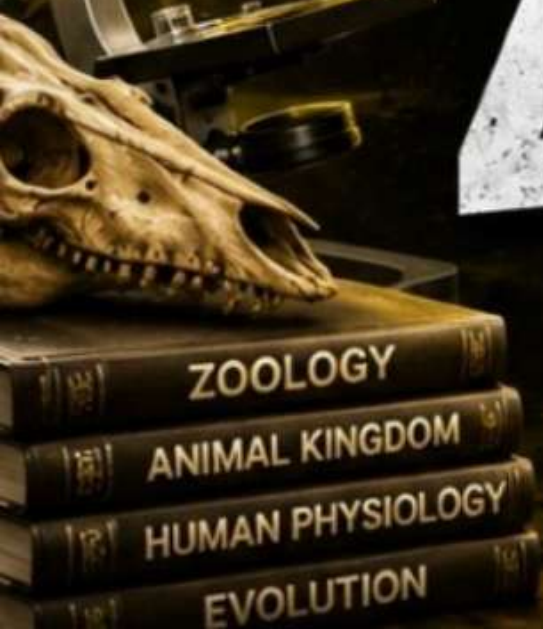
ZOOLOGY

Oneshot

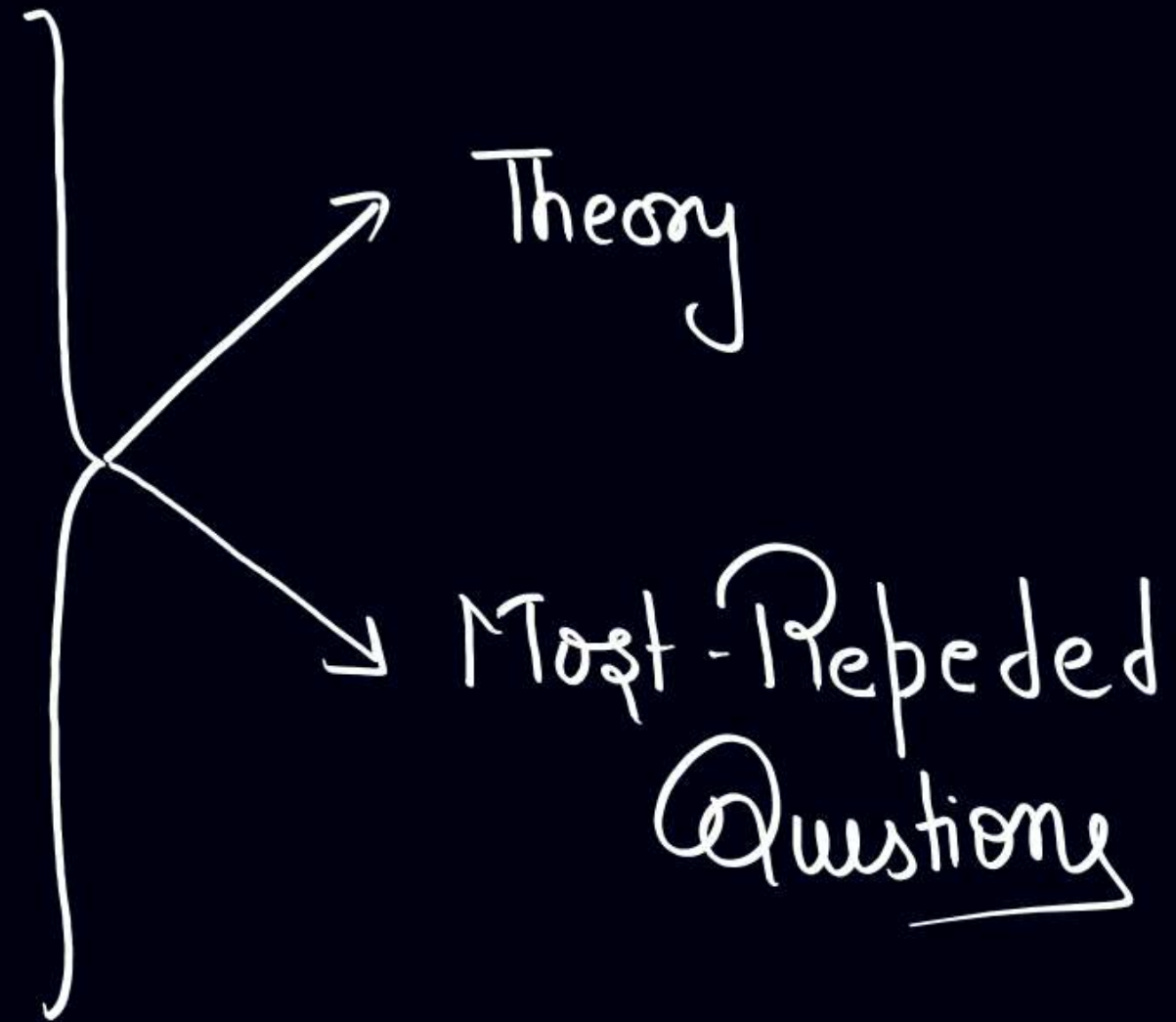


MD Sir

1
PART



Human Physiology
+
Biomolecules
+
Animal Kingdom
+
Frog



Breathing and Exchange of Gases

Respiratory organs

Mechanisms of breathing vary among different groups of animals depending mainly on their habitats and levels of organisation

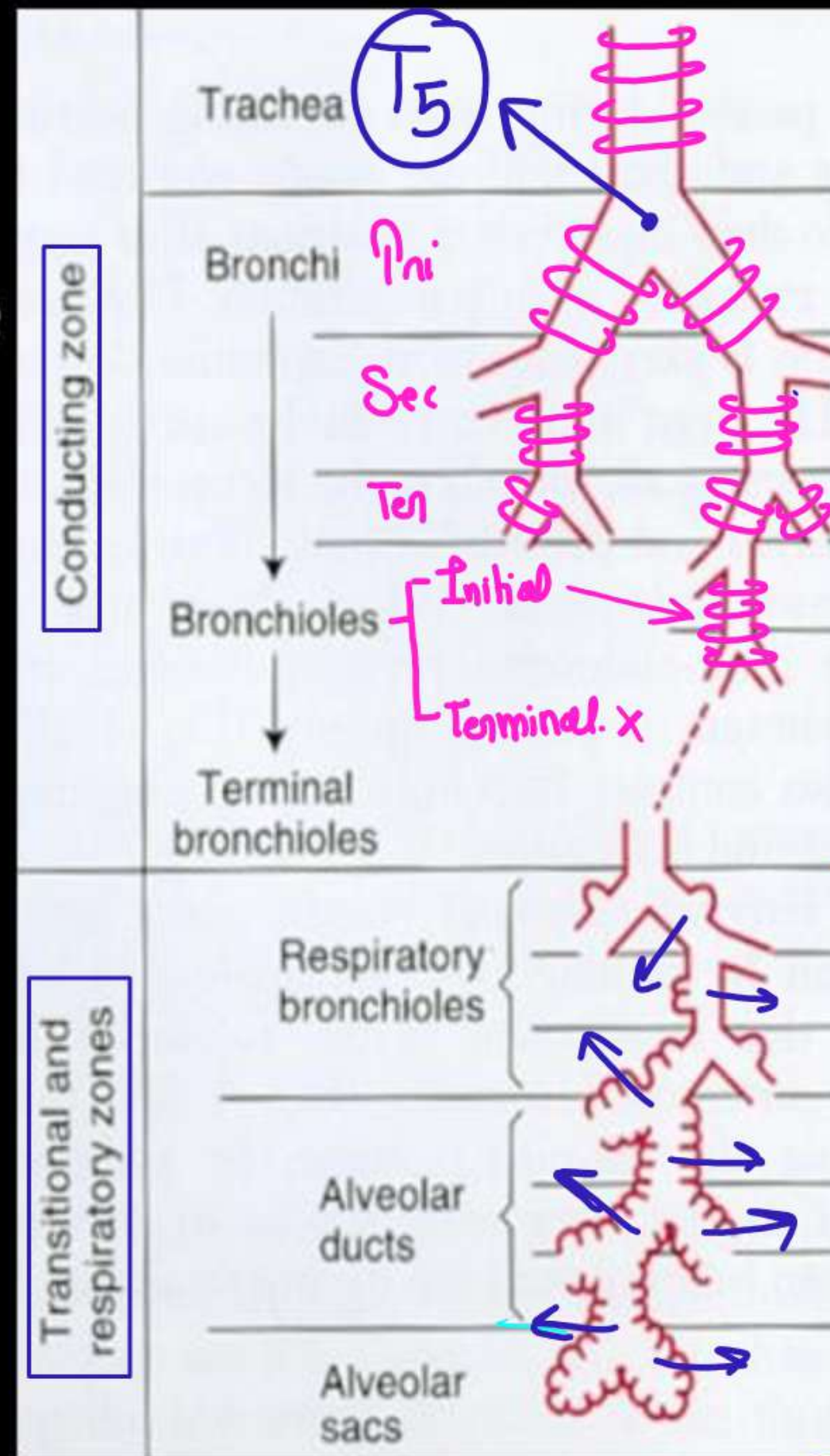
Examples of Animals	Respiratory organs
Sponges, coelenterates, flatworms	Body surface
Earthworms	Moist cuticle
Insects	Network of tubes (tracheal tubes)
Aquatic arthropods, molluscs, Fish	Special Vascularised structures called Gills (Branchial respiration)
Terrestrial forms (amphibians, reptiles, birds and mammals)	Vascularised bags called Lungs (Pulmonary respiration).
Amphibians like Frogs	Moist skin (cutaneous respiration)

Function of Conducting zone-

1. Conduction of air from Ex. Nostrils to Respiratory Zone
2. Clear conducting air from foreign particle
3. Humidification of air
4. Maintain temp. of conducting air according to body temp.

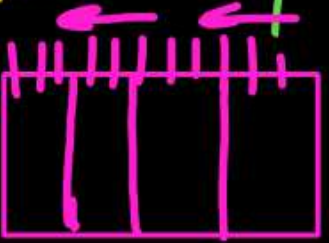
Respiratory Zone -

1. Diffusion of gases ✓



Epithelium in -
Conducting Zone →

- Simple Ciliated Columnar Epithelial Tissue.



Respiratory Zone

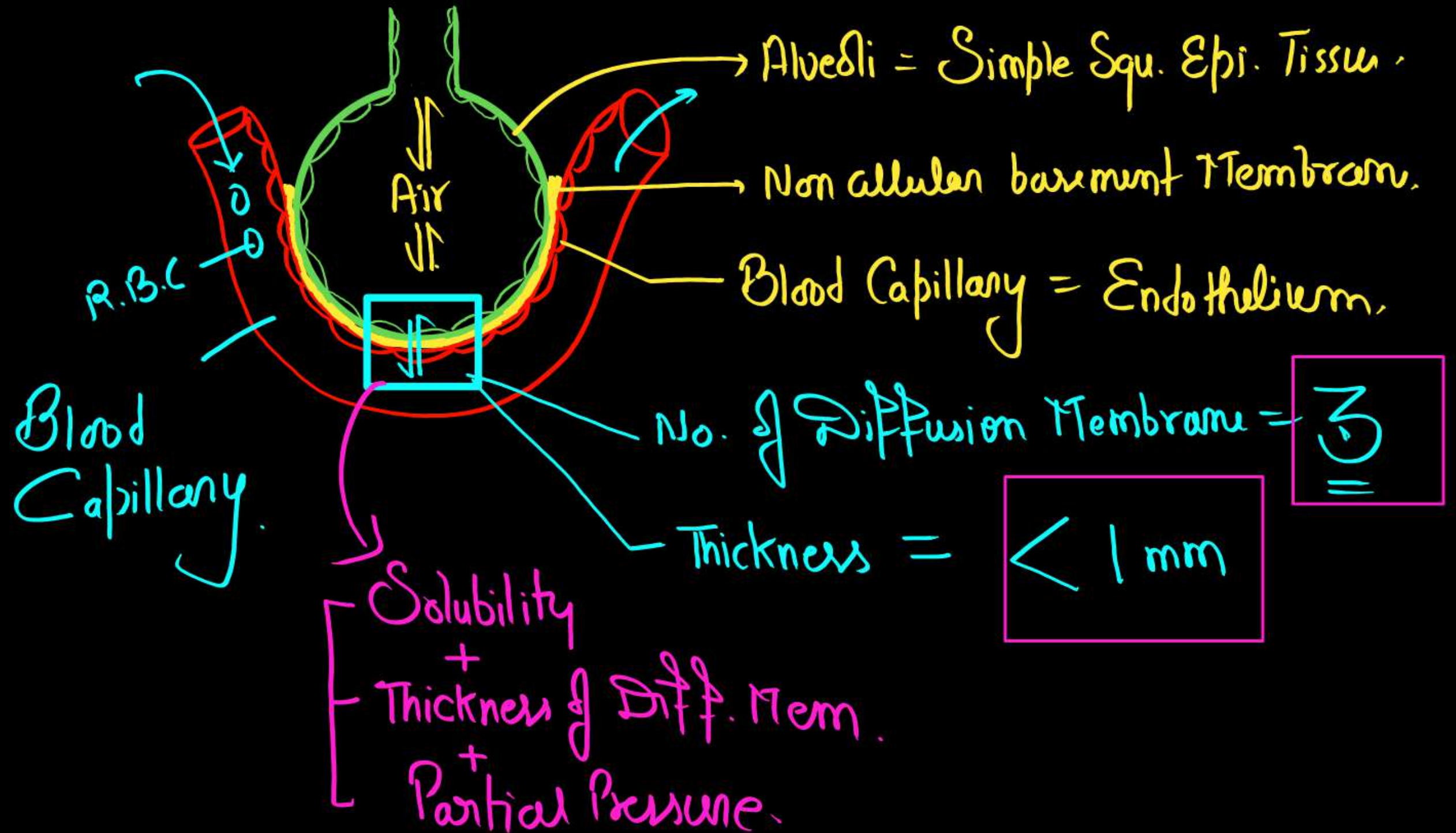
- Simple Squamous Epi. Tissue.



C-Shape Hyaline Cartilage
Trachea to Initial bronchiole
Incomplete Dorsally

Structure of Alveoli

Thin, irregular-walled and vascularised bag-like structures



Inspiration (Active)

Contraction in Diaphragm
(↑ - Thoracic Volu. → A-P axis)



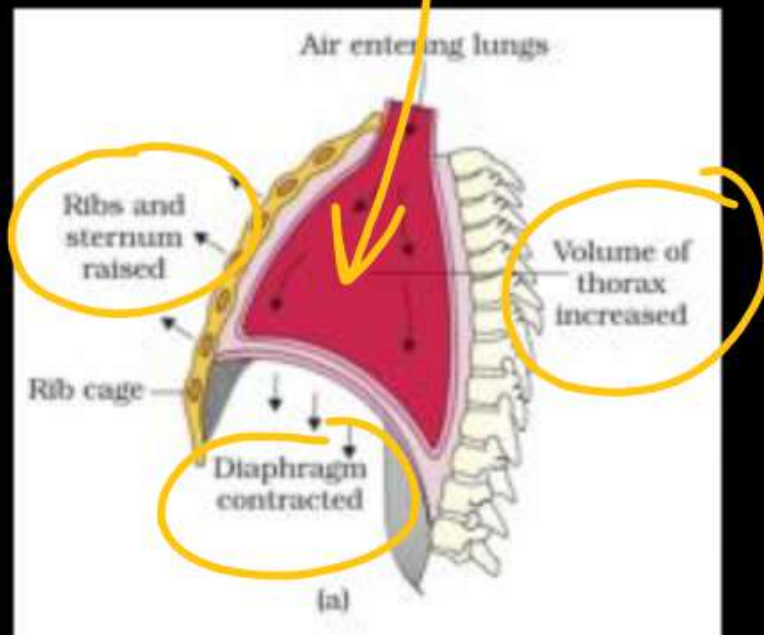
Contraction in E.I.C.M
(↑ - Thoracic Volu. - D-V axis)



↑ - Total Thoracic Volu.

↓ - Intrapulmonary Pressure

↓
Atm. Air enters in lungs



Expiration (Passive)

Relaxation of Diaphragm
(↓ - Thoracic Volu.)



Relaxation of E.I.C.M
(↓ - Thoracic Volu.)



↓ - Total Volume

↓
↑ - Intrapulmonary Pressure

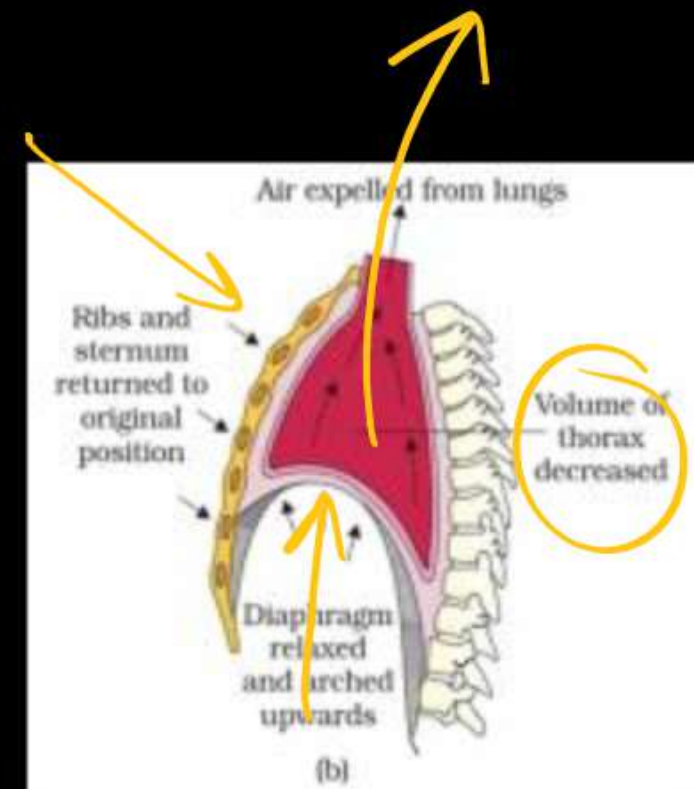
(Expiration)

For full Expiration

Contraction

D-I.C.M ✓

2) Abdominal Mus ✓



Respiratory Volumes and Capacities

= Spirometer

1 Q = 100%

Lung volumes

Spirogram

Pulmonary Capacities

I.R.V = 2500 - 3000 ml

Inspiratory reserve volume

Tidal volume T.V. = 500 ml

Expiratory reserve volume

ERV = 1000 - 1100 ml

Residual volume

R.V. = 1100 - 1200 ml

I.C. = T.V + IRV = 3500 ml

EC = T.V + ERV = 1600 ml

V.C. = T.V + ERV + IRV = 4600

F.R.C. = ERV + RV = 2300

T.L.C. = $\underbrace{T.V + IRV + ERV + RV}_{V_C} = 5800$

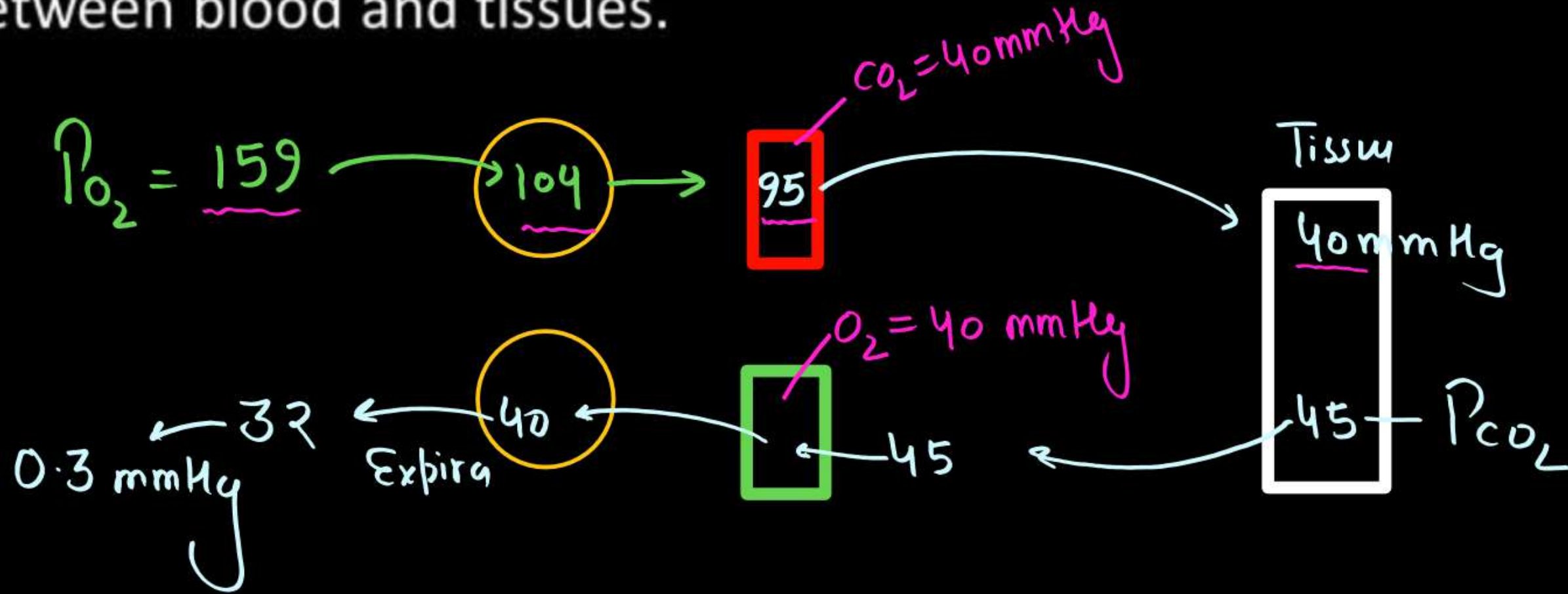
Note –

- Spirometry cannot be used to measure the residual volume or any capacities which incorporate the residual volume such as functional residual capacity (FRC) and total lung capacity (TLC).
- Pulmonary Ventilation -
$$= T.V \times \text{Rate of Breathing}$$
$$= 500 \times 12-16/\text{minute}$$
$$= \underline{6000-8000 \text{ ml}/\underline{\text{minute}}}$$

Exchange of Gases

Q//

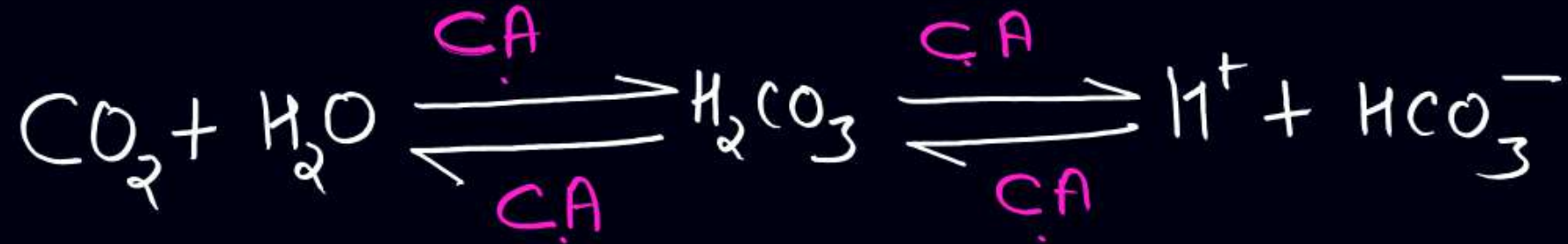
Alveoli are the primary sites of exchange of gases. Exchange of gases also occur between blood and tissues.



Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

vvfmp

(A) = Near tissue.
- Loading of CO_2
→ R-Shift (Bohr's Eff.) = Dissociation of Hb-O_2



← (B) = Near lungs (unloading of CO_2)
- L-Shift = Haldane Eff. (Association of Hb-O_2)

Co-factor
(Zn^{++})

Carbonic Anhydrase } R.B.C
Plasma

Oxygen Haemoglobin Dissociation Curve

Sigmoid Curve / Bohr's Effect

vvvvmp
@

Favourable

Unfavourable

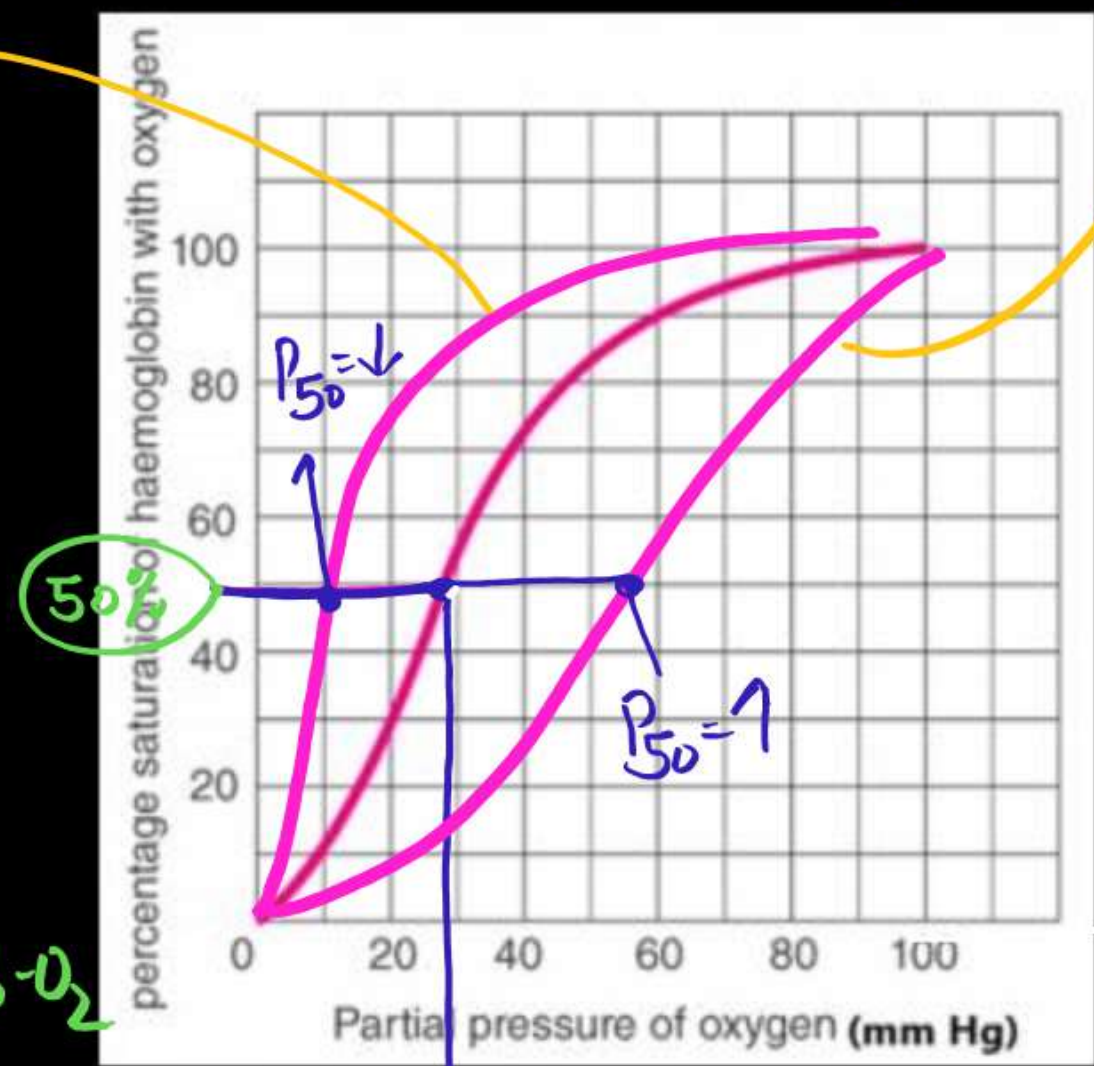
Left Shift
(Haldem Eff.)

$P_{O_2} \uparrow$ & $P_{CO_2} \downarrow$

$H^+ \downarrow \rightarrow \uparrow pH$

$\uparrow$ Association of $Hb-O_2$

$P_{50} \downarrow$ Temp. $\downarrow$
(Near lungs)



Right - Shift (Bohr's Eff.)

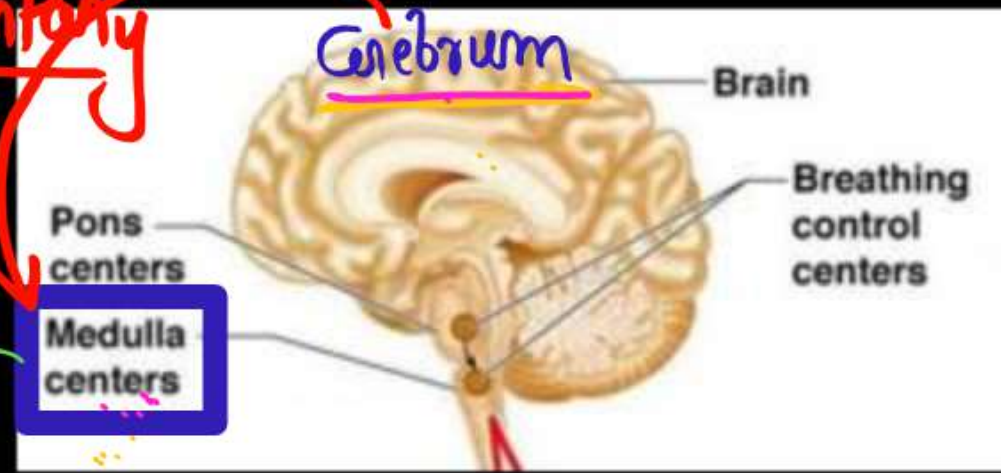
$P_{O_2} \downarrow$ & $P_{CO_2} \uparrow$

$\uparrow H^+ \rightarrow pH \downarrow$

Dissociation of $Hb-O_2$

$P_{50} = \uparrow$
Near Tissue
Temp = $\uparrow$

Nervous Regulation of Breathing-



Control only
Involuntary Action

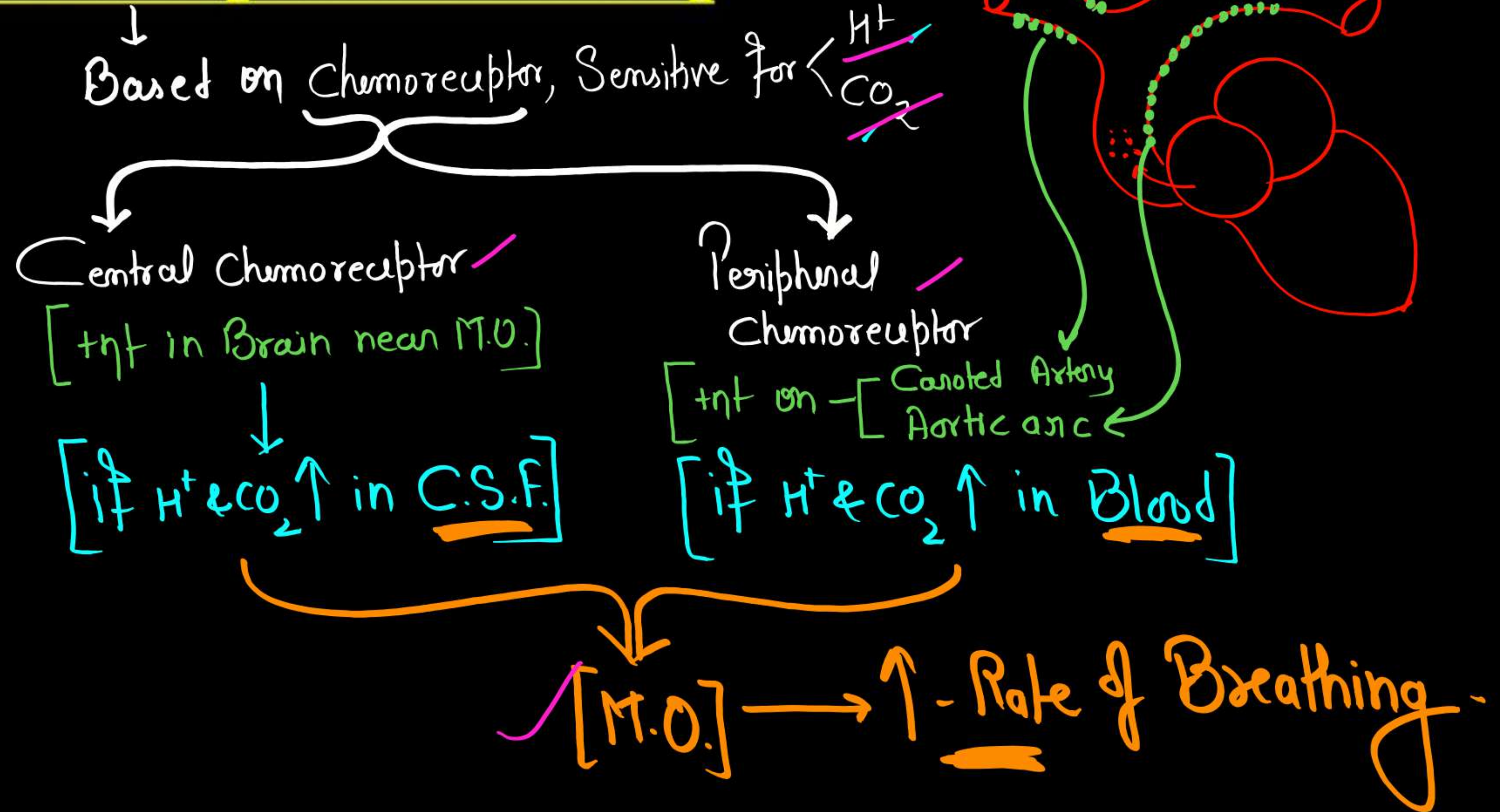
Vamp

M.O.
↓
(Respiratory Rhythm Center)

Control = Contraction
of [EICM
Diaphragm] } Normal Breathing

Pons → Pneumotaxic Center } Switch off → Stop inspiration $\approx 1-4$ sec.
↓ - Duration of Inspiration
↑ - Rate of Breathing

Chemical Regulation of Breathing



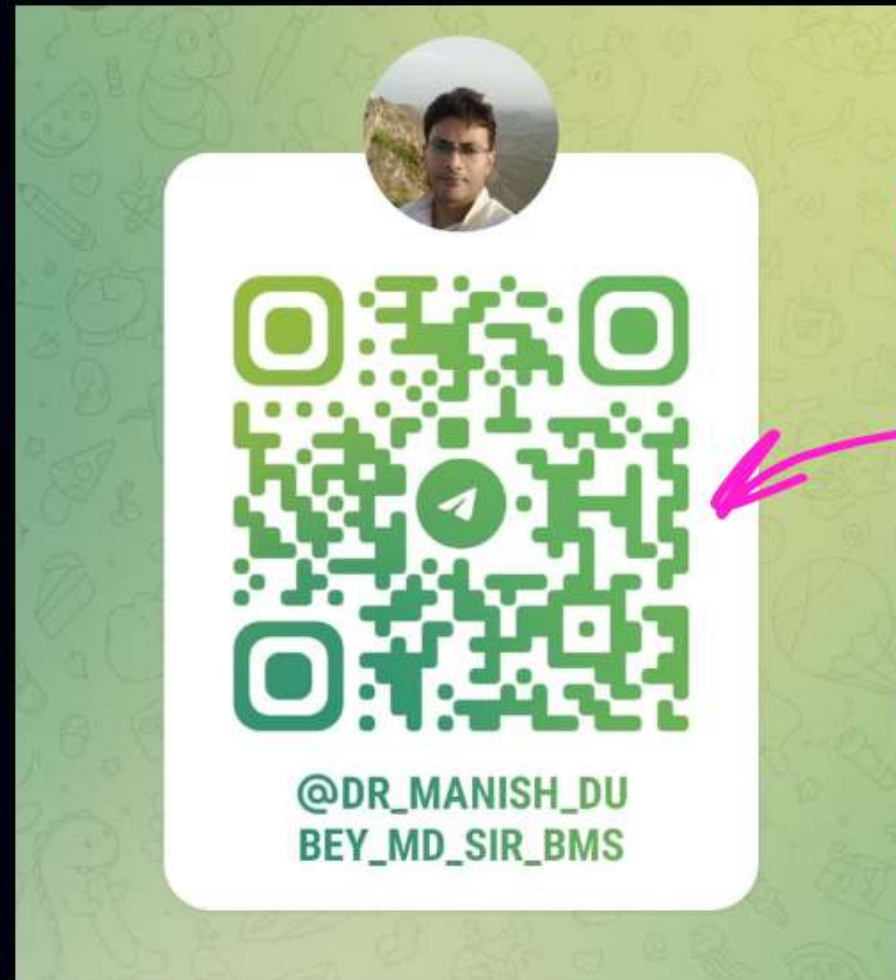
DISORDERS OF RESPIRATORY SYSTEM

(1) Asthma is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

(Allergy)

(2) Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking.

3) - Occupational Respiratory Disorders → Inflammation leading to Fibrosis.
[Proliferation of Fibrous tissue]
(Lungs Damage)



Body Fluids and circulation

18.1 BLOOD

Origin = Mesodermal

Blood is a special connective tissue consisting of a fluid matrix, plasma, and formed elements.

[Composition of Blood]

[Plasma = 55%]

- 1) - Water = 90-92% NCERT
- 2) - Inorganic Sub = 1%
- 3) - Organic Sub. = 8-9% NCERT
 - Protein = 6-8%

[Formed elements = 45%]
(Blood Cells)

Erythrocyte
(R.B.C)

5-5.5
million/mm³

Leucocyte
(W.B.C)

6000-8000
/mm³

Thrombocyte
(Platelets)

1.5-3.5 Lakh
/mm³

Leucocytes (WBC) –

Number of WBC – $6000 - 8000/\text{mm}^3$

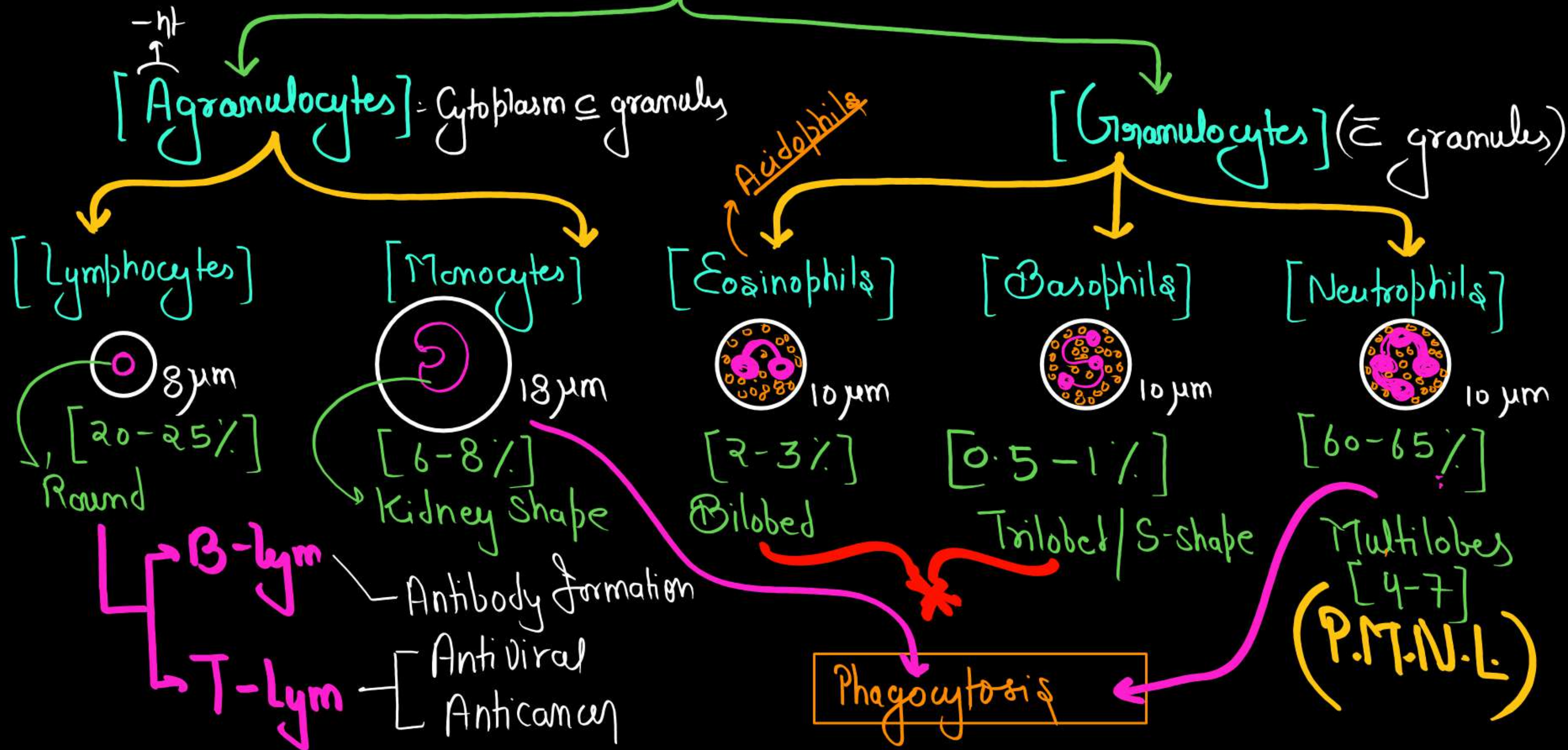
Leucocytosis = $\uparrow$ - No. of W.B.C $\Rightarrow$ During infection ✓

Leukaemia = $\uparrow$ - No. of W.B.C $\Rightarrow$ During Blood Cancer ✓

Diapedesis = Amoeboidal Movement of W.B.C.

Life span = hours ———— Year

Types of Leucocytes (WBC)





[Eosinophils]



1) → Antiallergic

2) - Anti-parasitic



Granules Contain
Lysosome like Enzymes

Resist
fungal infection

Granules Contain
Antiallergic Enzyme -
Histaminase

Associated With
Allergic Reaction

[Basophils] = Blood



Granules Contain



Heparin → Anticoagulant

Histamine → Inflammatory Reaction.

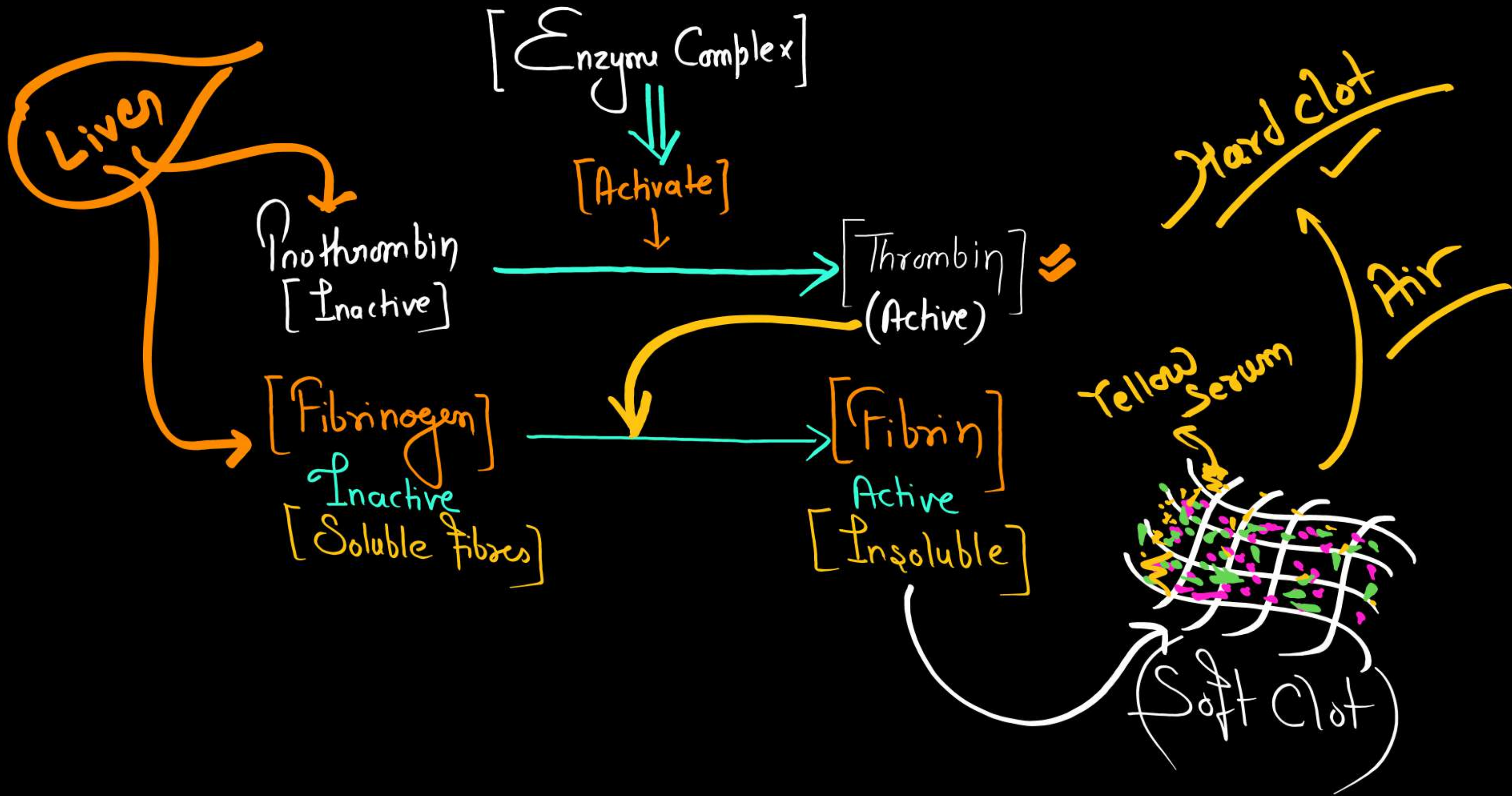
Serotonin → Vasoconstrictor.

etc

Involve in Inflammatory
Reaction

Mast cells

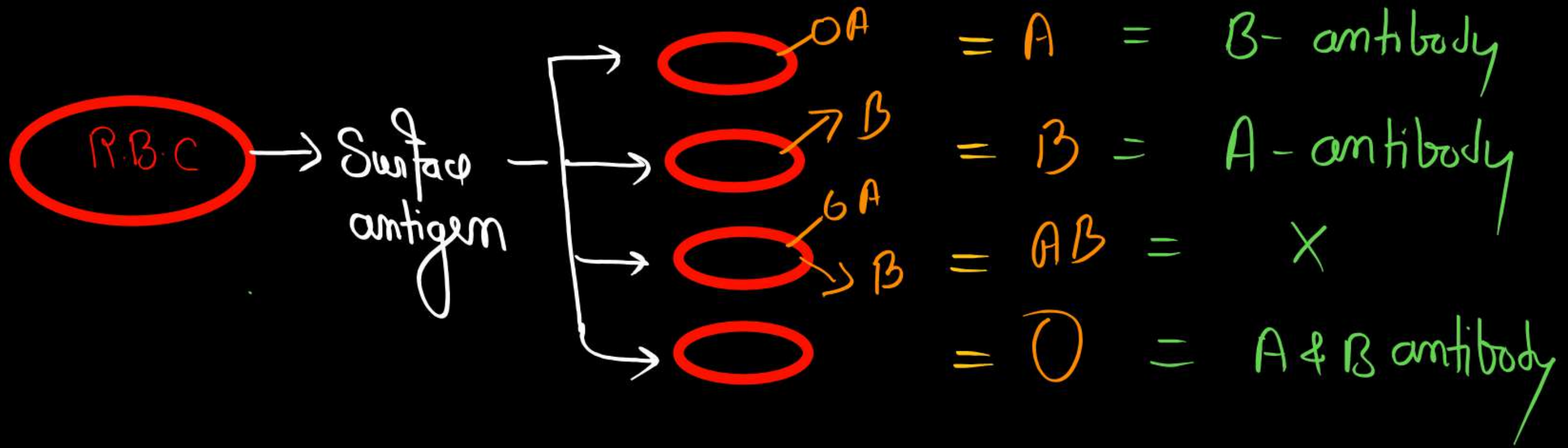
Conn. Tissue



Blood Groups

(1) ABO grouping-

- ABO grouping is based on the presence or absence of **two surface antigens** (chemicals that can induce immune response) on the RBCs namely **A and B**.
- Similarly, the plasma of different individuals contain two natural antibodies (proteins produced in response to antigens)



[Donor] → Antigen → Antibody ← [Recipient]

(Mismatching)

Donor	Antigen		Antibody	Recipient
BY	(Ag)			BY
A	A			A
B	B			B
AB	AB			AB
O	X			O

(U.D)

(both antigen - nt)

U.R
↓
(both Antibody - nt)

(2) Rh Grouping

- Rh antigen similar to one present in Rhesus monkeys (hence Rh), is also observed on the surface of RBCs of majority (nearly 80 per cent) of humans.

Human population — $\rightarrow$ if Rh antigen +ve on R.B.C. = $Rh^{+ve} = \underline{80\%}$
 $\rightarrow$ if Rh antigen -ve on R.B.C. = $Rh^{-ve} = \underline{20\%}$

Note = Naturally antibody -nt in case of
Rh blood Group.

Rh incompatibility (mismatching)

A special case of Rh incompatibility (mismatching) has been observed between the Rh-ve blood of a pregnant mother with Rh+ve blood of the foetus.

Pregnant Mother = Rh-ve
Foetus = Rh+ve } → 1st baby = Normal
But other show = jaundice & Anaemia
[Erythroblastosis]
(Rhogum)

Note → This can be avoided by administering Anti-Rh antibody to mother just after the delivery of 1st child.

CONDUCTION OF HEART BEAT (Myogenic Heart)

Origin of Heart beat $\Rightarrow$ Modified Cardiac Muscles

$\Downarrow$
[Nodal Tissue]

Types of Nodal Tissue $\rightarrow$

- 1) $\rightarrow$ S.A. Node (Pace Maker) $\rightarrow$ Largest, Responsible for
- 2) $\Rightarrow$ A.V. Node (Pace Setter) $\rightarrow$ Potential Max^m Rhythmicity of Heart
- 3) $\Rightarrow$ Bundle of His (Bundle of Purkinje fibres / A.V Bundle)
- 4) $\Rightarrow$ Purkinje fibres

$\rightarrow$ Show = Autorhythmicity

1) - S.A. Node

- * Position = Upper Right Corner of R-Atrium
- * Potential = 70-75 beat/minute

2) - A.V. Node =

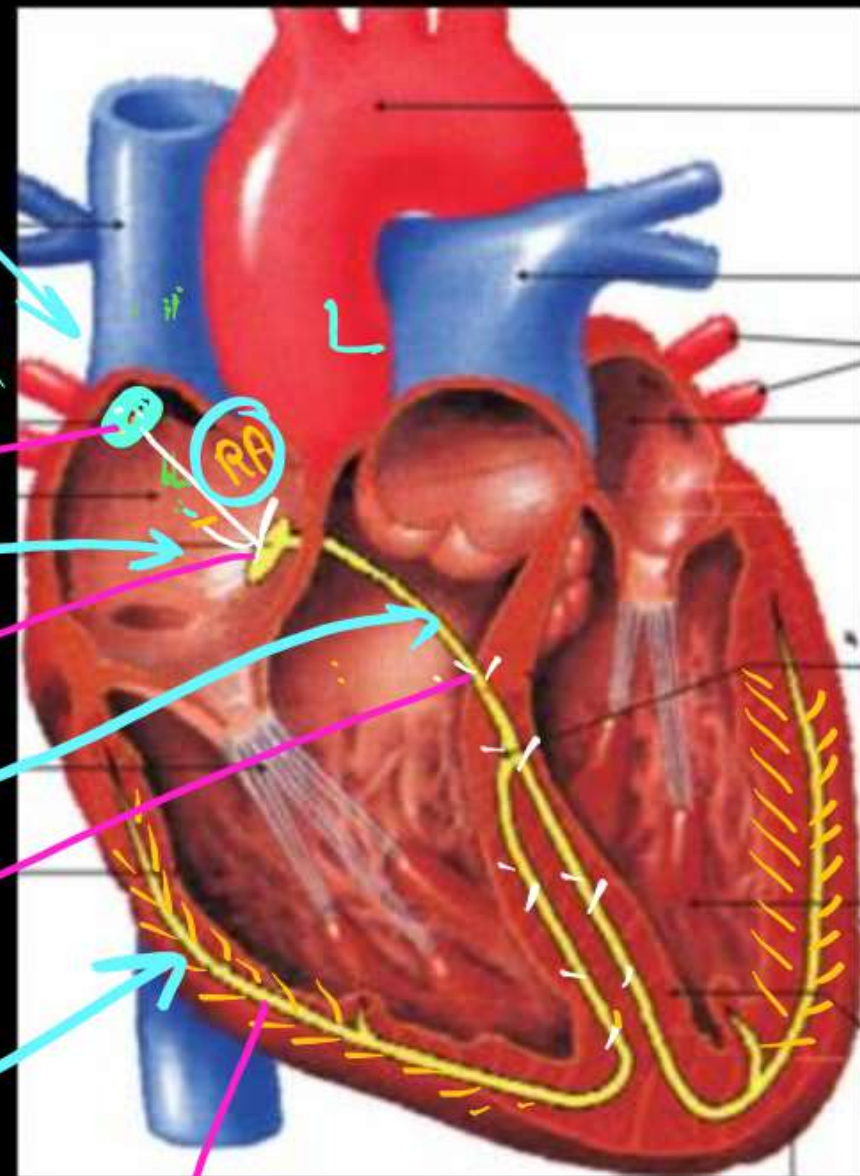
- * Position = Lower Left Corner of R-Atrium
- * Potential = 60-65 impulse or beat/minute

3) - Bundle of His

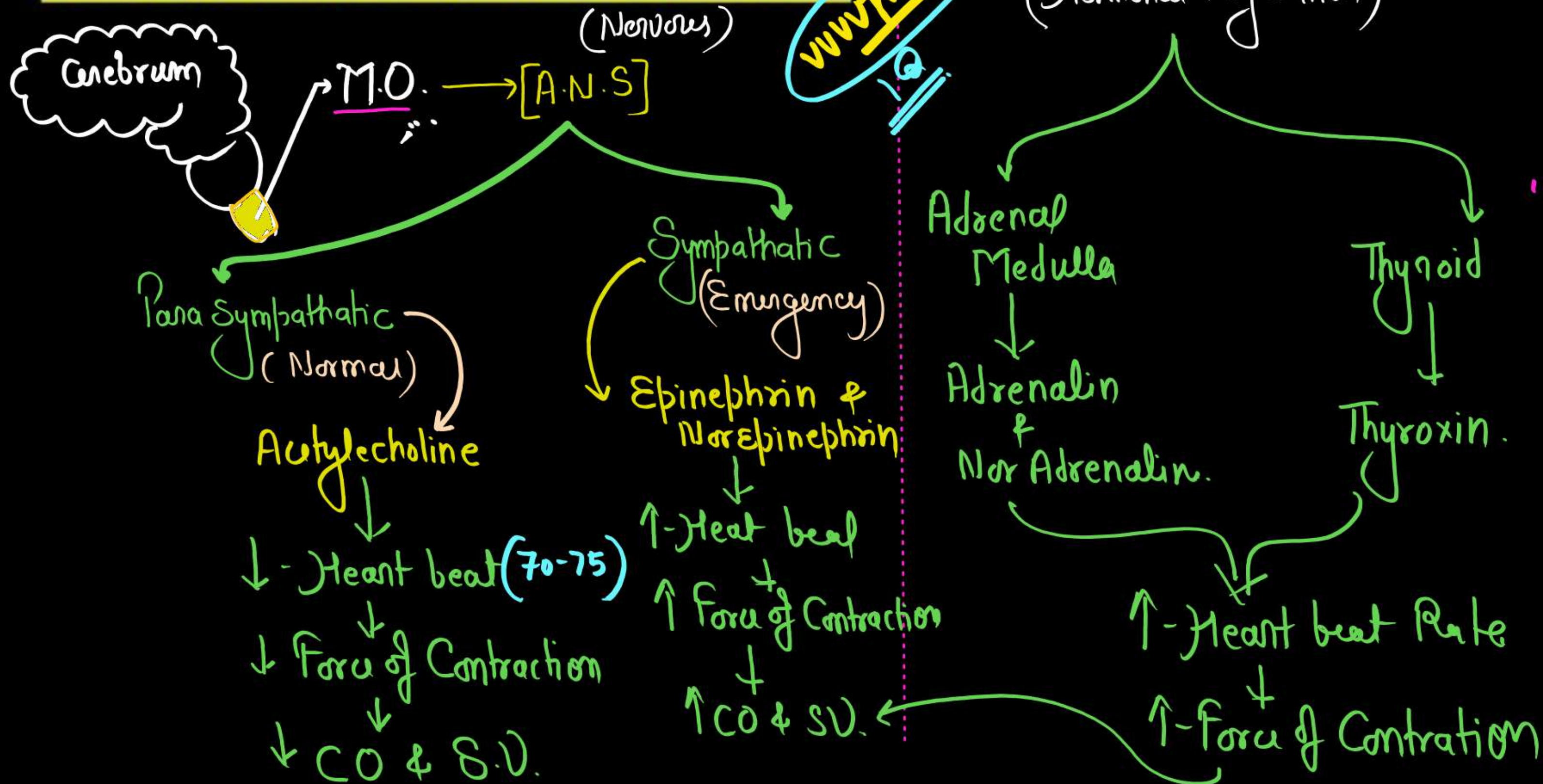
- * Position = InterVentricular Septum
- * Potential = 40-45 impulse/min.

4) - Purkinje fibres

- * Position = Wall of Ventricle
- * Potential = 20-25 impulse/min.



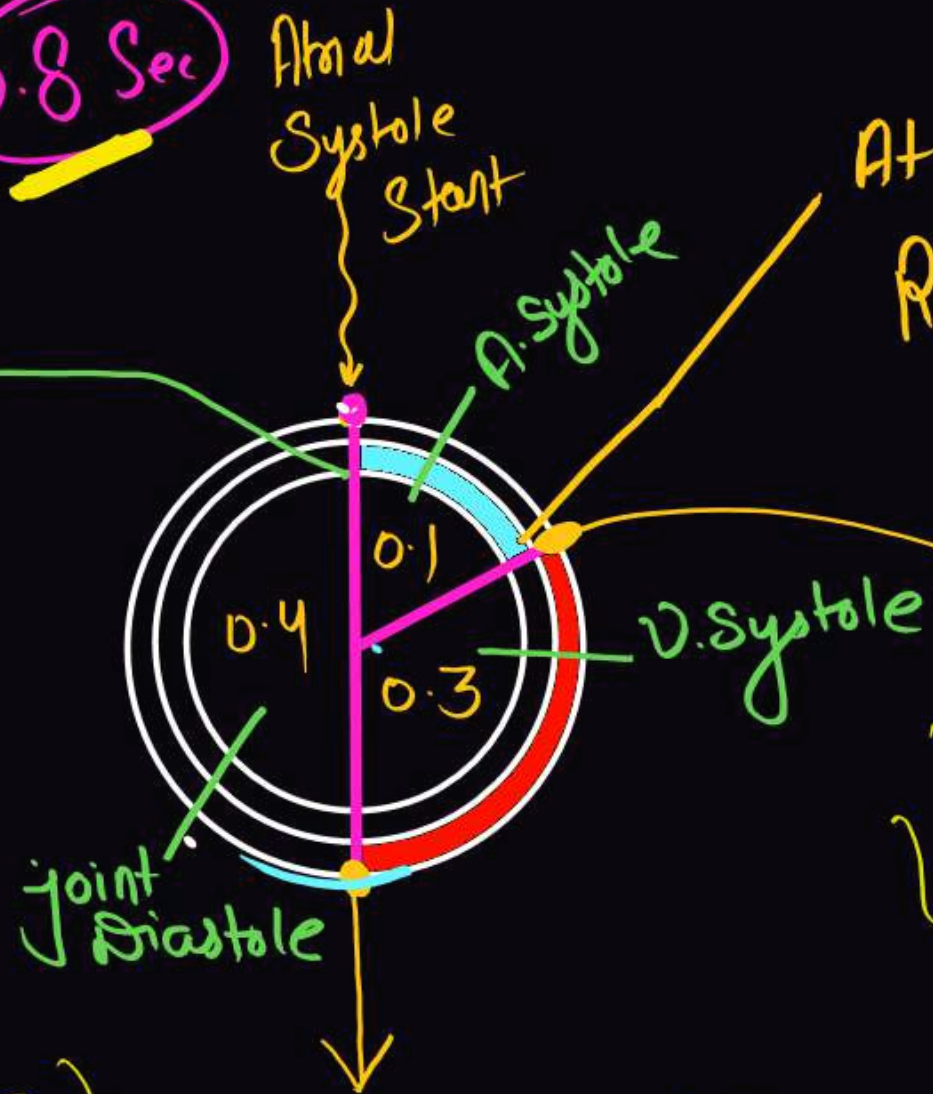
(Hormonal Regulation)



Cardiac Cycle

0.8 Sec

At the end of joint diastole $\Rightarrow$
70% ($\frac{2}{3}$) Ventricle
fill by blood Passively
in 0.4 Sec



At the end of Atrial Systole
Remaining $\frac{1}{3}$ (30%) Ventricle
fill by blood (Active)
= 0.1 Sec

Ventricle Systole Start
 I^{st} Heart Sound = LUB
Due to closing of A.V. Valve

Ventricular Diastole
[II^{nd} Heart Sound = Dub]
Due to closing of Semilunar Valve.

Duration b/w -

• I & II - Sound = 0.3

• II & I - Sound = 0.5

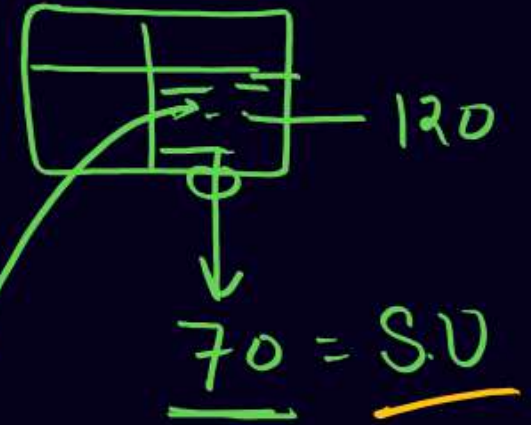
① **EDV** = (120 ml)
Each Ventricular

EDV $\propto$ Size of Ventricle
(EDV $\propto$ SV)

② **ESV** = 50 ml (Fixed)

E.S.V. = E.D.V. - S.V.

120 - 70 = 50 ml



③ **SV** = 70 ml

S.V. = EDV - ESV

= 120 - 50 = 70 ml

④ **CO** = S.V. $\times$ HBR/min.

= 70 $\times$ 72

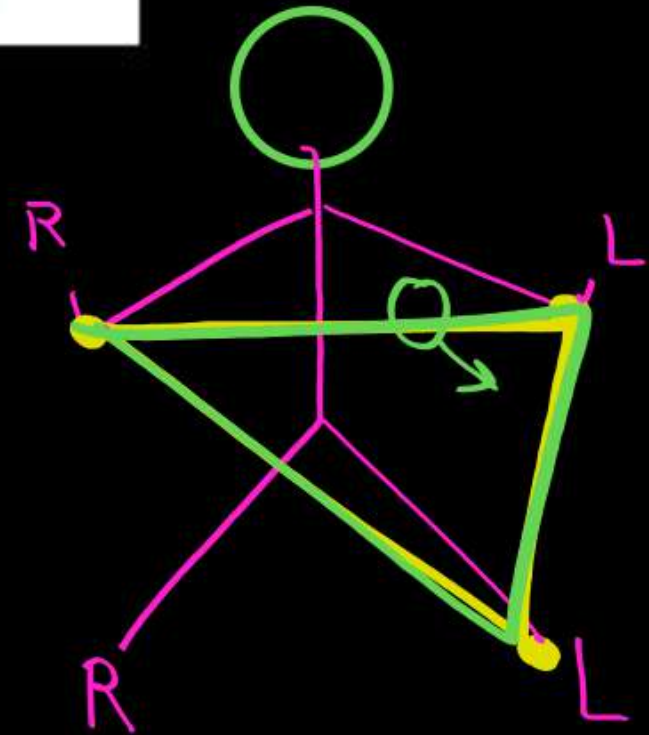
= 5040 ml/min.

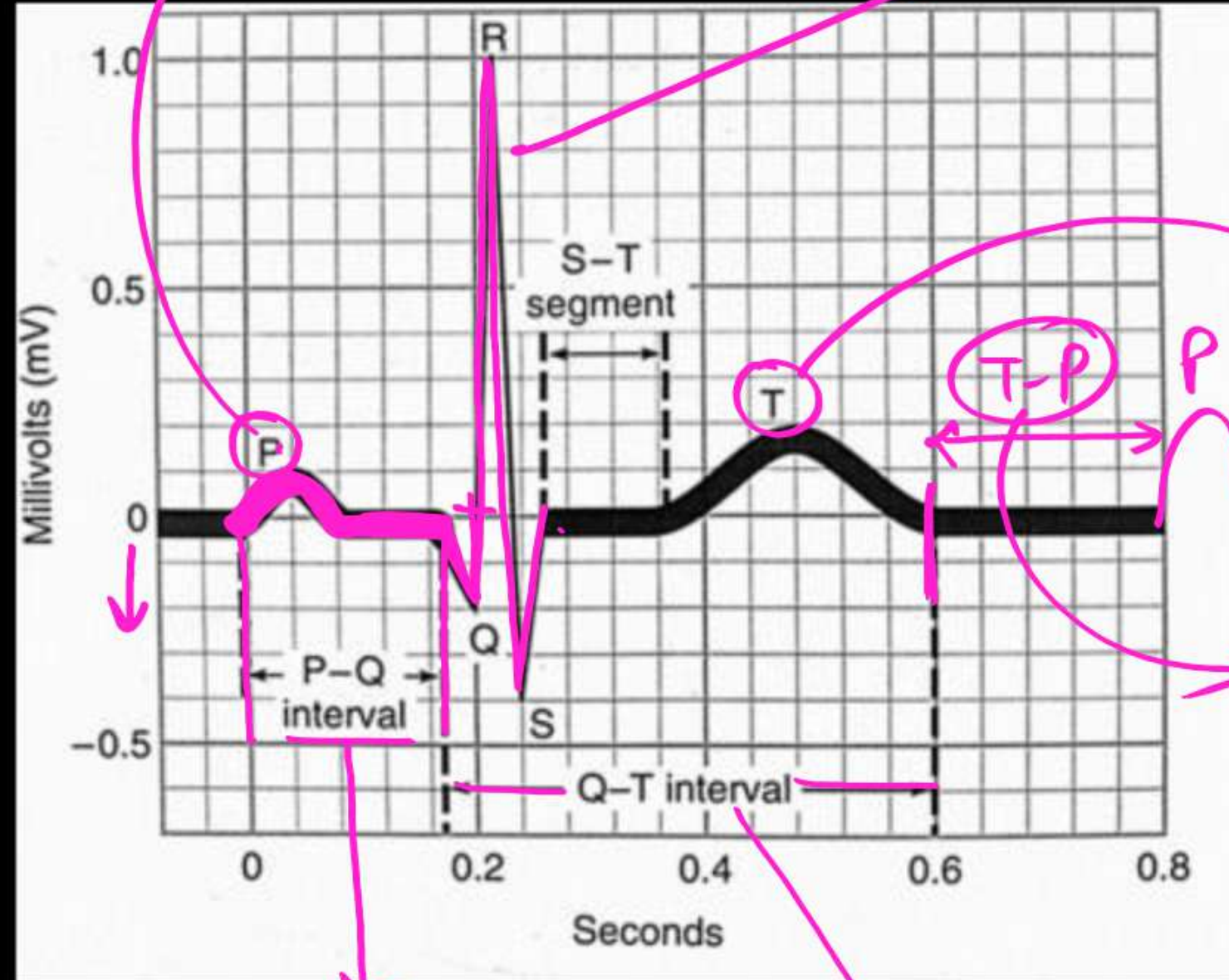
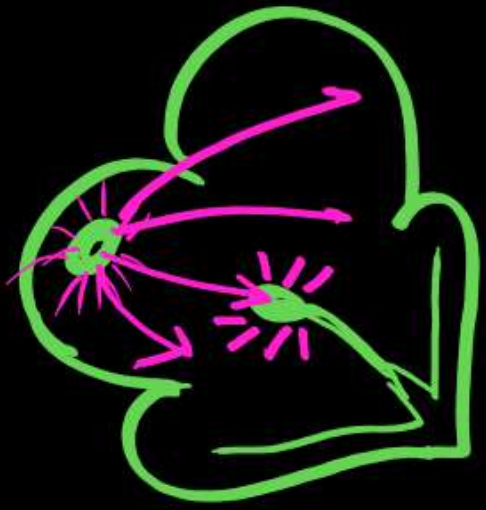
5 l/min.

15.3.3 Electrocardiogram (ECG)

You are probably familiar with this scene from a typical hospital television show: A patient is hooked up to a monitoring machine that shows voltage traces on a screen and makes the sound "... pip... pip... pip..... peeeeeeeeeeeeeeeeeeeeeee" as the patient goes into cardiac arrest. This type of machine (electro-cardiograph) is used to obtain an electrocardiogram (ECG). ECG is a graphical representation of the electrical activity of the heart during a cardiac cycle. To obtain a standard ECG (as shown in the

v v v p m p





Depolarisation
in Atria / S.A. Node

QRS = Depolarisation in
Ventricular / AV Node
(Start after = Q)

Repolarisation / Relax
in Ventricular
(Relaxation of Ventricular)

Depolarisation &
Contraction in Atria.

Depolarisation &
Contraction in Ventricular } at the end
of T wave

DISORDERS OF CIRCULATORY SYSTEM

(1) High Blood Pressure (Hypertension):

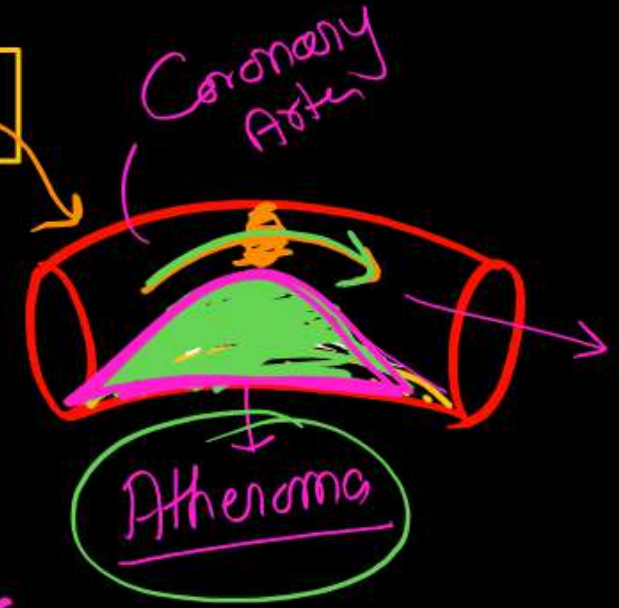
- Hypertension is the term for blood pressure that is higher than normal (120/80). If repeated checks of blood pressure of an individual is 140/90 (140 over 90) higher, it shows hypertension.
- High blood pressure leads to heart diseases and also affects vital organs like brain and kidney.

$$\left. \begin{array}{l} \text{Systolic} = \underline{120} \text{ mmHg} \\ \text{Diastolic} = \underline{80} \text{ mmHg} \end{array} \right\} 120 / 80 - 90$$

Handwritten notes: The number 140 is circled in orange with an arrow pointing to the 120. The number 90 is circled in orange. There are also some orange and green scribbles.

@ (2) Coronary Artery Disease (CAD):

- Coronary Artery Disease, often referred to as atherosclerosis, affects the vessels that supply blood to the heart muscle.
- It is caused by deposits of calcium, fat, cholesterol and fibrous tissues, which makes the lumen of arteries narrower. blood flow = ↓



(3) Angina:

- It is also called 'angina pectoris'. A symptom of acute chest pain appears when no enough oxygen is reaching the heart muscle.
- Angina can occur in men and women of any age but it is more common among the middle-aged and elderly.
- It occurs due to conditions that affect the blood flow.

(4) Heart Failure:

- Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.
- It is sometimes called congestive heart failure because congestion of the lungs is one of the main symptoms of this disease.

(5) Heart attack

Death of a part of Cardiac Muscles

(6) Cardiac arrest → Complete Death of Cardiac Mus.
CO = 0
Pip-pip-pip-peeEEEE ✓



@DR_MANISH_DU
BEY_MD_SIR_BMS

Locomotion

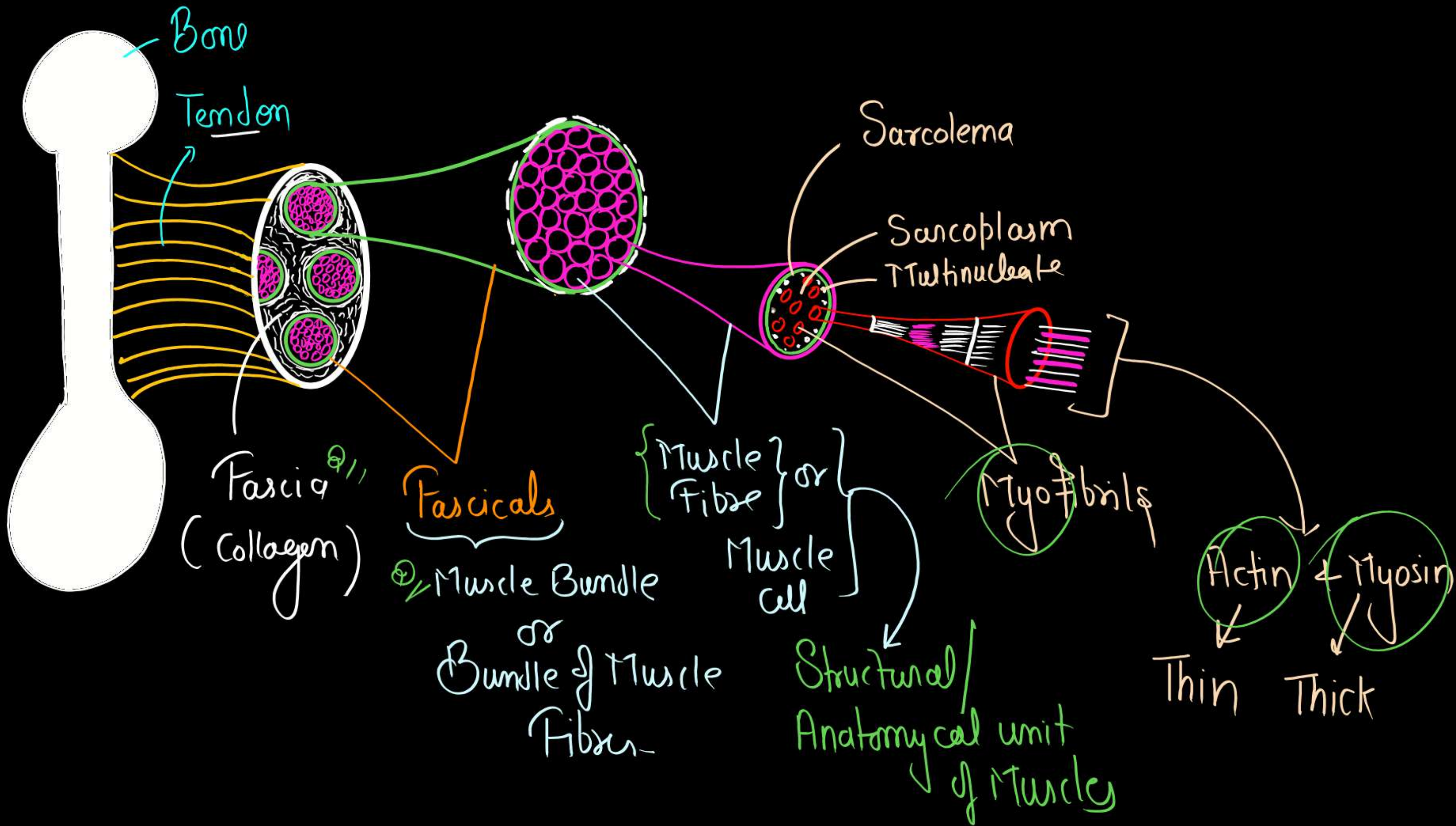
Skeletal Muscles

- Voluntary ✓
- Striated
- Striped
- Multinucleated

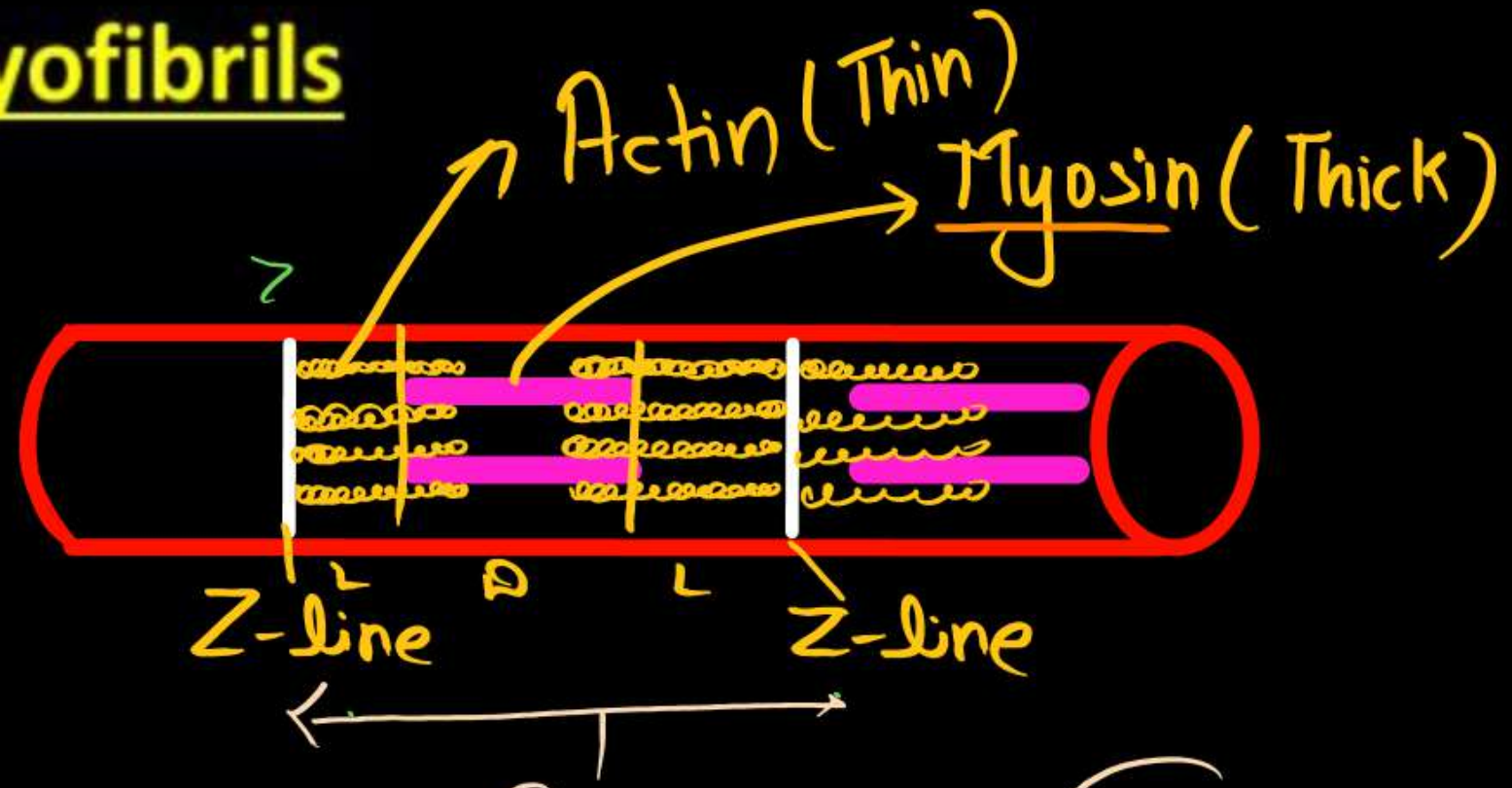
Light & Dark band

Transverse. band //

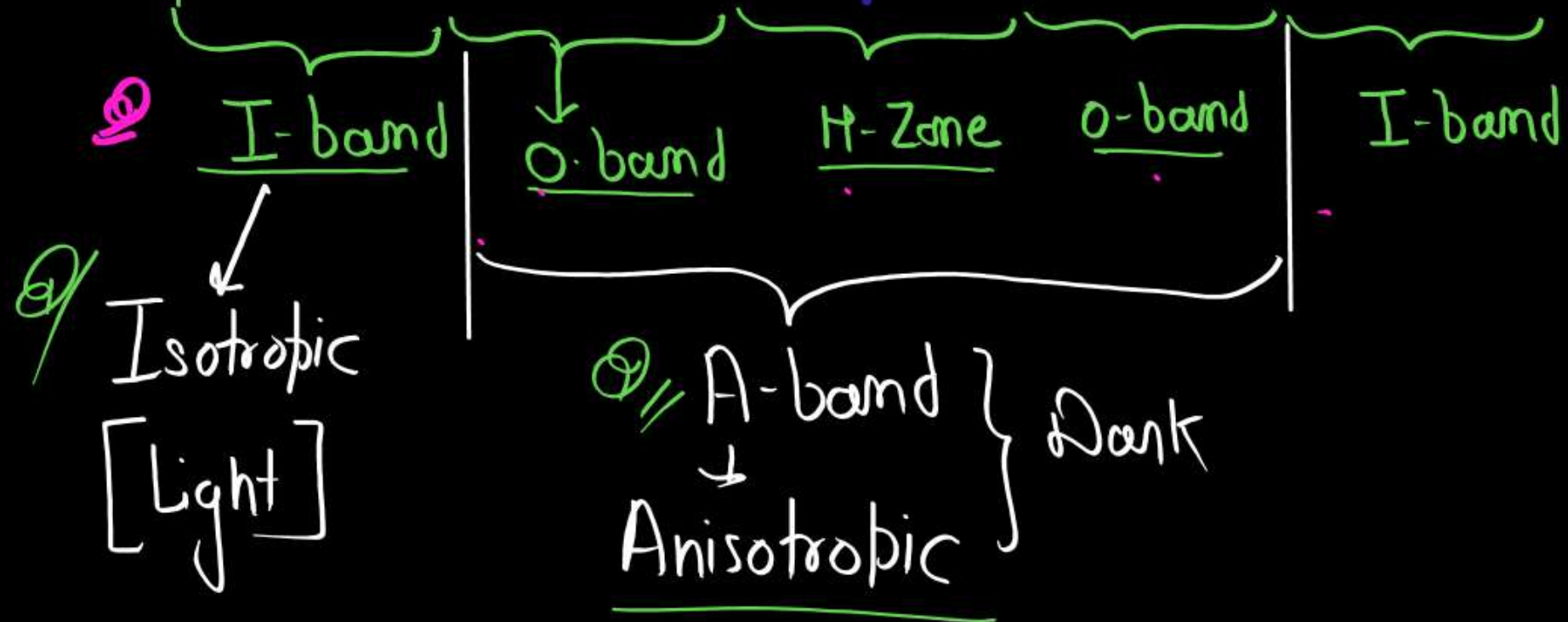
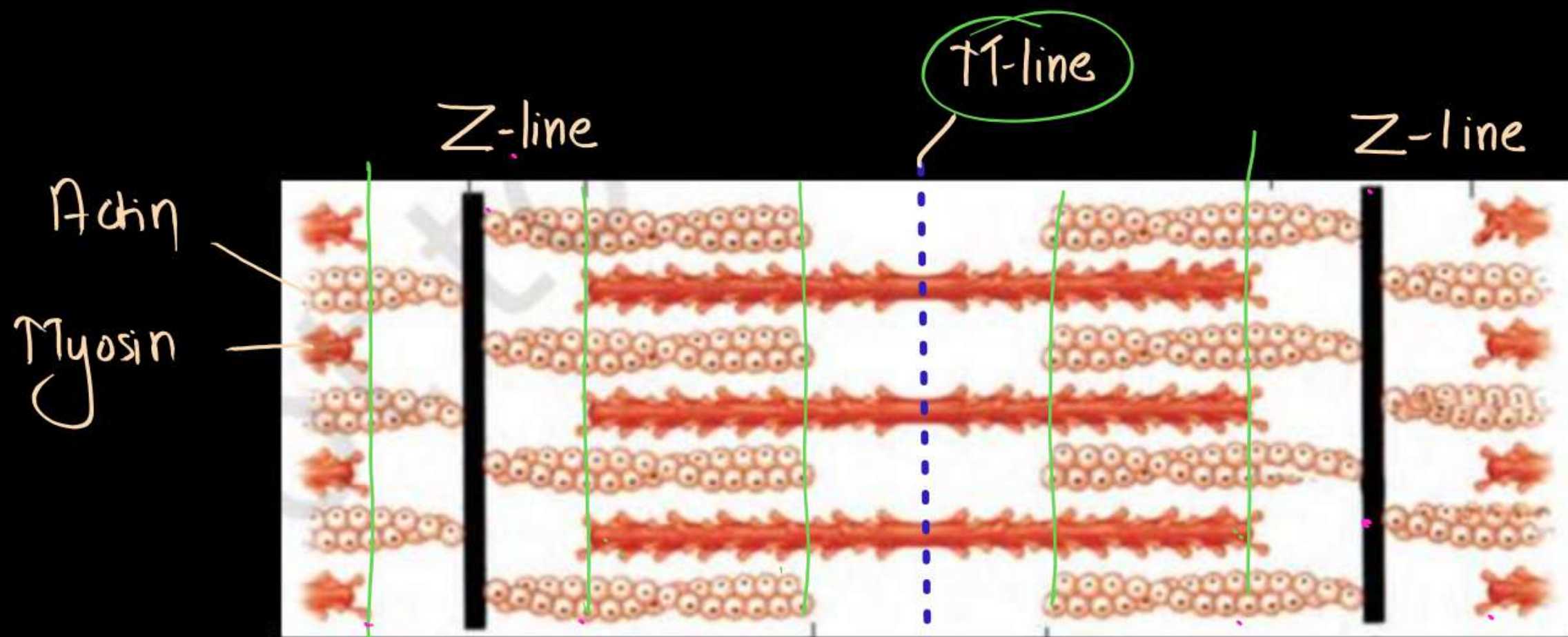




Structure of Myofibrils



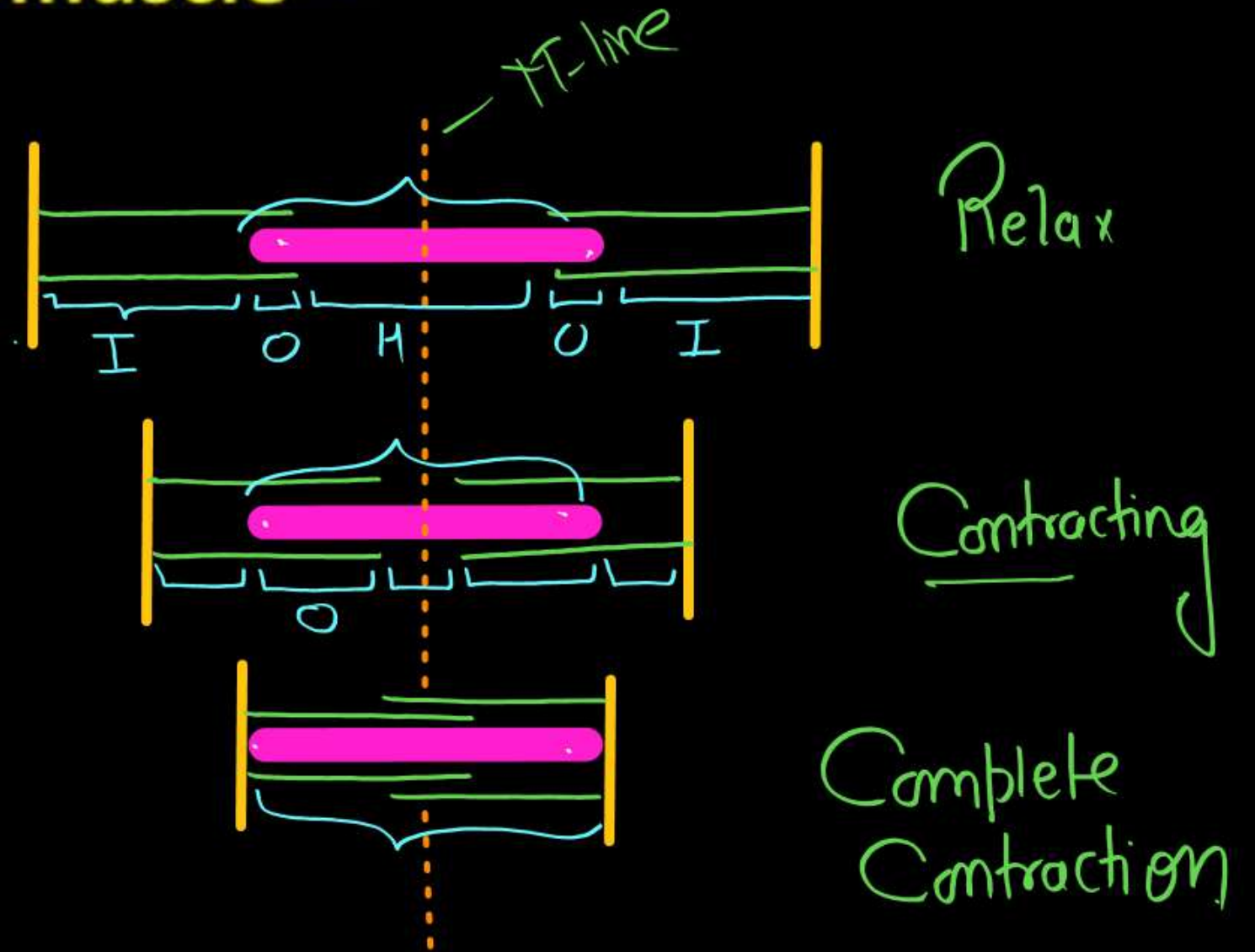
Sarcomere = Functional
unit of Muscle.



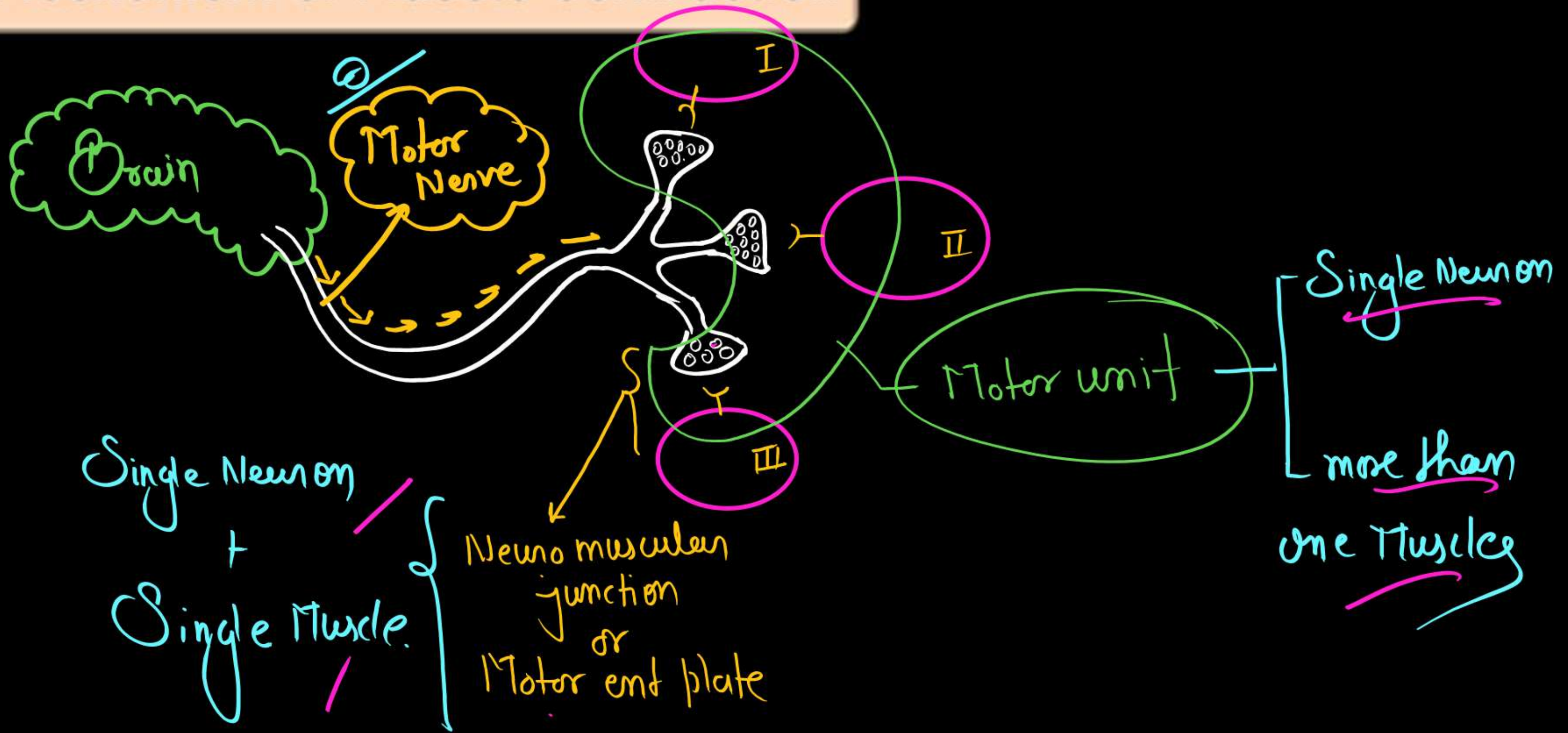
M-line + ht b/w =
 A-band / H-Zone
 Z-line + ht b/w
 I-band

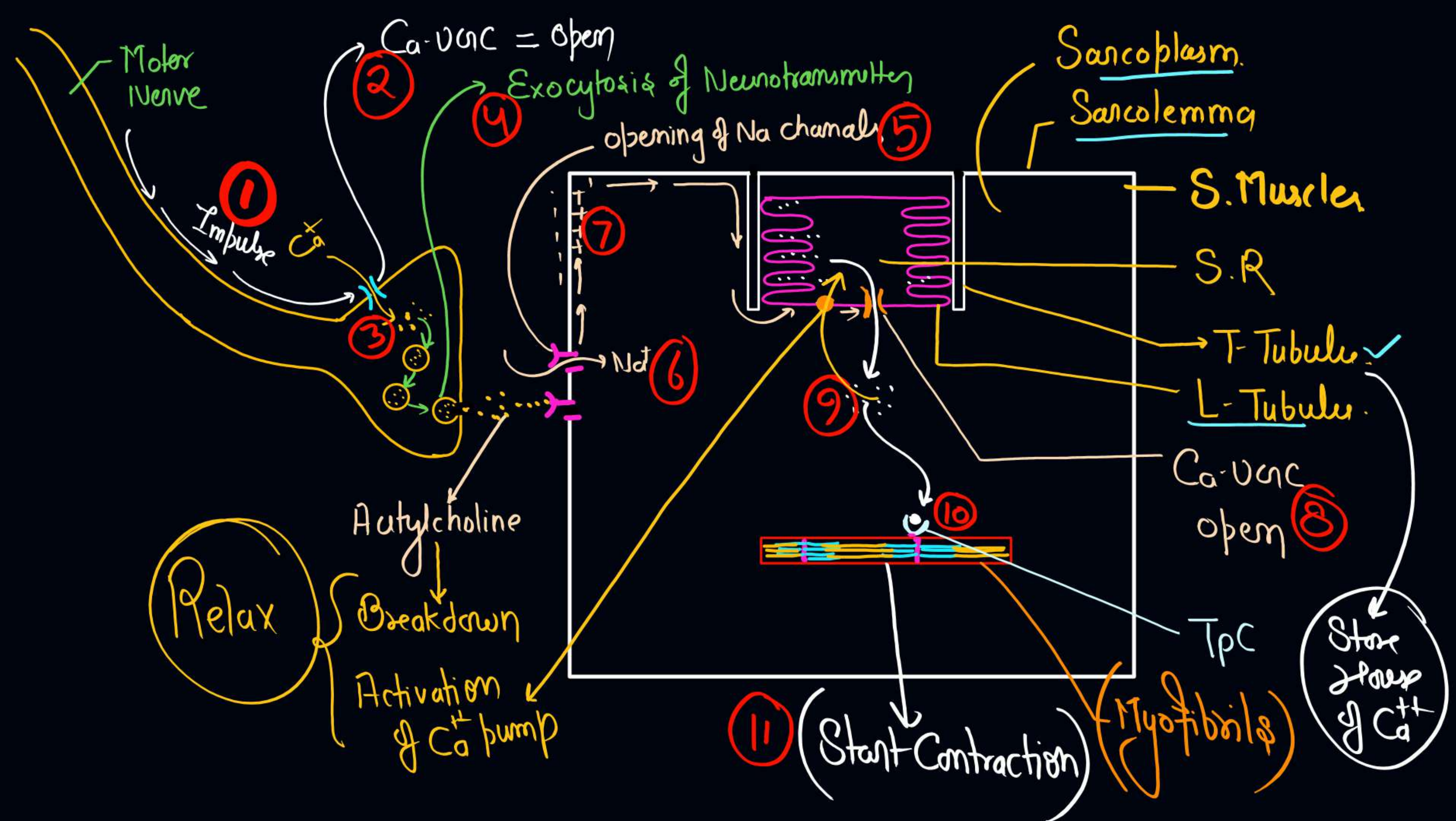
During Complete contraction of muscle –

- Length of Sarcomere = ↓
- Length of I Band = ↓
- Length of H Zone = ↓
- M-line = Disappear
- Length of O band = ↑
- Length of A band = Unchange

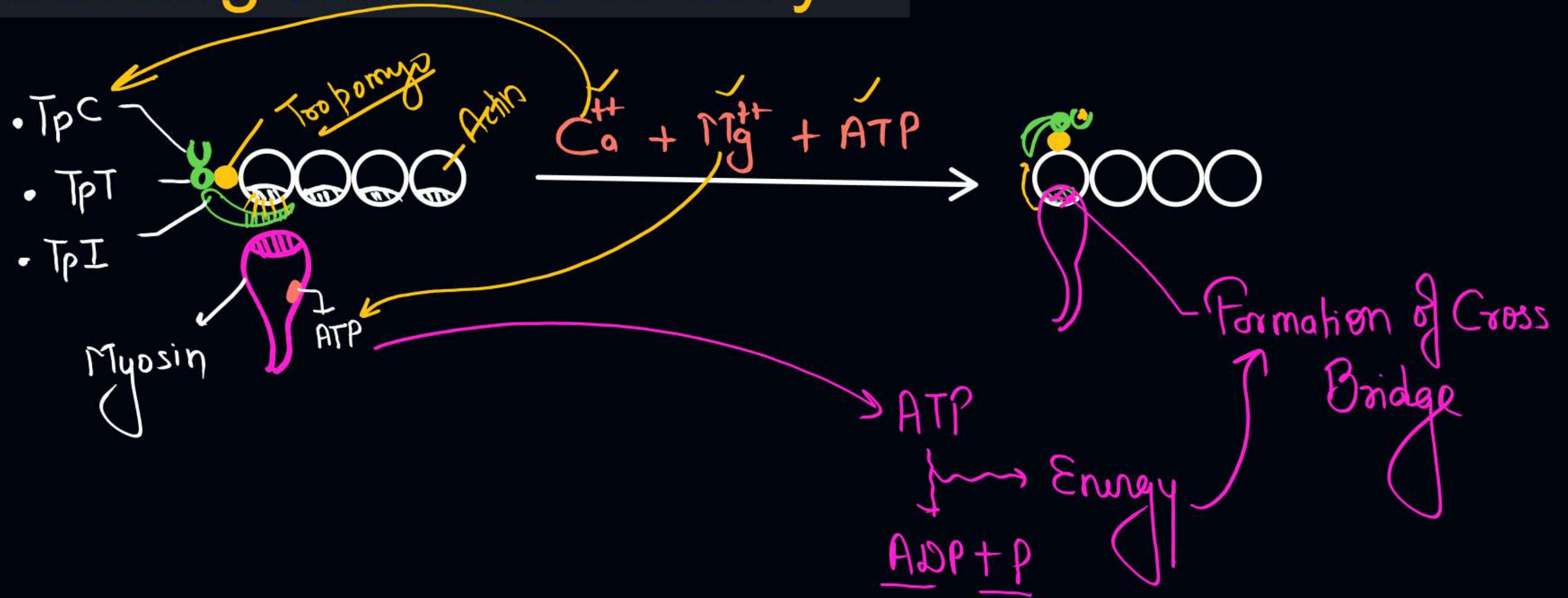


Mechanism of Muscle Contraction

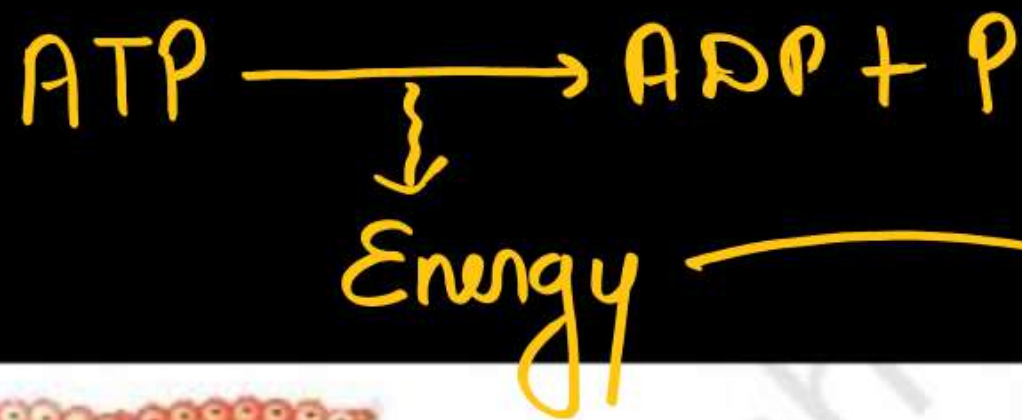




Sliding filament Theory

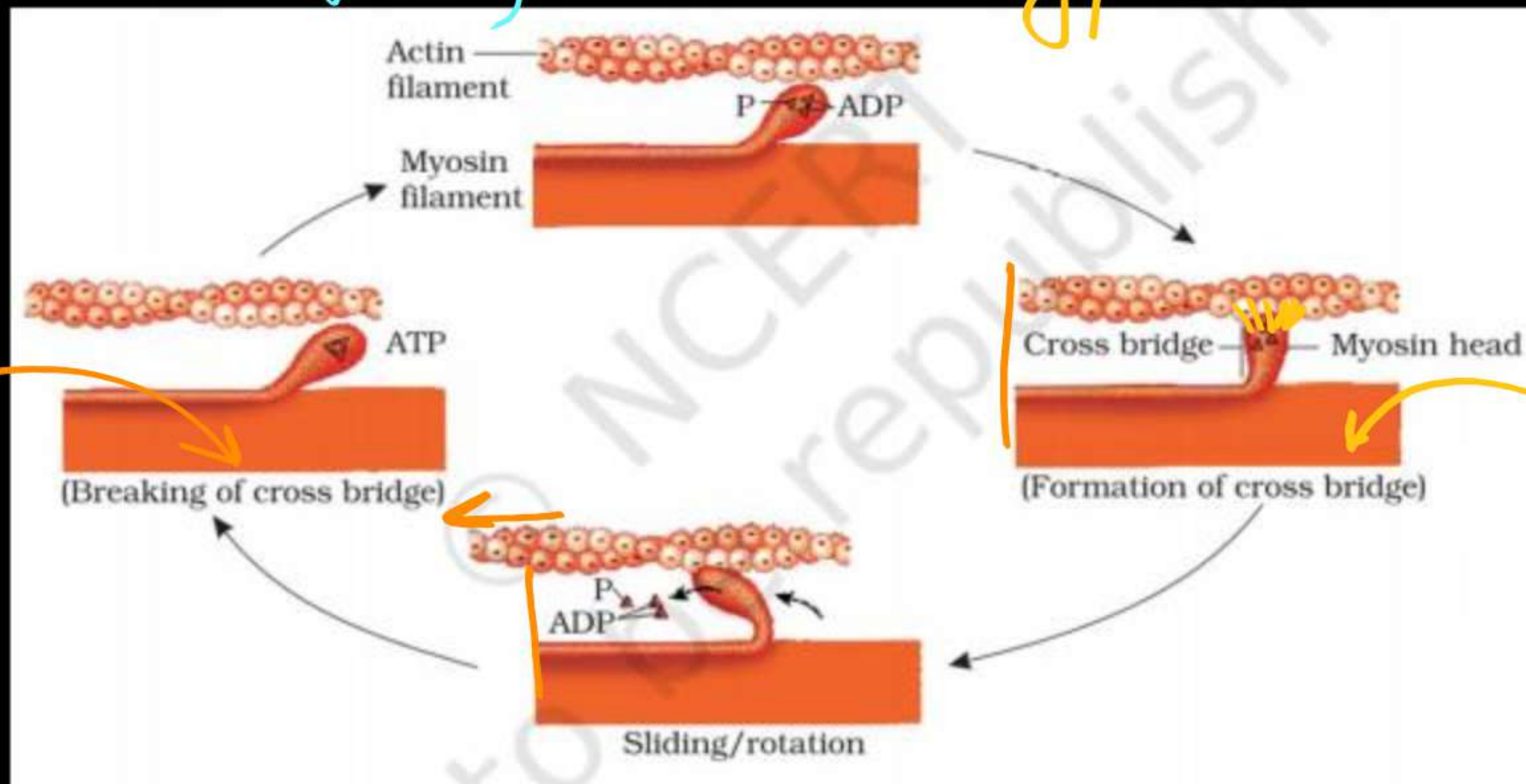


Crossbridge Formation = Breakdown of ATP
Crossbridge Breakdown = Formation of ATP



Breakdown of Cross bridge

Formation of Cross Bridge



ATP

ADP
+ High Energy Phosphate

Types of Muscles

skeletal

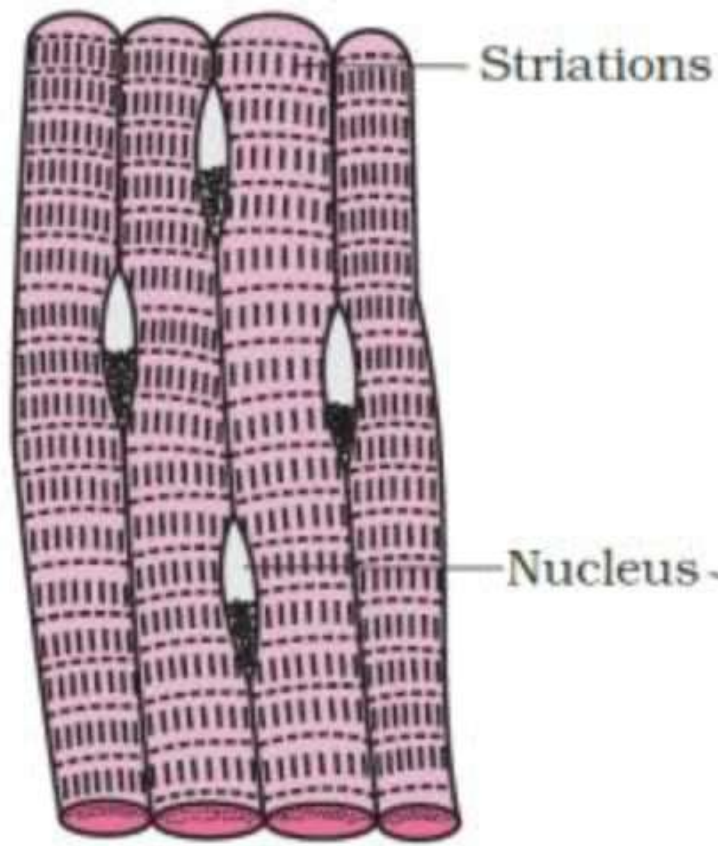
smooth

Cardiac

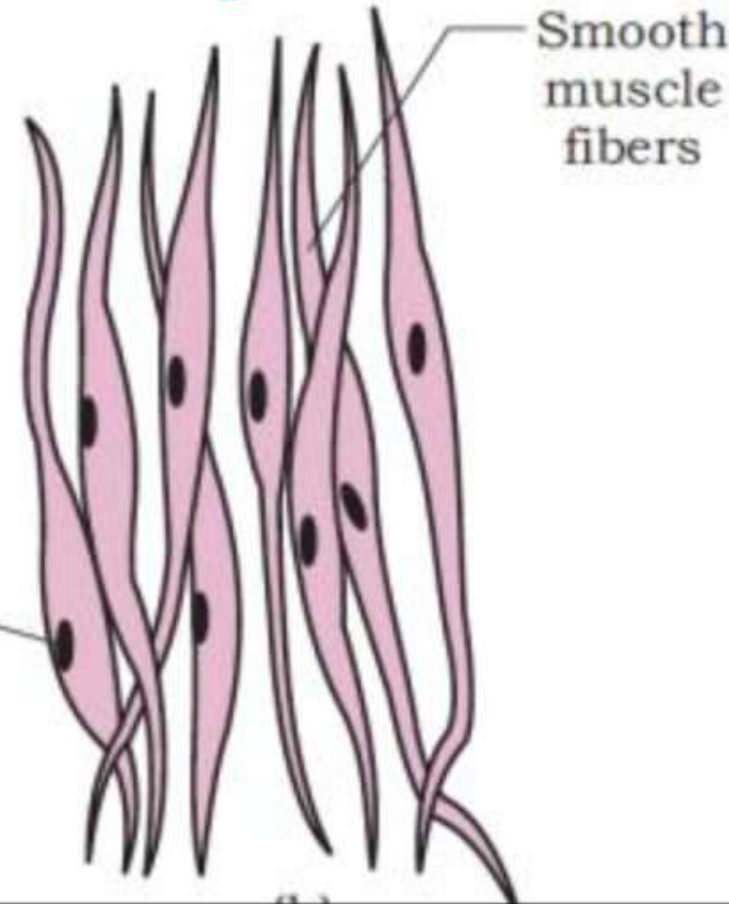
Limbs
Muscles

organs

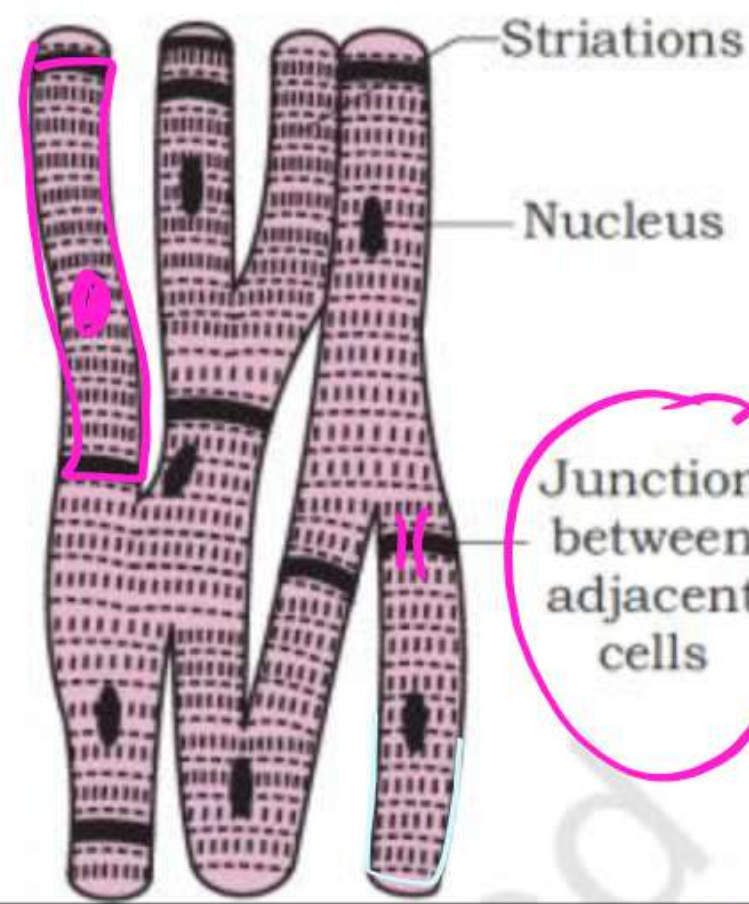
Heart



Cylindrical / unbranched
Multinucleated
Striated
Voluntary



Spindle Shape
Uninucleated
Unstriated
Involuntary



Cylindrical / Branched
Uninucleate
Striated
Involuntary

Human Skeleton (206)

Axial (80)

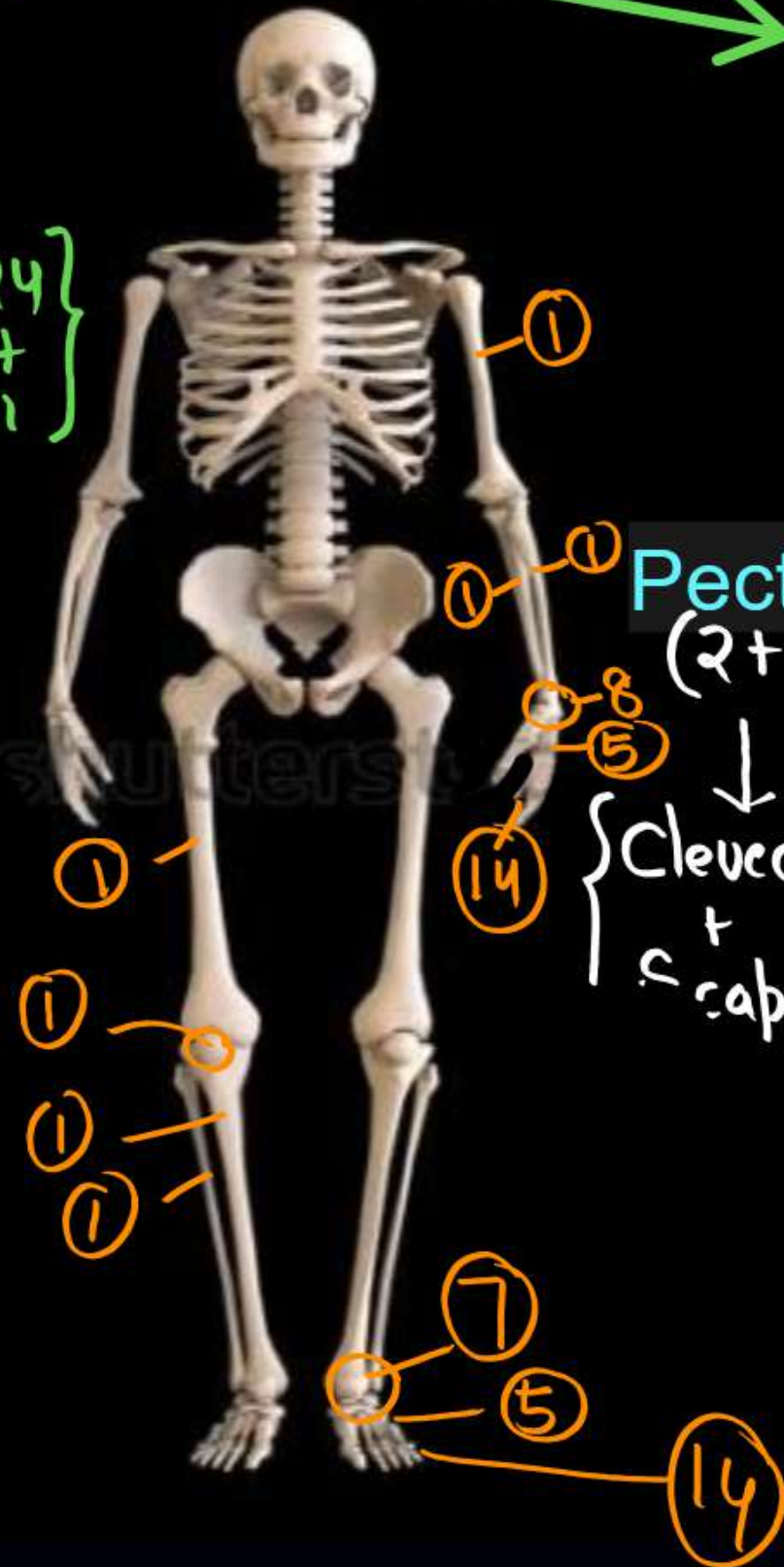
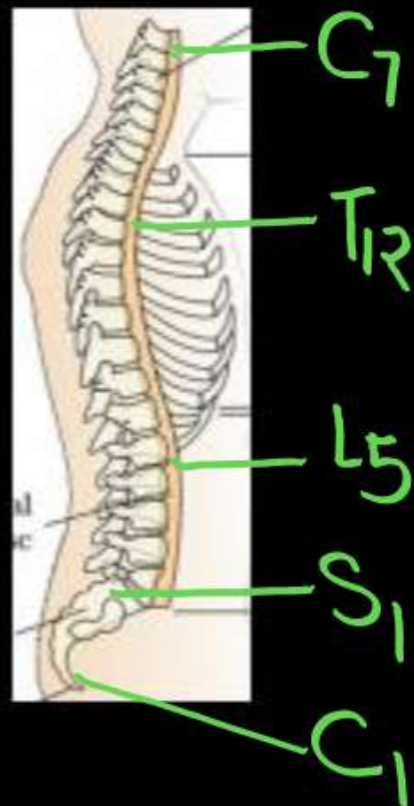
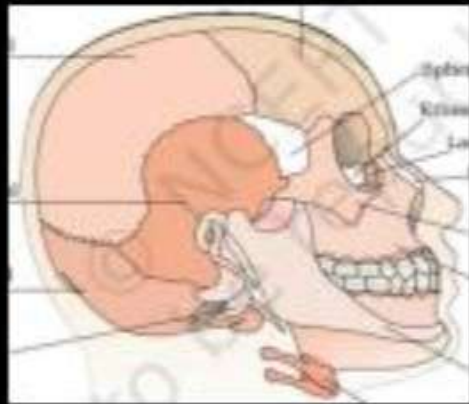
Appendicular (126)

Head (22)

V. Column (26)

Ribs + Sternum (24 + 1)

Skull (22)
+ Hyoid (1)
+ Ear Ossicles (3+3)
Cranium (8)
Facial (14)



Girdle (6)

Limb bones (120)

$$30 \times 4 = 120$$

Pectoral (2+2)

Cleavage + scapula

Pelvic (1+1)

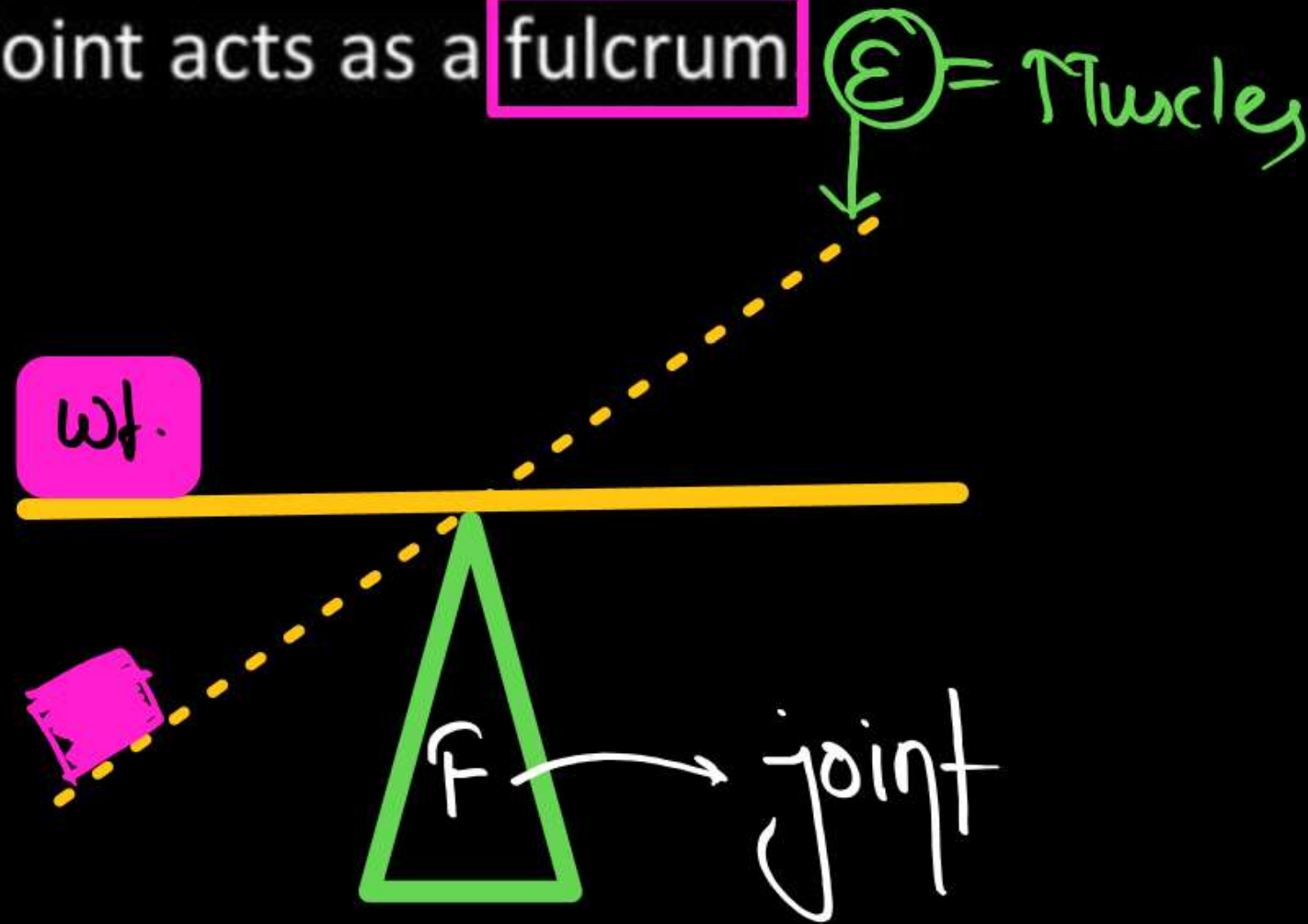
Coxal

Joints

vvvvvmp $1Q = 100\%$

Joints are **points of contact** between bones, or between bones and cartilages.

Force generated by the **muscles** is used to carry out movement through joints, where the joint acts as a **fulcrum**



Joints

Fibrous joints

(Immovable)

eg → Sutures of Skull

Cartilaginous joints

(Semimovable)

eg → Pubic Symphysis
→ Intervertebral Disc

Synovial joints

(Movable)

Pivot b/w Atlas & Axis

Gliding joints • Wrist & Ankle

Saddle joint  • Carpal & Metacarpal of Thumb

Hinge

• Knee & Elbow
• b/w Phalanges

Ball and Socket joints

• Shoulder
• Hip

Condyloid joints

• Skull & Atlas
• Metacarpal & Phalanges

Disorders

20.5 DISORDERS OF MUSCULAR AND SKELETAL SYSTEM

Myasthenia gravis: Auto immune disorder affecting neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

Muscular dystrophy: Progressive degeneration of skeletal muscle mostly due to genetic disorder.

Tetany: Rapid spasms (wild contractions) in muscle due to low Ca^{++} in body fluid.

Arthritis: Inflammation of joints.

Osteoporosis: Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.

Gout: Inflammation of joints due to accumulation of uric acid crystals.

Hormones

I.R

Hypothalamus

Secretory Neurons (Nuclei)

R.H.

PRh
GnRH
CRh
TRh

Portal System

A. Pituitary

PP

Supra optic

Para Ventricular

GHRH
Somatotrophin

① Somatotrophic (STH)

or Growth Hormone

GH

② T.S.H

T₃/T₄ Thyroxin

③ ACTH

Cortisol

④ Gonadotrophin

MSH

⑤ PRL

L.H

F.S.H

Antrogen Sperma.

ovulation
Corpus luteum
Prog.

SMP

Development of Folli
Estrogen.

Vasopressin
[ADH]

[Oxytocin]

⑥ { Contraction of B.U
Absorption of water
from - [ACT
CD

• Contraction in
1) Mammary glands
[Milk Ejection]
2) - Uterus
[Birth]

(Both Synthe. by Hypothalamo.)

• Lactation
• D. Mammary

Disorders

~~vvvmp~~

Hyposecretion of G.H.

Hypertecretion of G.H.

Diabetes Insipidus
(Hyposecretion of ADH)

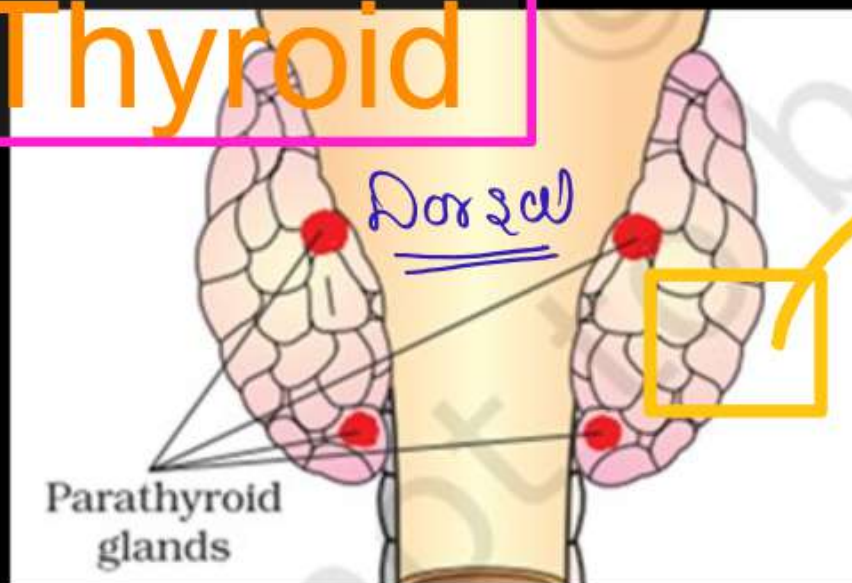
Pituitary Dwarfism
(Midget) (↓ of G.H.)

Acromegaly → Hypertecretion in late stage
↳ Premature Death if unchecked

Gigantism — Hypertecretion in Early stage

Diabetogenic effect → ↑ - G.H.
↳ ↑ - Blood Glucose level.

Thyroid



Thyroid

B.O.



Stromal Tissue

Follicles

[Simple Cuboidal Epi. tissue]

Thyroglobulin [Store form of Thyroxine]

Parafollicular Cells

Release - **T.C.T.**

[Hypocalcemic Hormone]

↓ - Blood Ca^{++} level

by —→ Adsorption of blood Ca^{++} in bone matrix
—→ Elimination of Extra Ca^{++} through kidney

✓ Thyroxine = Iodinated form of Tyrosine Amino acid.

- ✓ • Tri-iodothyronine [T_3] → Most active form
- ✓ • Tetraiodothyronine [T_4] = Thyroxine ✓

Function ⇒

- 1) - Regulation of BMR ✓
- 2) - ↑ Erythropoiesis = ↑ Formation of R.B.C
- 3) - ↑ Blood Glucose level, Breakdown of — [Fat, Protein, Carbo.]
- 4) - Maintain water-Electrolyte Balance

Disorders

~~vvvvvvvvvv~~

Hyposecretion

of thyroxine

1) - Simple Goiter

2) - Cretinism
(During Pregnancy) { Deaf Mutism
Abnormal Skin
Low I.Q.

3) - Irregular Menstrual ✓
In adult Female.

Hypersecretion

of Thyroxine.

✓ • Exophthalmic Goiter
or

• Grave's Disease

• Diabetogenic Effect
= ↑ - thyroxine

Parathyroid = (2 pairs)

Release = P.T.H
[Hypercalcemic]

↑ - Blood Ca^{++} & ↓ Blood Phospho.

1) - Stimulate = Osteoclast
(↑ - Breakdown of Bone)

2) - Stimulate = J.G. Cells of Kidney
Calcitriol = ↑ absorption
of Ca^{++} from gut

Disorders

Hyposecretion
of P.T.H

↓
[↓ - Blood Ca^{++} level]

↓
(Tetany)

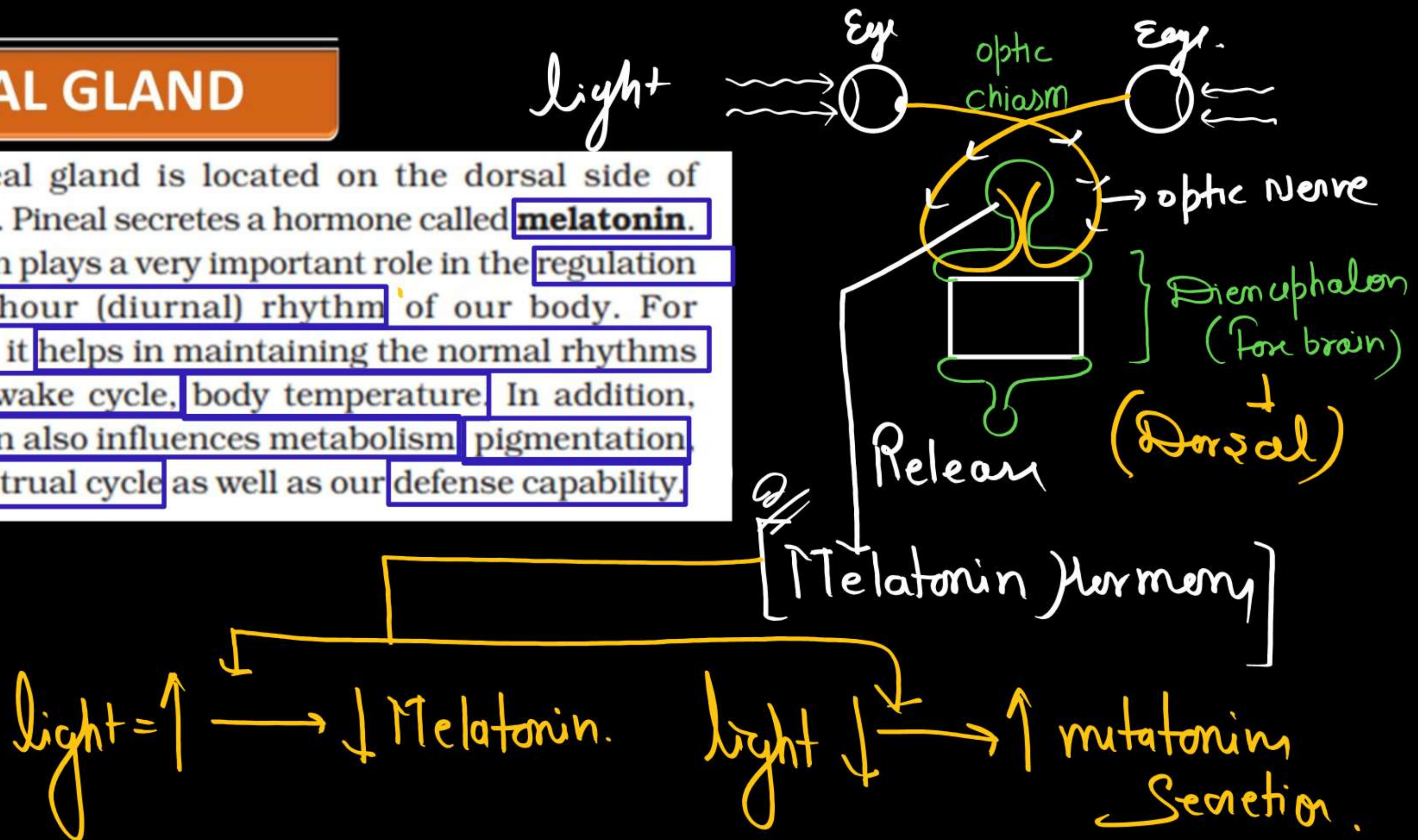
Hypersecretion
of P.T.H

↓
↑ - Breakdown of bone

↓
(Osteoporosis)

PINEAL GLAND

The pineal gland is located on the dorsal side of forebrain. Pineal secretes a hormone called **melatonin**. Melatonin plays a very important role in the regulation of a 24-hour (diurnal) rhythm of our body. For example, it helps in maintaining the normal rhythms of sleep-wake cycle, body temperature. In addition, melatonin also influences metabolism, pigmentation, the menstrual cycle as well as our defense capability.



[Thymus]

Release

[Thymosin Hormone]

Cell Mediated Immunity (CMI)

or

T-lym. Mediated Immunity

T_H-cell

T_K-cell

T_S-cell

T-Memory cell

Maturation

&
Differentiation
of T-lym

Humoral Immunity

or

B-lym Mediated Immunity (B.M.I)

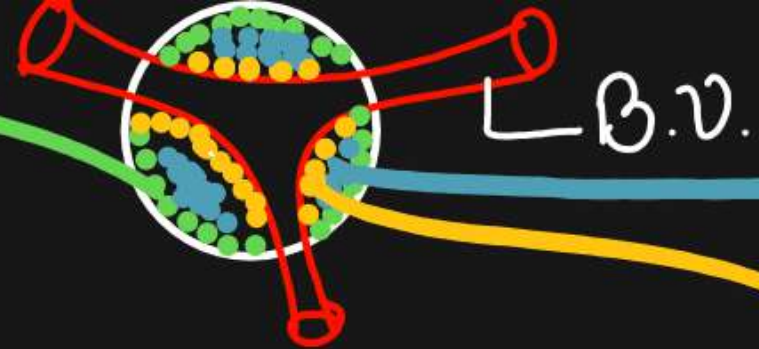
Plasma cells

Memory

Production
of

(Antibody)

Pancreas



[α-Cells]

Release - Glucagon [Hyperglycemic]

↑ - Blood Glucose by -

1) - Gluconeogenesis = Fat/Protein > Break

2) - Glycogenolysis = Glycogen

3) - ↓ Cellular uptake of Glucose

[β-Cells]

Release - Insulin [Hypoglycemic]

↓ - Blood Glucose level by -

↑ - Cellular uptake of Glucose

[δ-cells]

Somatostatin

Inhibit ↓ α-cell, β-cell

Disorders

Diabetes Mellitus

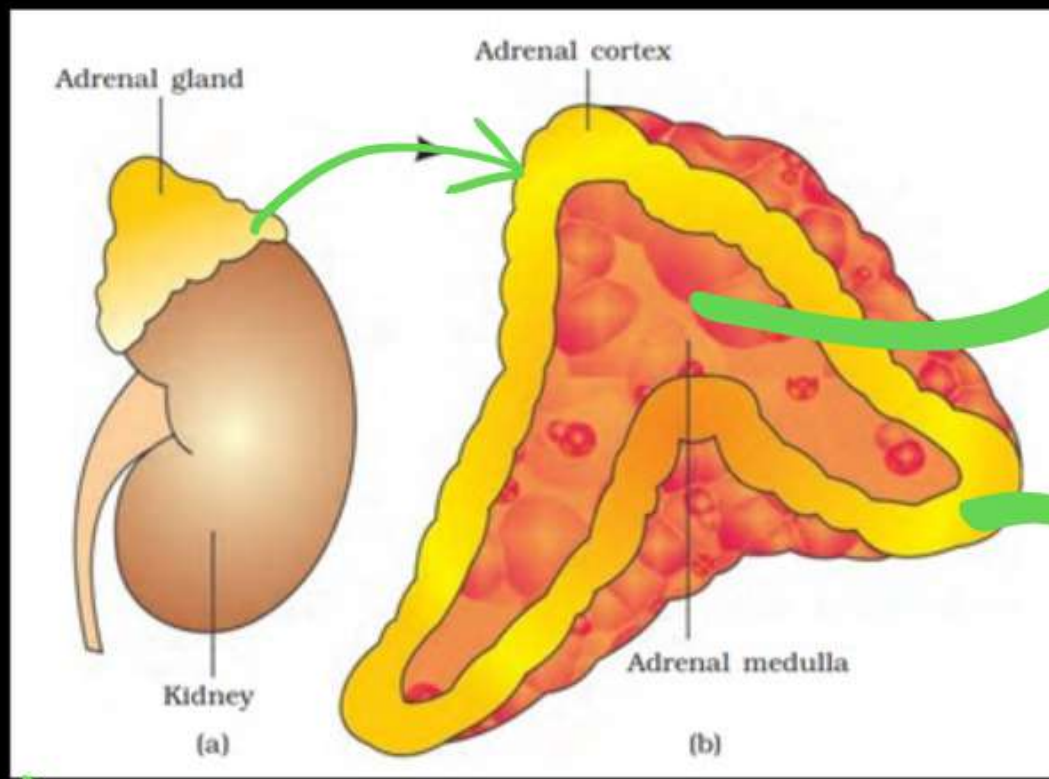
↑ Blood Glucose

↑ - Ketone bodies

↑ - Glucosuria

} Hyposecretion of Insulin

~~vulgar~~



Medulla [Inner]

- Origin = Ectodermal
- Nature of Hormone = Protein / Emergency Hormones or F.F. Hormones or Catecholamines

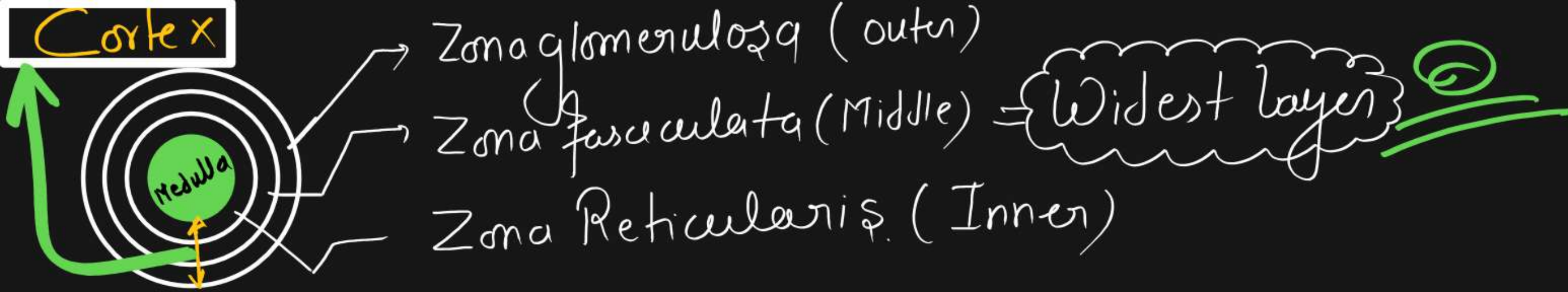
① Adrenalin [80%]

② Noradrenalin [20%]

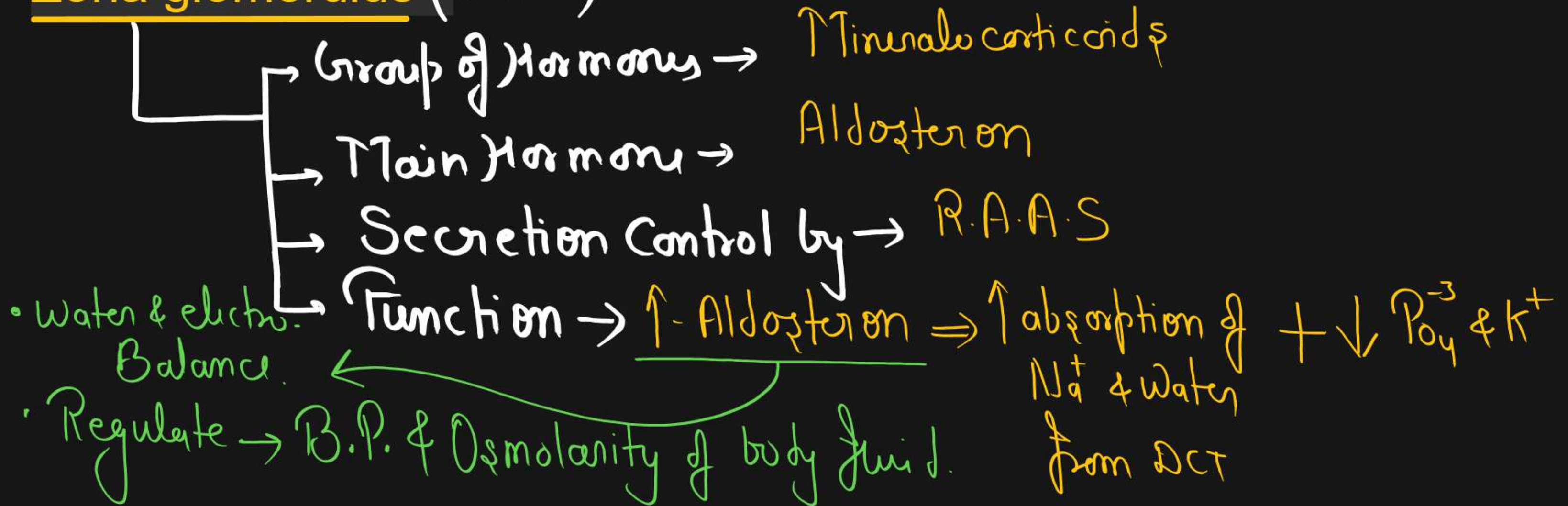
- 1) - ↑ alertness
- 2) - Pupillary dilation, ③ piloerection
- 4) - Sweating etc
- 5) - ↑ heart beat, Rate of Respiration
- 6) - Break down of Glycogen (↑ Blood Sugar)
- 7) - " " of Lipid & Protein.

Cortex [Outer]

- Origin = Mesodermal
- Nature of Hormones = Steroids
- Vitally Imp. for Life



Zona glomerulus (outer)



Zona fasciculata (Middle)

- 1) - Group of Hormone → Glucocorticoids
- 2) - Main Hormone → Cortisol (Hydroxycortisone)
- 3) - Secretion Control by → A.C.T.H A. Pituitary (Life Saving Hor.)
- 4) - ~~Spoke~~ Function
 - Stimulate - Formation of R.B.C
 - Antistress Hormone.
 - ↑ - Blood glucose level.
 - Act as Immuno suppressor
 - Used in ⇒ Autoimmune disorders ✓
 - ⇒ Organ Transplantation ✓
 - ⇒ Allergy ✓
 - ⇒ Inflammation ✓

Zona reticulosis (Inner) → Sex Corticoids

Female Sex Hormone } ⇒ Estradiol

Male Sex Hormone } ⇒ [D.H.E.A]
(Androgen)

- Responsible
- Body Hair
 - Pubic Hair
 - Facial Hair at Puberty

Disorders →

Hyposecretion

⇒ Addison's (↓ Gluco. Miniralo.)
Re-NEET

Hypersecretion ⇒ Cushing (↑ Glucocorti.)
Conn's (↑ Mineralocor.)



Hormone of - vvvmp

Imorga.

1) - Heart = ANF

↳ Vasodilator -
inhibit RAAS -
Diuretic

2) - Kidney (J.G. cells) = Erythropoietin = ↑ Erythropoiesis

3) - GIT = Gastrin ✓ = ↑ Gastric juice

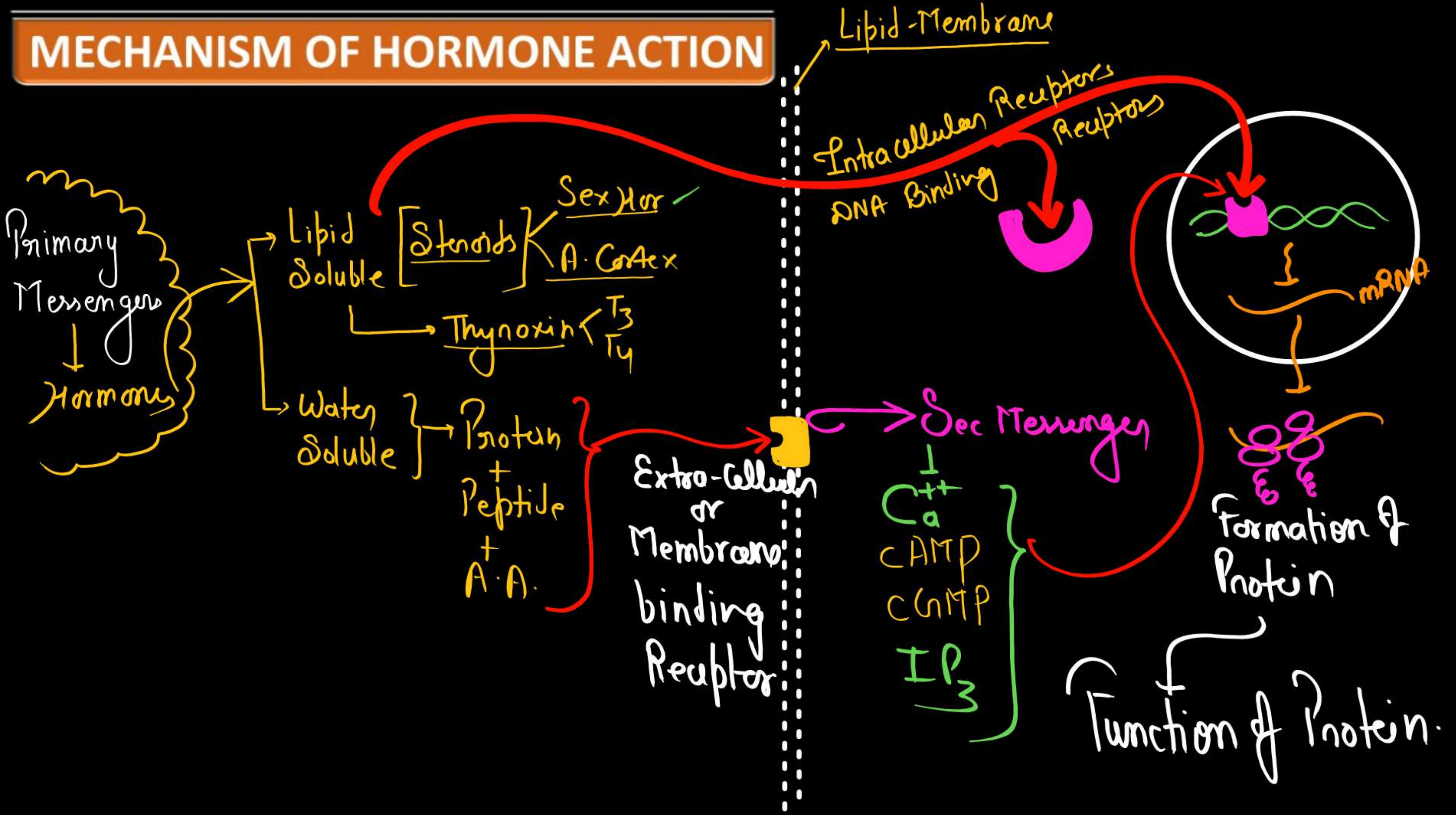
GIP ✓ = Inhibit Gastric Motility & HCl

Secretin = ↗ Stimulate Secretion of Alkaline fluid from - [Liver
Pancreas

CCK = Stimulate ↗ Pancreatic juice
Bile juice.

mp

MECHANISM OF HORMONE ACTION





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Excretory

Types of nitrogenous waste

Ammonia

- Pri. Nitrogenous Waste
- Highly Toxic → Not Storable
- Readily Soluble → For removal require lot of water
- Ammonia diffuse through Skin & Gills = NH_4^+
- Kidney do not play any significant Role in its Removal

Ammoniotelic

eg -
Aq. Arthropods ✓
Aq. Amphibia (larva) ✓
Bony fish ✓

Types of nitrogenous waste

Urea

- Produced in Live from Ammonia (Ornithine Cycle)

- Less Toxic = Small amount Storable

Blood = 30 mg/100ml Kidney = Maintain

Osmolarity Gradient (Counter Current)

- Less Soluble → For removal require less water

Ureotelic

- eg- Terrestrial Amphibia (Adult)
- Marine fish (Carti.)
 - Mammals

Types of nitrogenous waste

Uric acid

- In Birds & Reptile → Major End Product of Protein Metabolism
- In Mammals uric acid also produced

by breakdown of Purines (A & G)

A + G → Xanthine → Uric acid Crystals

- Uric Acid — Least Toxic ✓
— Least Soluble → for removal
Require least water

Uricotelic

eg - Reptile ✓
Birds ✓
❌ Land Snail ✓
Cockroach (Arthropods) ✓

Gouty
Arthritis

Types of Nephron

Cortical Nephron

85%

Small

L.H = Short

U.R = -nt

Max^m part = Cortex

Work during
Normal Condition

Juxtamedullary Nephron

15%

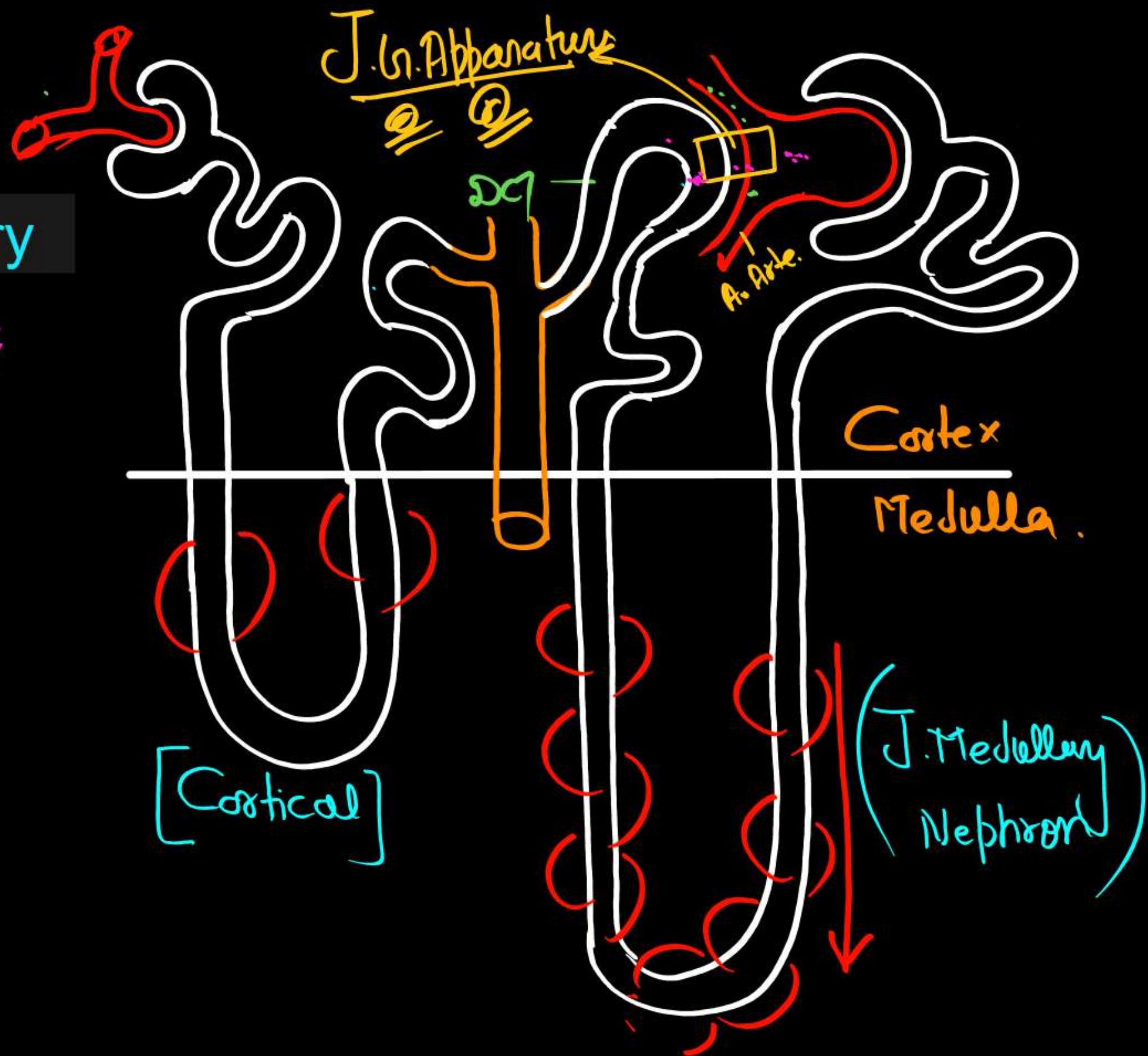
Large

Long

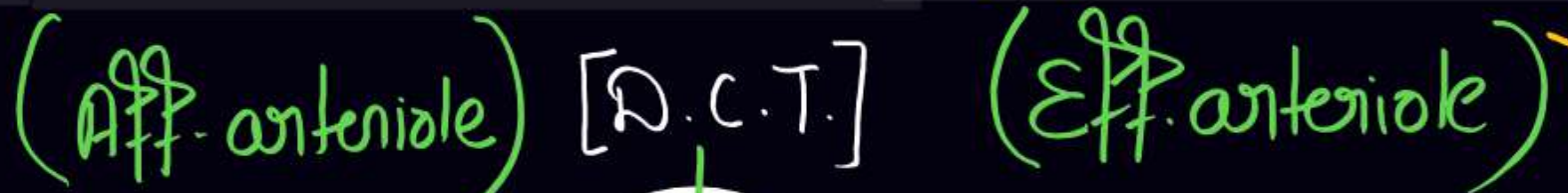
+nt

= Medulla

Emergency



Juxta Glomerular apparatus



J. Cr. Cells →

During Emergency Release $\rightarrow$

1) Renin = $\uparrow$ G.F.R / $\uparrow$ B.P.

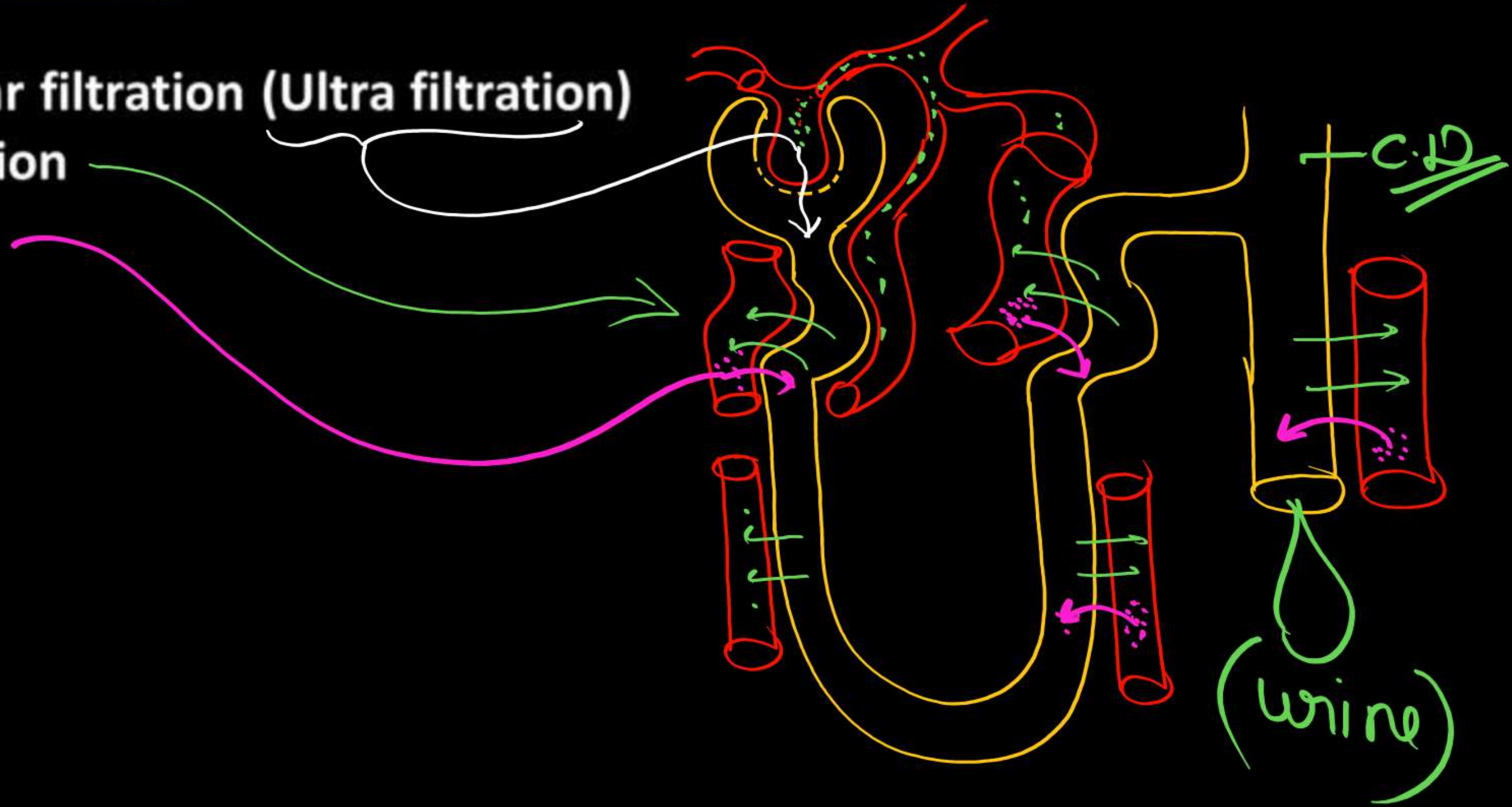
2) - Erythropoietin = $\uparrow$ Erythropoiesis $\rightarrow$ Formation of R.B.C

3) - Calcitriol (Vit D₃) = ↑ absorption of Ca⁺⁺ from gut food.

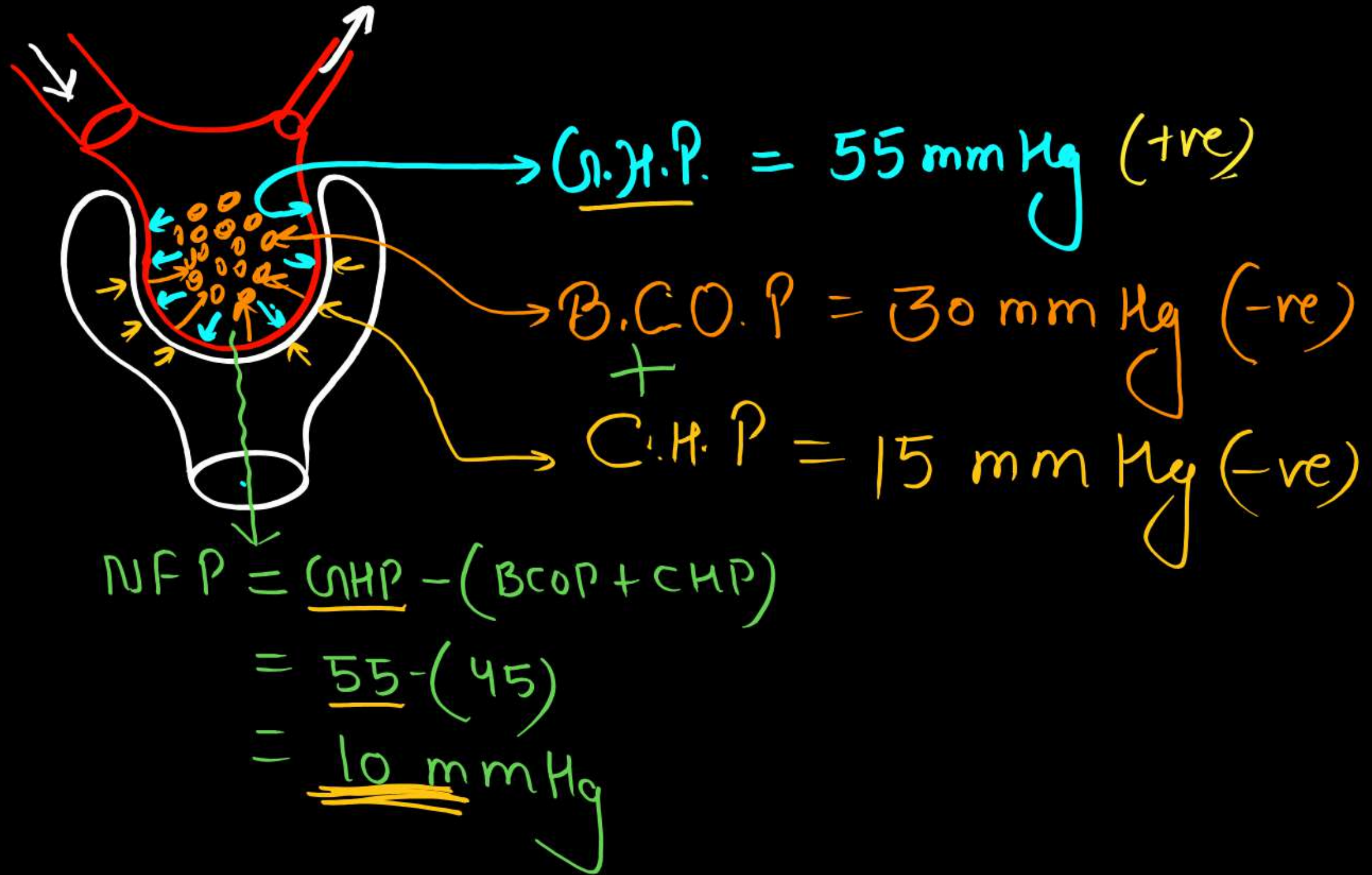
Macula
cells of DCT
+
J.G. Cells
↓
Modified Smooth
Muscles of Aff. arterioles

Urine Formation

- (1) Glomerular filtration (Ultra filtration)
- (2) Reabsorption
- (3) Secretion



(1) Glomerular filtration (Ultra filtration)



2) GFR

vvv fmp
NEET

1100-1200 ml blood
flow by both kidney
Per minute

or
 $\left\{ \frac{1}{5} \text{ of Cardiac out put} \right\}$

99% Reabsorbed
in Blood



Blood Corpuscles - not
Plasma protein - not

vvv fmp

Filtrate

125 ml/minute [180 l/Day]

{ G.F.R }

Only 1% Convert into urine

= 1.8 l/Day (1.5 l/Day)

Tubular reabsorption

PCT → Brush Bordered Cuboidal

Max^m = 70-80% → Ions & water

Glucose
Amino acids
Vit-C
+ve ions (Na^+)
-ve ions
Water

100% } Active Transport

Passive Transport

Osmosis

LH ← = Squamous
↑ = Cuboidal

Min^m = 5%

↓ = H_2O Passive
↑ = Thin = Na^+
Thick = Na^+ + Active

DCT = Cuboidal

Conditional (15%)
- Na^+
- Water

Hypo.

ADH
↓
DCT/CD
+ Water

Adrenal glands

Aldosterone
↓
DCT

+ Na^+ } Active

CD = Cuboidal

Water = ADH
Ureter.

Absorb by Diffusion.

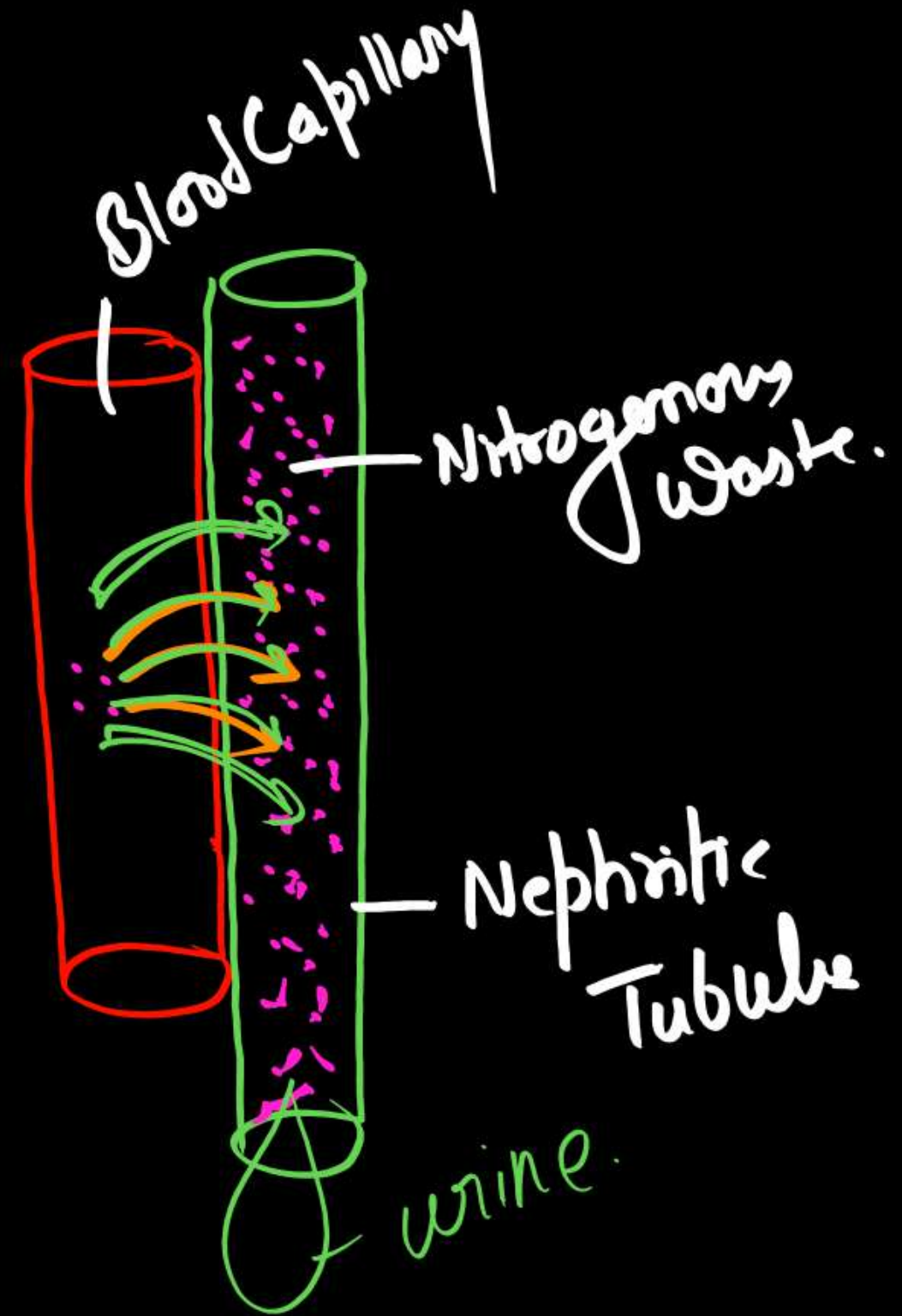
(3) Secretion - (Active)

a) - P.C.T. = H^+ , NH_3 , Drugs, Toxic Sub. etc.

b) - L.H. $\left\{ \begin{array}{l} \downarrow = X \\ \uparrow = \text{urea} \end{array} \right.$

c) - D.C.T = H^+ , NH_3 , K^+

d) - C.D = H^+ & K^+

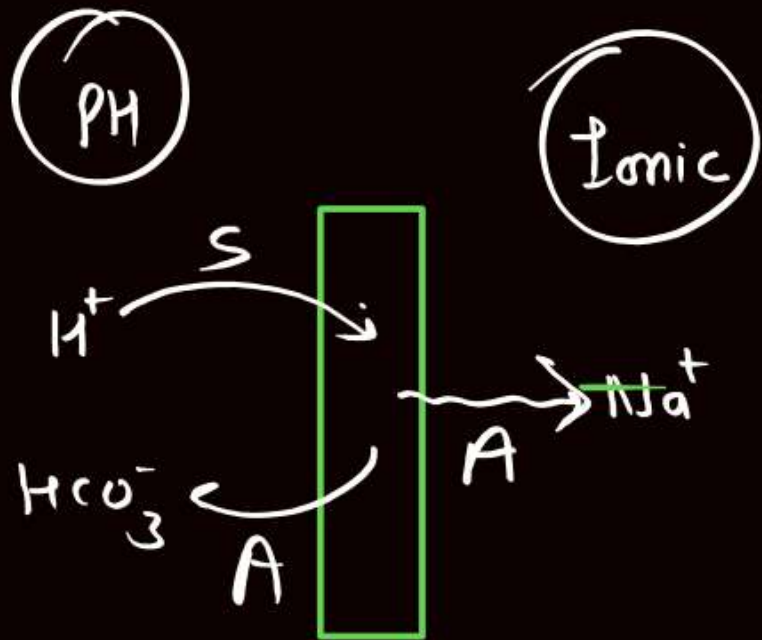


Function of Tubules

vvvvvfmjz

PCT

Maintain $\rightarrow$ pH & Ionic Bala.



LH

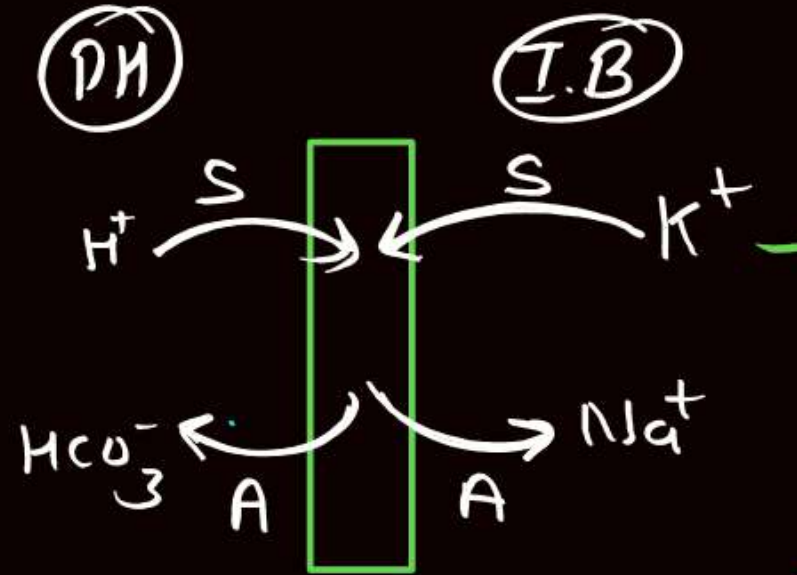
X

Absorption $\leftarrow H_2O$
 $\leftarrow NaCl$
 +
 Secætion = urea
 $\downarrow$
 Maintain
 Osmolarity Gradient



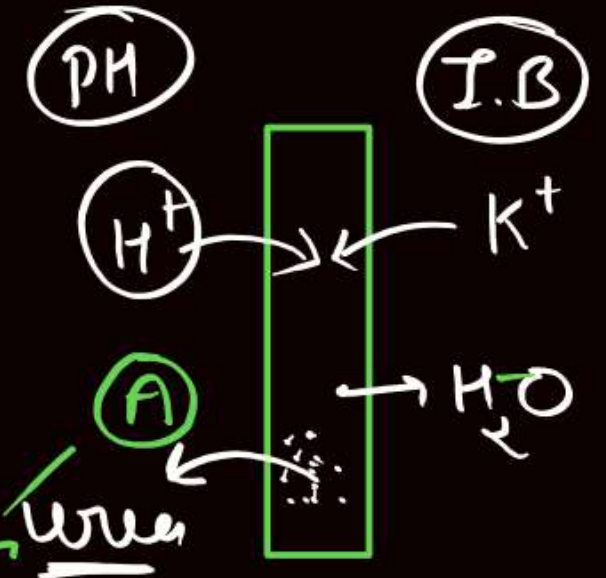
DCT

Maintain = pH & Ionic Bala.

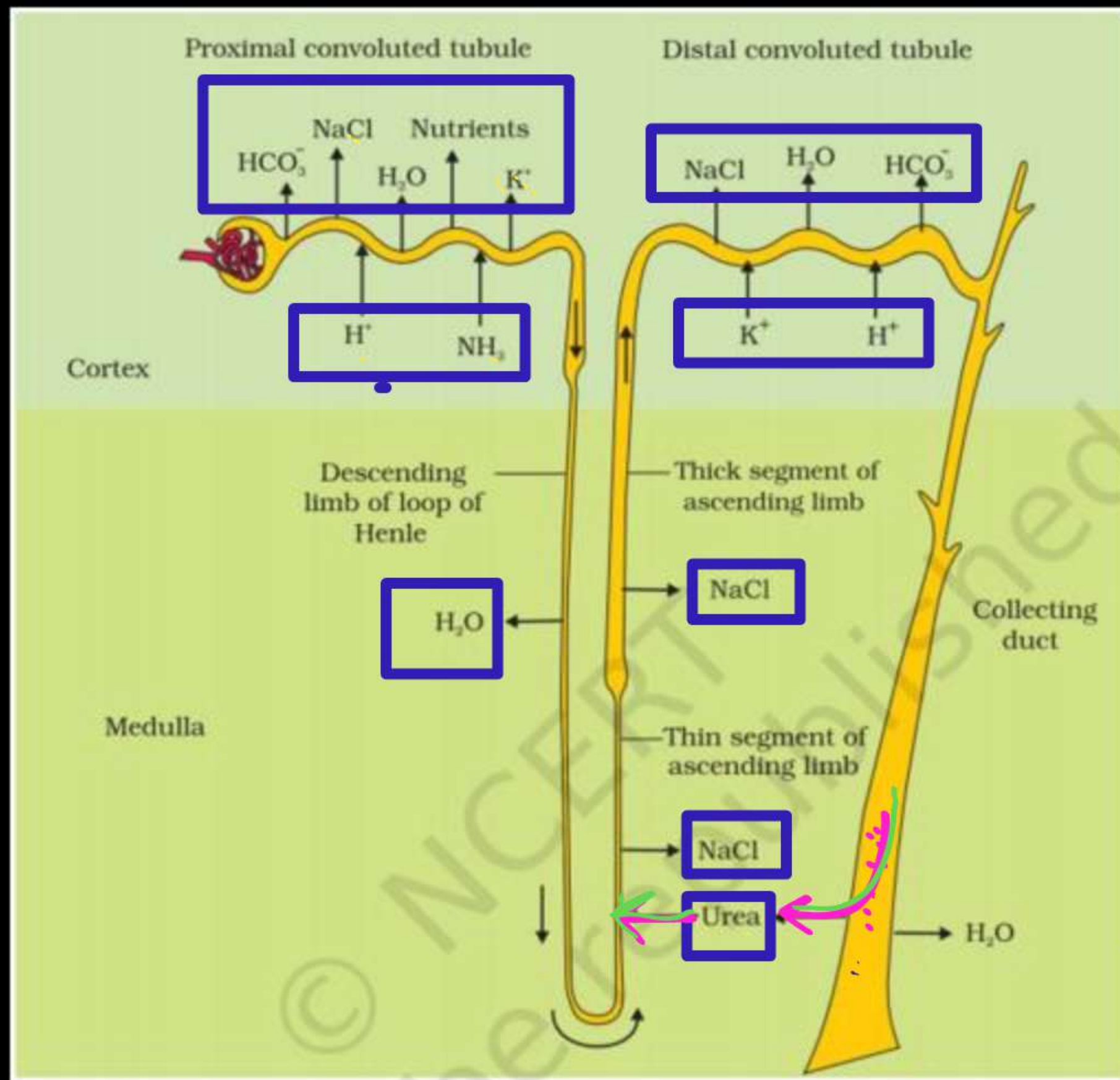


CD

Main = pH & Ionic



Absorption of -
 water = Cons. Urine
 Urea = Maintain
 Osmolarity



Counter Current Mechanism

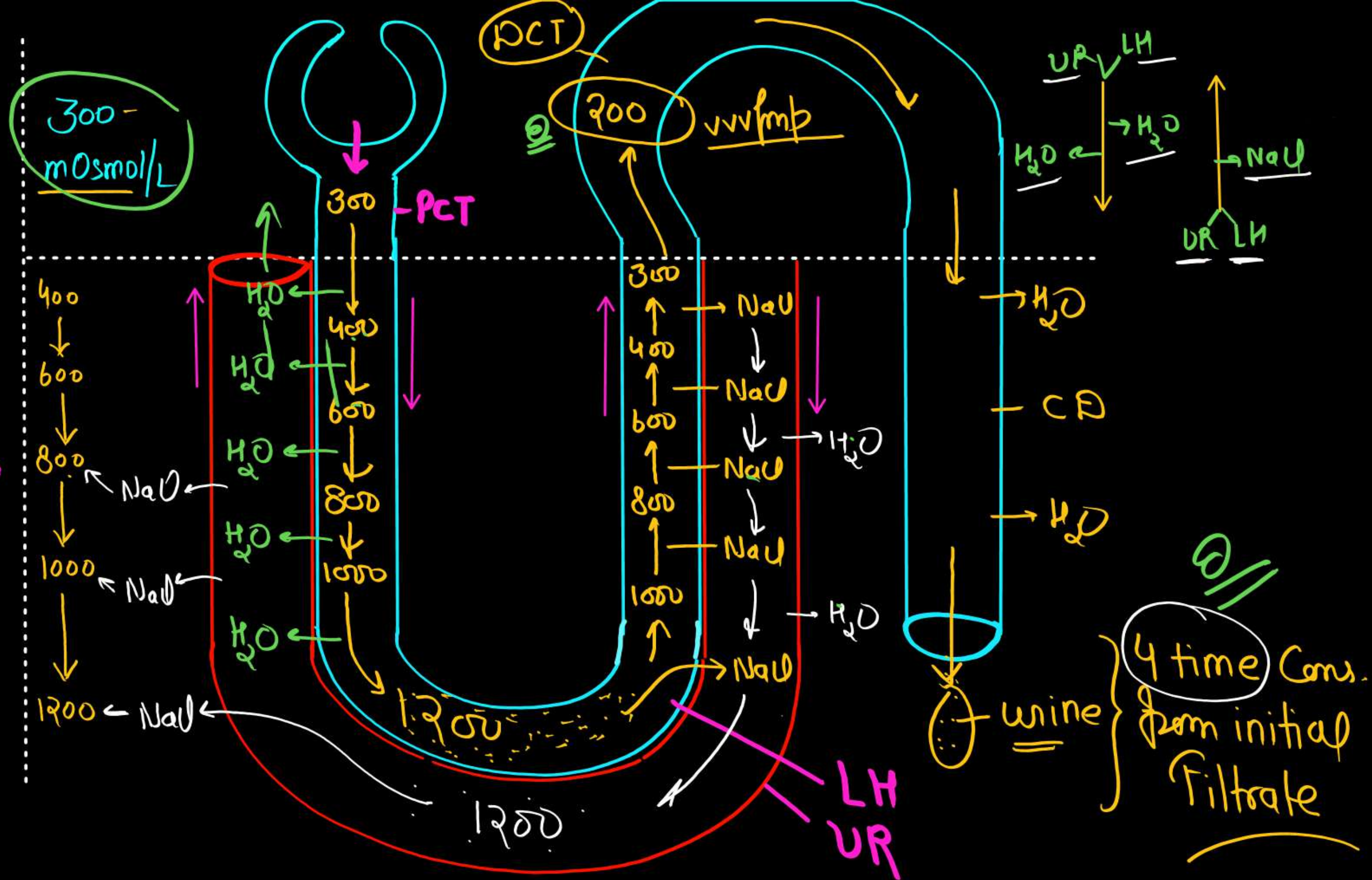
- Mammals have the ability to produce a concentrated urine. The Henle's loop and vasa recta play a significant role in this.
- Osmolarity gradient towards the inner medullary interstitium mainly caused by NaCl and urea.
- Human kidneys can produce urine nearly four times concentrated than the initial filtrate formed.

Cortex

300-
mOsmol/L

Medulla

NaCl
+
Urea



Regulation of Kidney Function

(1) Regulation By Hypothalamus

if G.F.R. = ↓
or
B.P. = ↓
or
Blood Water = ↓
or
Blood Osmolarity = ↑
or
Dilution of urine = ↑

Stimulate

Hypothalamus
Osmolarity Receptors

A/P Pituitary

ADH (Vasopressin)

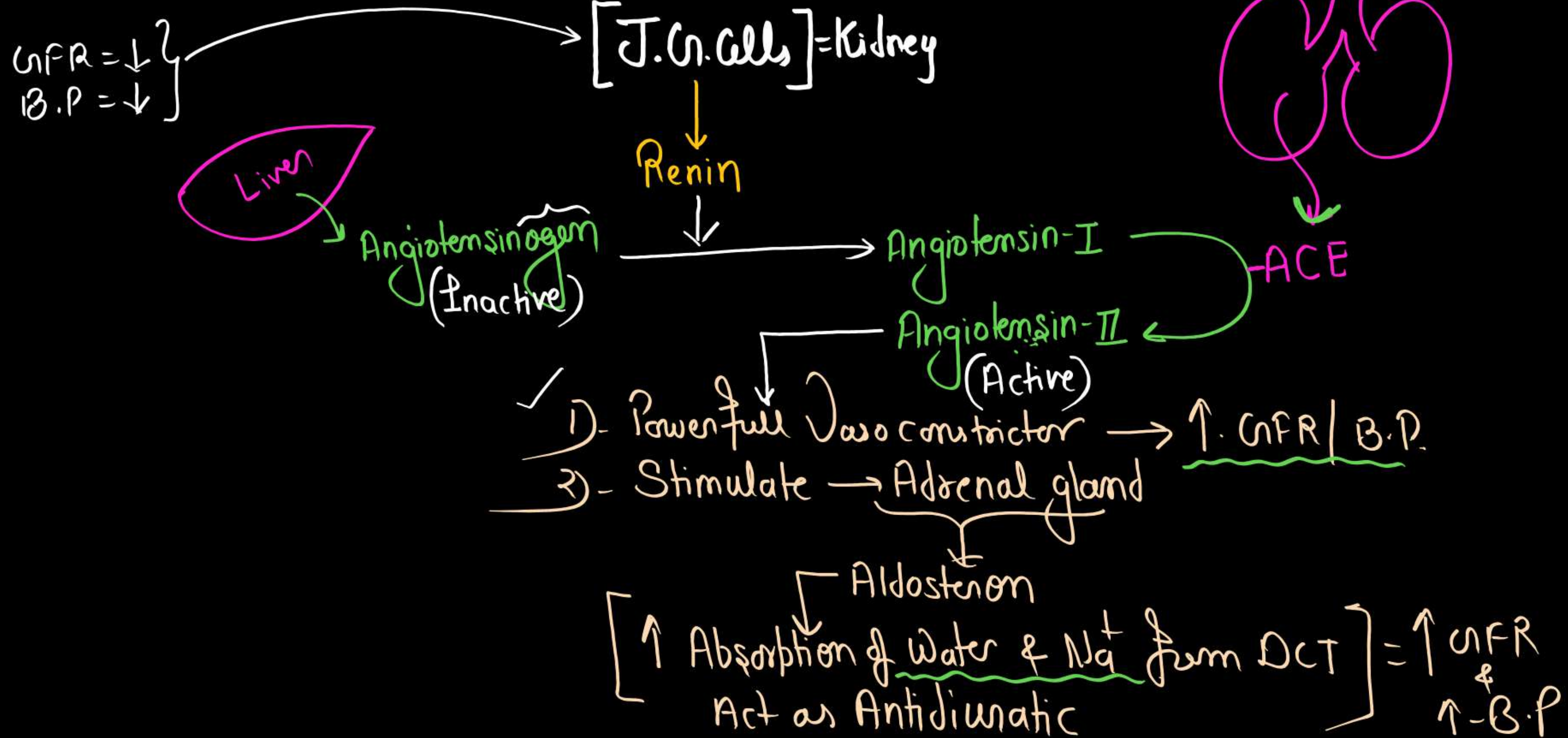
✓ ↑ absorption of
Water from DCT & CD

Constriction in
Blood Vess. = ↑ B.P
&
↑ GFR

↑ Cons. of urine &
↑ GFR / B.P etc

Feed back inhibition of ADH ✓

(2) Regulation By J.G. Cells of kidney (R.A.A.S.)



(3) Atrial Natriuretic Factor (ANF)

[Antagonistic of R.A.A.S.]

↑ B.P.

↑ Pressure on wall of Heart

↓
= [Wall of Atria] //

} Release

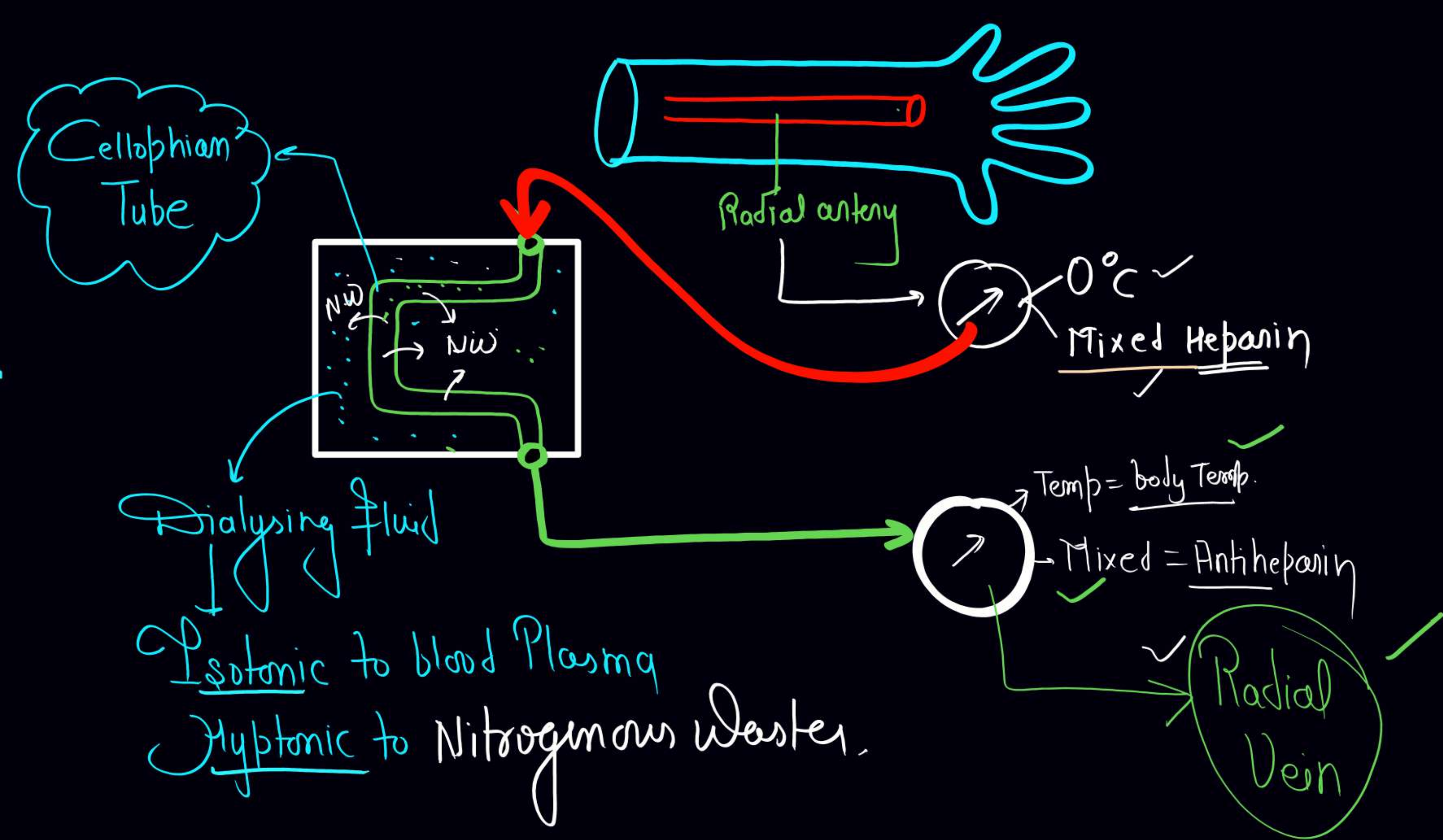
= [ANF] //

- 1) Powerful = Vasodilator
- 2) Inhibit = R.A.A.S & Aldosterone secretion
- ✓ 3) Diuretic = ↑ Dilution of urine.

DISORDERS OF THE EXCRETORY SYSTEM

- Malfunctioning of kidneys can lead to accumulation of urea in blood, a condition called **uremia**, which is highly harmful and may lead to **kidney failure**. In such patients, urea can be removed by a process called **hemodialysis**. During the process of haemodialysis, the blood drained from a convenient artery is pumped into a dialysing unit called **artificial kidney**.





} Diabetes

- Glycosuria : - Presence of glucose in urine.
- Ketonuria : - Presence of ketone bodies in urine
- Renal calculi: - Stone or insoluble mass of crystallized salts (oxalates, etc.) formed within the kidney.
- Glomerulonephritis: - Inflammation of glomeruli of kidney



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BEY_MD_SIR_BMS

Neural Control.....

Human Brain - Central information Processing Organ

(1) Fore Brain

→ Telencephalon

→ Diencephalon

Cerebrum

olfactory lobes

Amygdala

Hippocampus

Thalamus

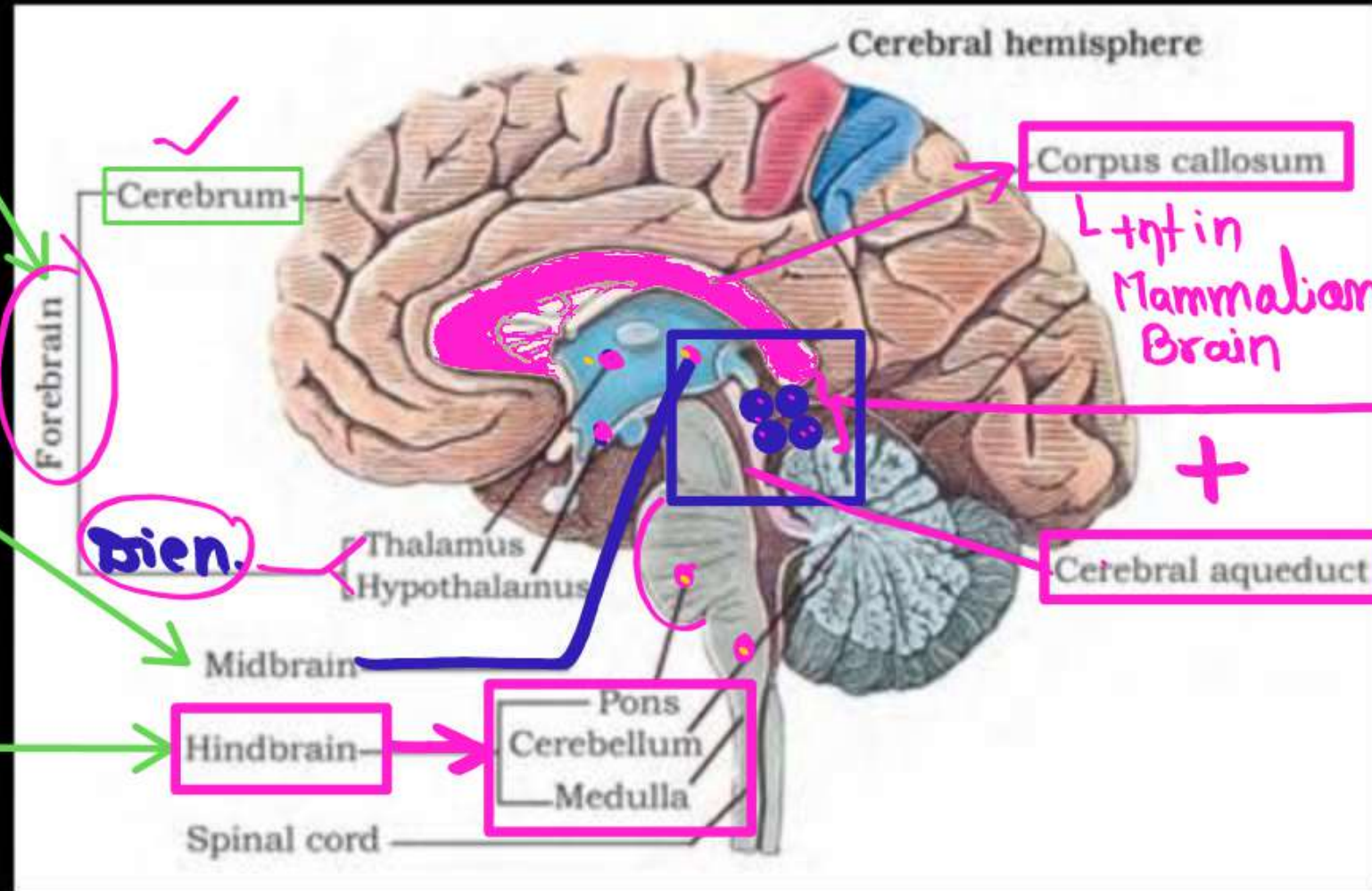
Hypothalamus

Epithalamus

Limbic System

(2) Mid Brain

(3) Hind Brain



Corpora Quadrigemina

[Dorsal]

Two Supp. Colliculi

Two inferior Colliculi

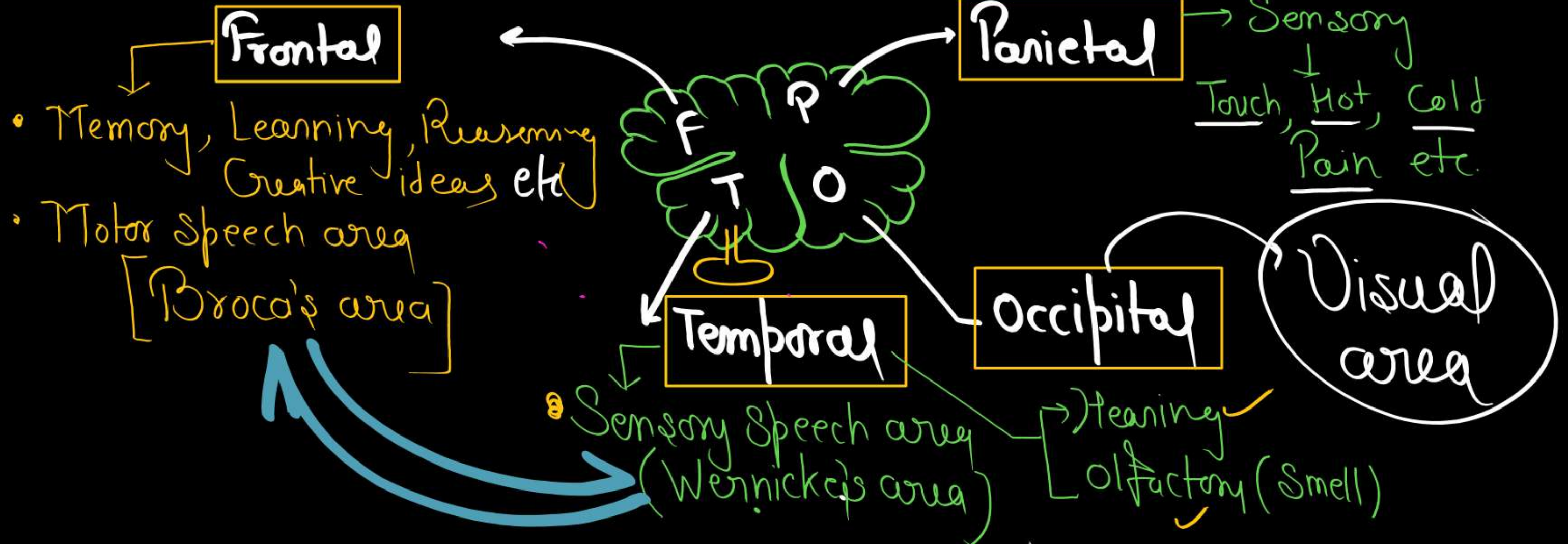
Vision

Auditory

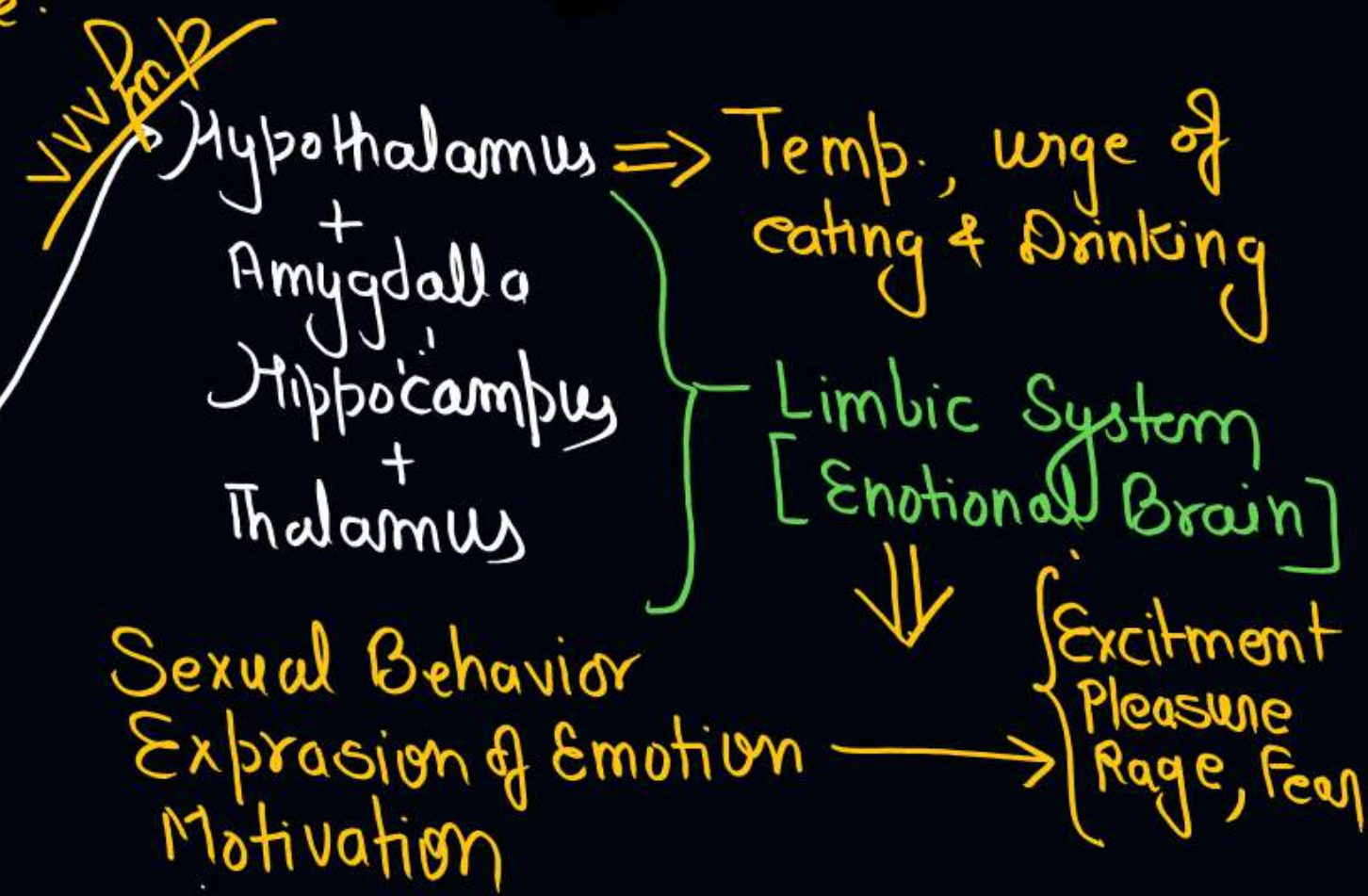
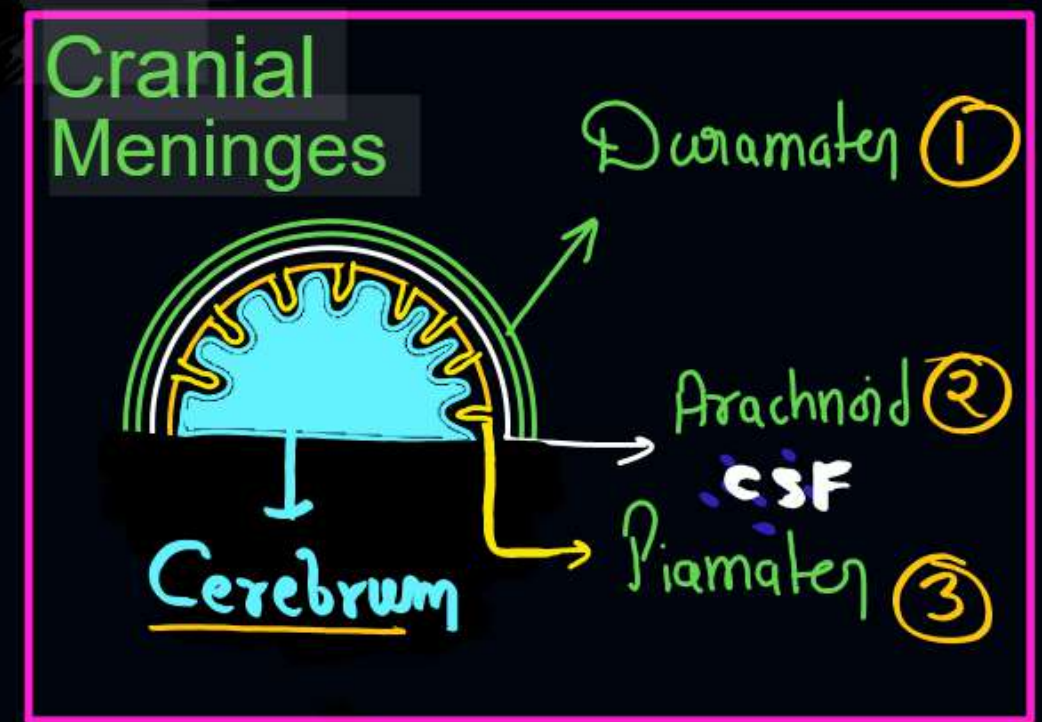
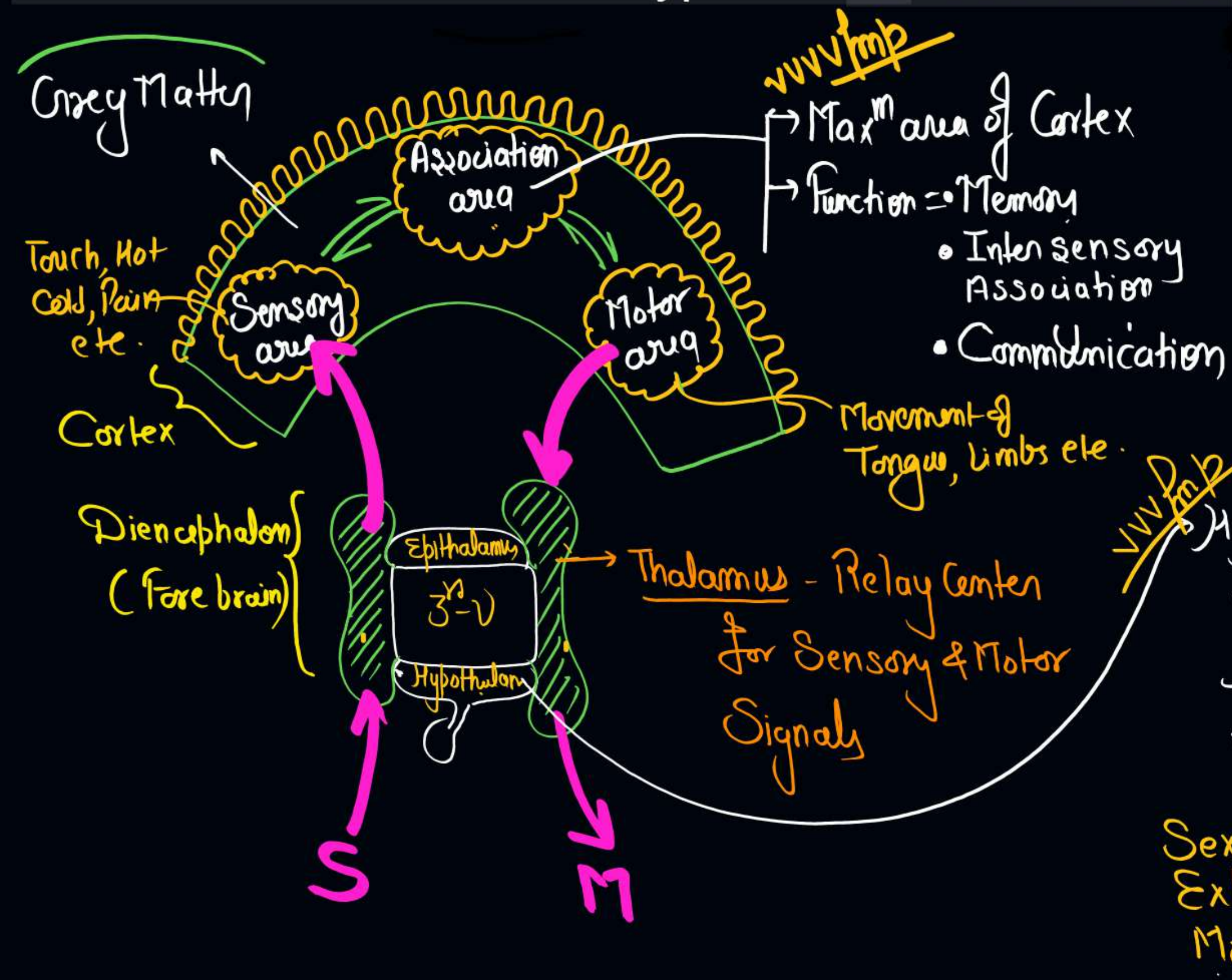
Corpus callosum

→ Thick bundle of Neurons, which Co-ordinate b/w both hemispheres
→ + not only in Mammals. Brain

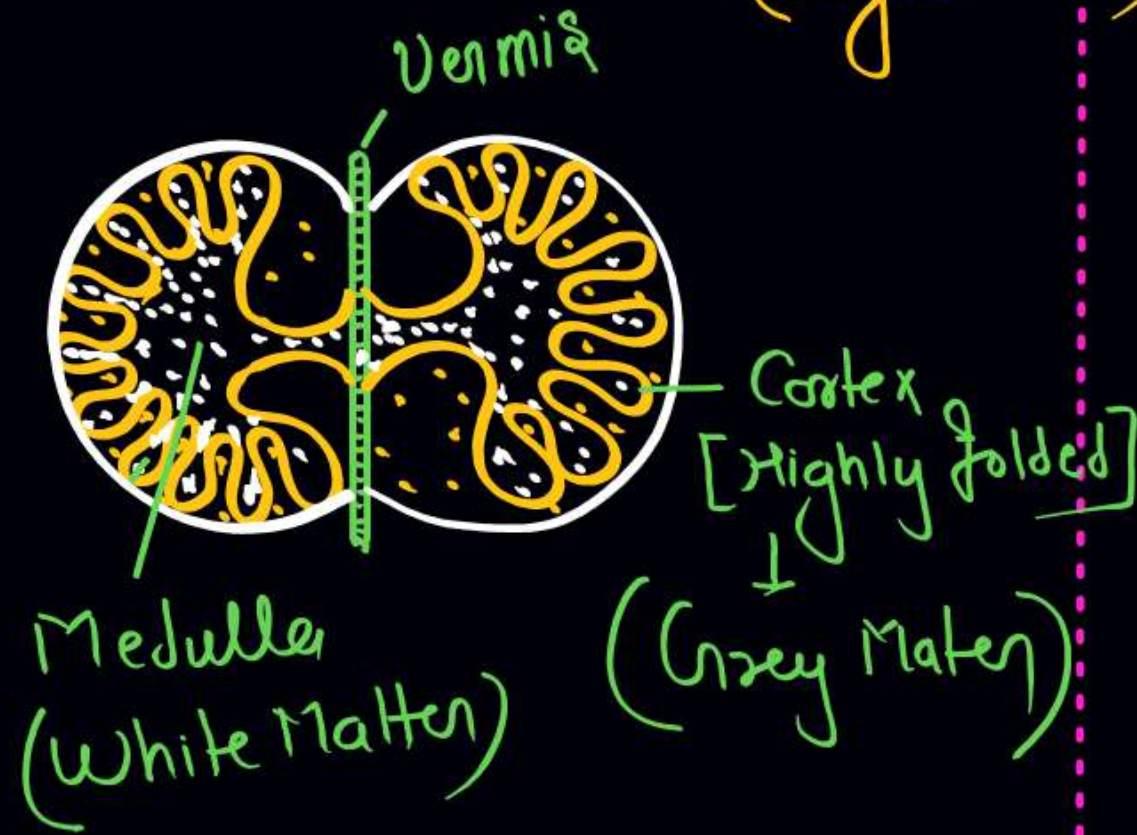
Lobes of Each Hemisphere →



Cerebral Cortex Contain 3 types of Functional area (Nucle)



Cerebellum = 2nd - (Largest Part)



Function = Balancing & Coordination of Voluntary Movement

Medulla oblongata

Control Involuntary action of body System

Control - Respiration

- Cardiovascular Reflex

- Gastric - Secretion

Pons

• Bundle of Nerve fibres tracts → Interconnect different Regions of Brain.

• Pneumotaxic Center →
↓ Duration of Inspiration →
↑ Rate of Respiration.

Brainstem



There is a wide diversity in living organisms in our biosphere. Now a question that arises in our minds is: Are all living organisms made of the same chemicals, i.e., elements and compounds?

Living Organism { Animals
Plants.
Microbes.
etc.

- Q //
- * C, H & O etc. +nt
 - * C & H most abundant
 - * Si Negligible

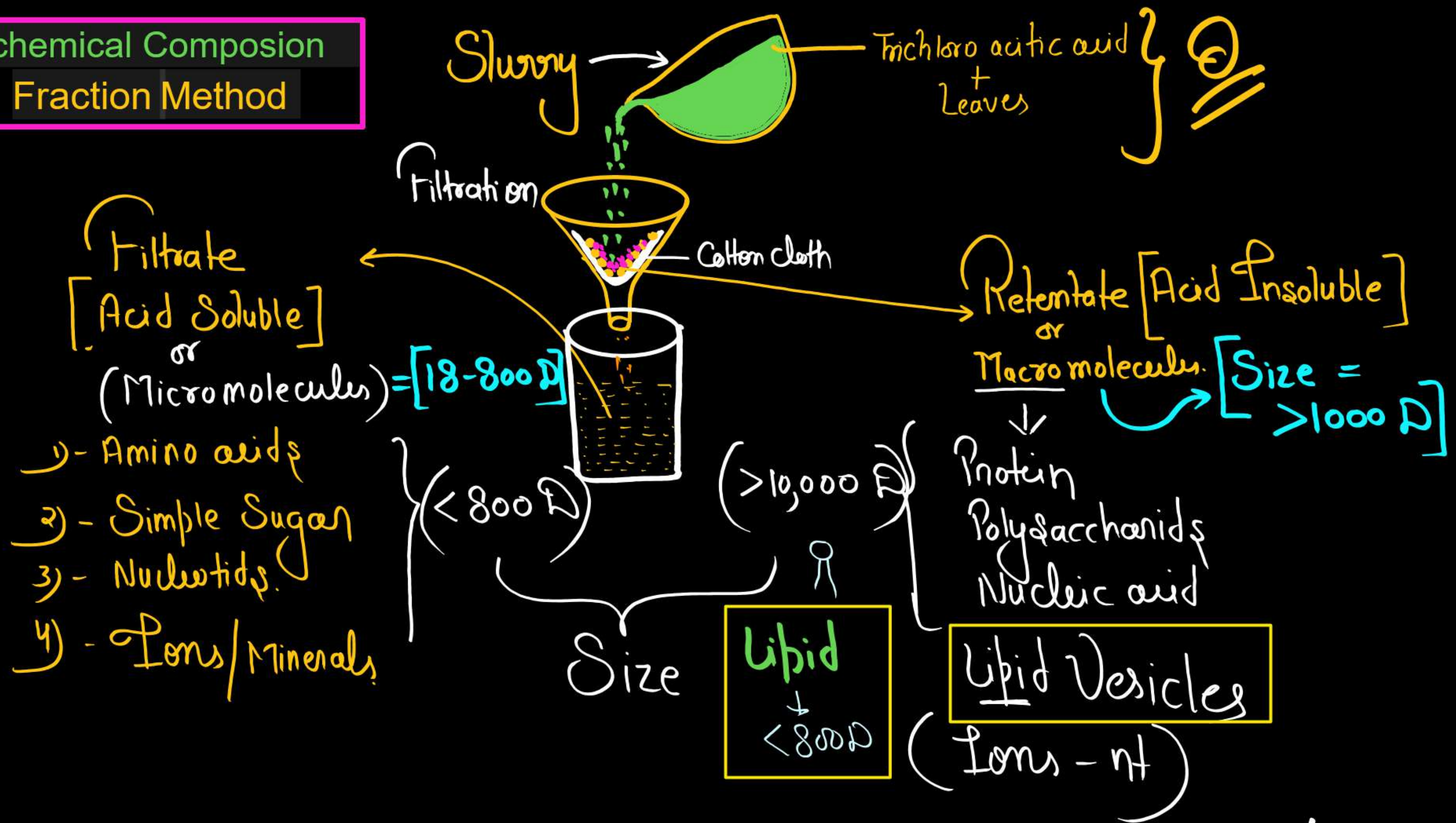
Earth Crust (Non living)

C, H & O etc +nt

Less.

most abundant

chemical Composition Fraction Method



Primary Metabolite.

→ Produce in all living Organism →
Animals, Plant, Bacteria & Fungus

- Involve in Metabolic Reaction
- Identifiable Role = Role in Growth, Development, Reproduction etc.
- Produce in large Quantity during Growth Phase.

eg - Amino acid
Lipid
Carbohydrate
Nucleic acid etc

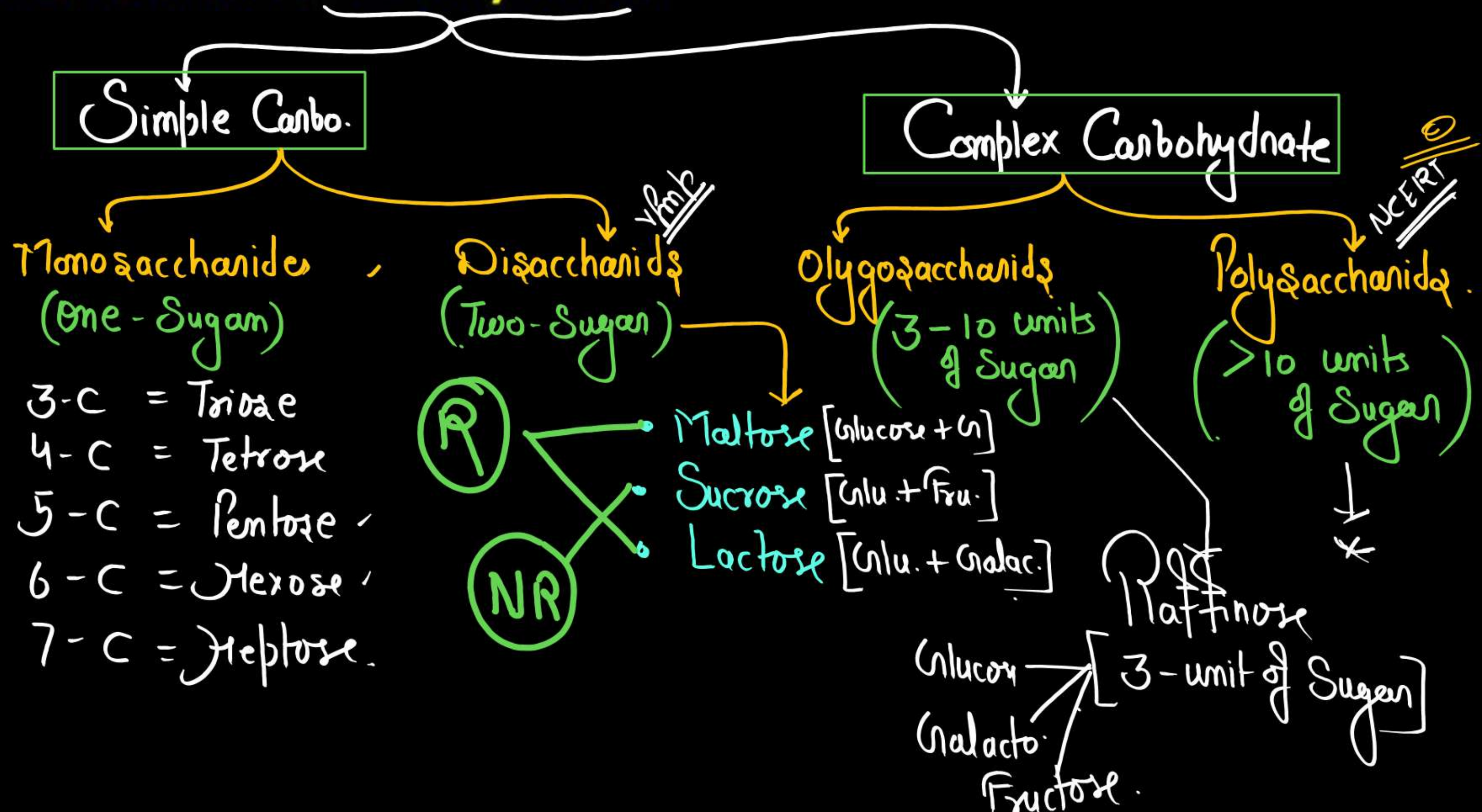
Sec. Metabolite

Produce in only - Plant, Bacteria & Fungus
(not in animals)

TABLE 9.3 Some Secondary Metabolites

Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoides	Monoterpenes, Diterpenes etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs @ //	Vinblastin, curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

Classification of Carbohydrates -



Poly saccharides $\rightarrow$ (Tastless, Water Insoluble)

(A) Homopolysaccharites

Starch $\rightarrow$ Plant Store Food
Monomer unit = α -Glucose.

$\downarrow$
Branched + Helical. Formed by =

Amylose (20%) + Amylopectin (80%)



Helix

Water Soluble
 α (1-4)

Blue colour
with I_2



Branched
Water insoluble
 α (1-4) & α (1-6)

(2) Glycogen -

Monomer = α Glucose

Animal Store Food $\rightarrow$ Liver
Skeletal Muscle

Branched $\rightarrow$ α (1-4)
 α (1-6)

Amylose - nt

(3) Cellulose

Monomer = β -Glucose

→ in cell wall of Plant & Algae

Digested by - Cellulase Enzyme

→ Bacteria & Protozoa

Wood Pulp - Cotton

- Paper formation

unbranched → β (1-4)

Chitin

Monomer = N-Acetylglucosamine (NAG)
↓
Derivative of Glucose

→ in - Fungal cell wall
- Exoskeleton of Cock.

Unbranched = β (1-4)

Inulin -

→ Monomer = β -Fructose

→ Store food → in root of Dahlia Plant

B) Hetero-polysaccharide

② Pepidoglycan

Hyaluronic acid

→ +nt in Connective Tissue

• +nt in = Bacterial Wall

• Polymer of → NAG + NAM

N-Acetyl Glucosamine

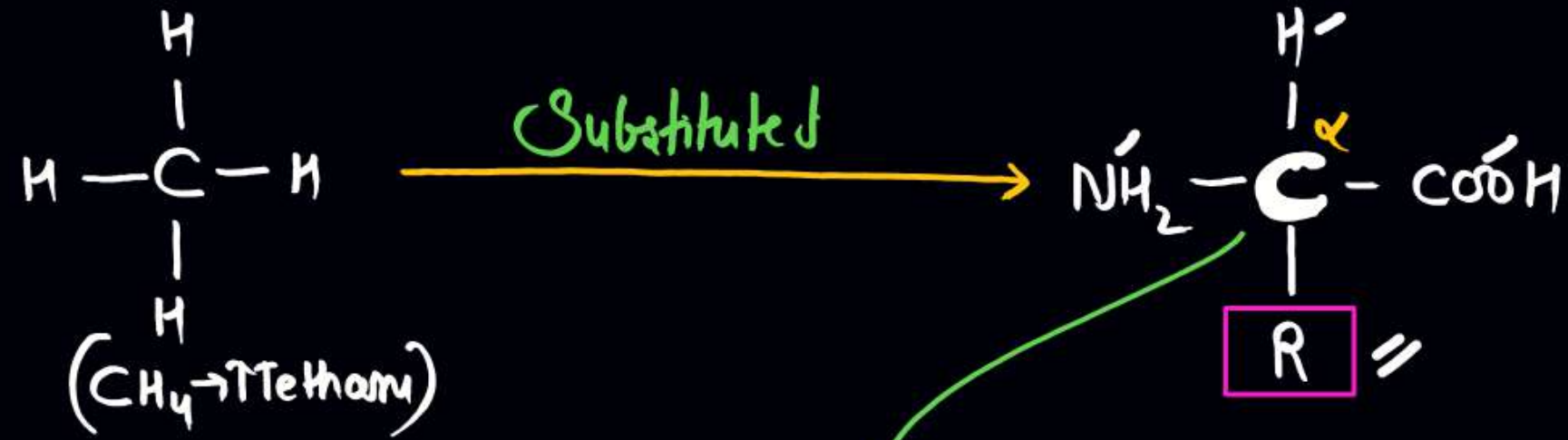
N-Acetyl Muramic acid

• linkage → $\beta(1-4)$ Q

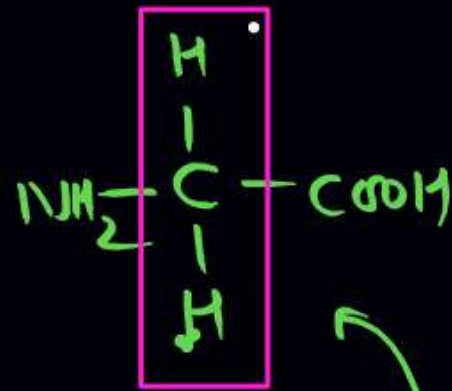
Imp

Amino Acids

- Mole.wt < 800 D
- +ve in acid soluble pool
- Synthesized by cells



[Chiral Carbon]



[Optically active]

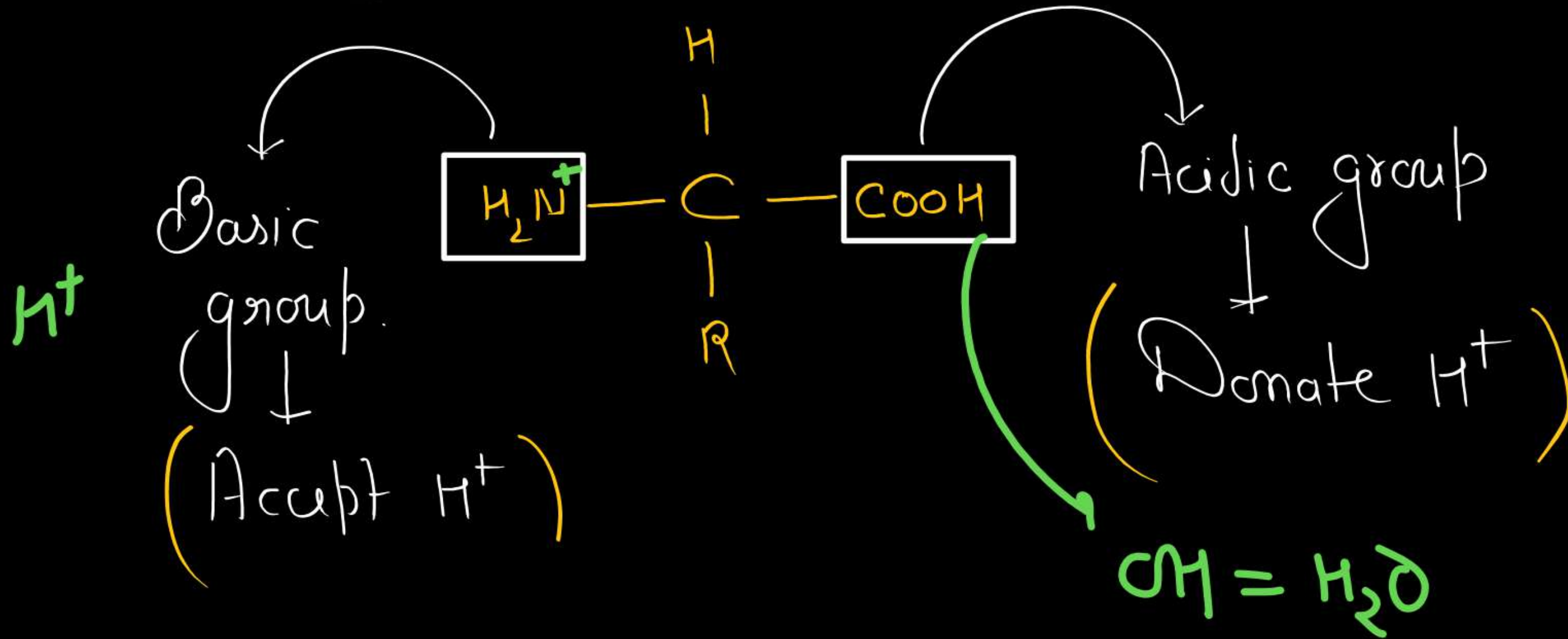
Exapt = Glycine

(Chiral Carbon
-H)

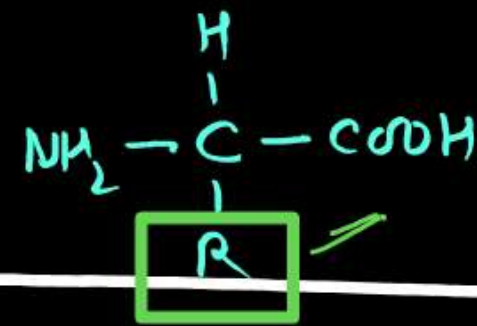
optically inactive

Zwitterionic form of amino acid

A particular property of amino acids is the ionizable nature of -NH_2 and -COOH groups. Hence in solutions of different pH, the structure of amino acids changes



Classification of amino acids



Acidic A.A.



-ve charge

(Polar)

Water Soluble
Hydrophilic

eg → Glutamic acid - 5-C
Aspartic acid - 4-C

Basic A.A.



+ve charge

(Polar)

Water Soluble
Hydrophilic

eg → lysine - 2N
Histidine - 3N
Arginine - 4N

Neutral A.A.



No - charge

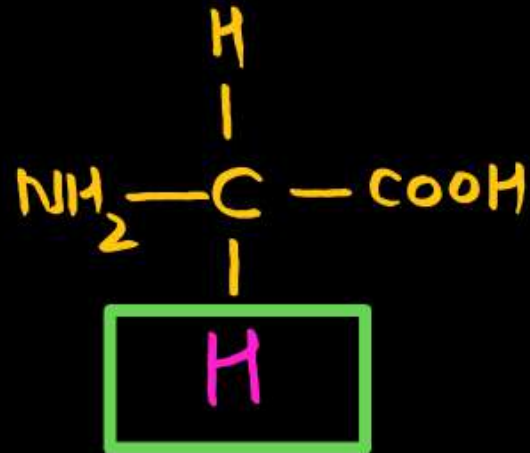
(Non polar)

Water insoluble
Hydrophobic

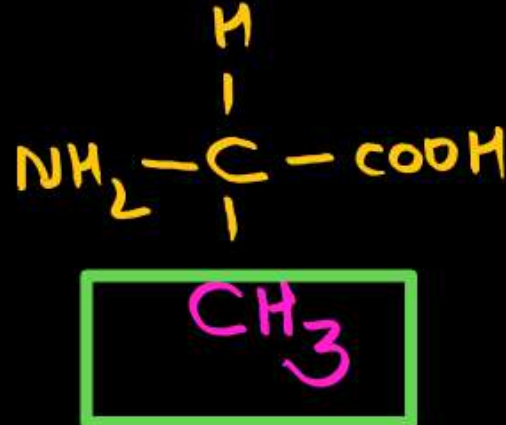
eg - Glycine
- Alanine
- Valine
- Leucine,
Isoleucine

Structure of Amino Acids

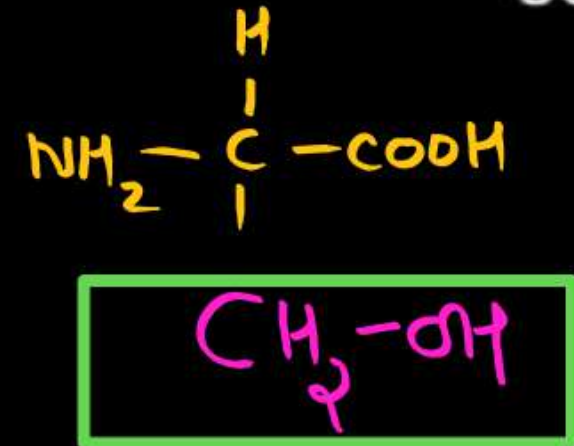
Glycine (Simple Str.)



Alanine

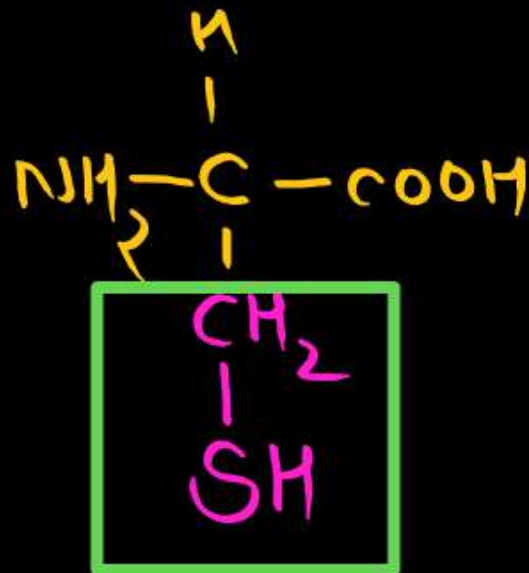


Serine

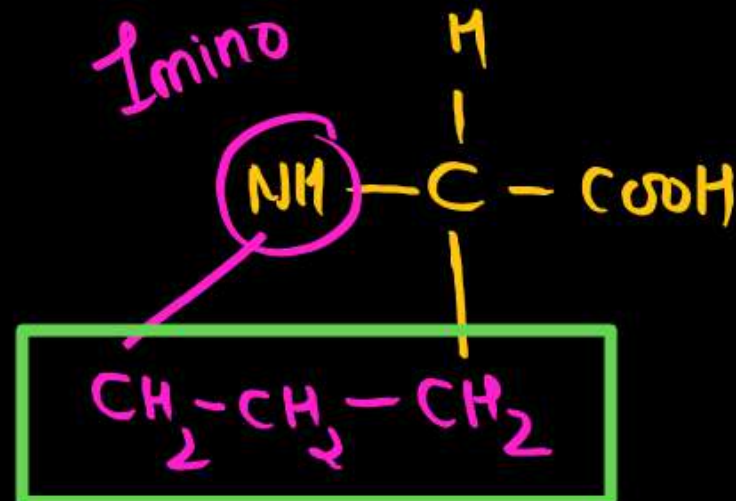


Complex Str.

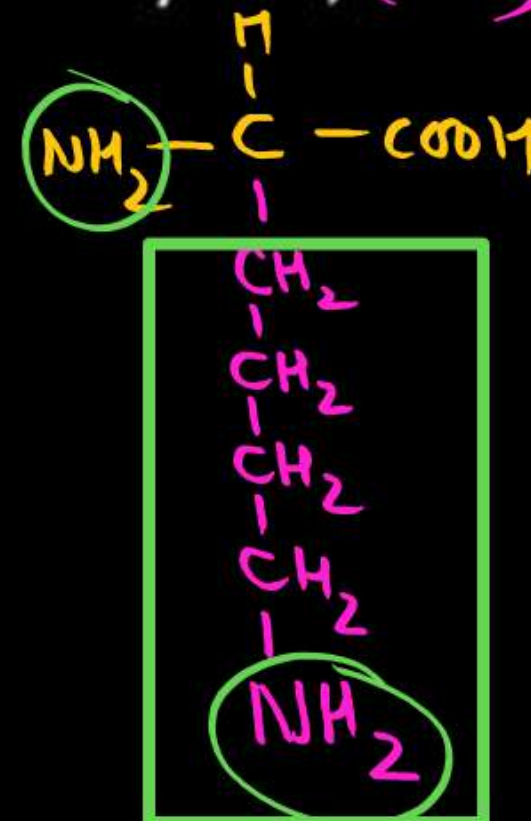
Cysteine, (S)



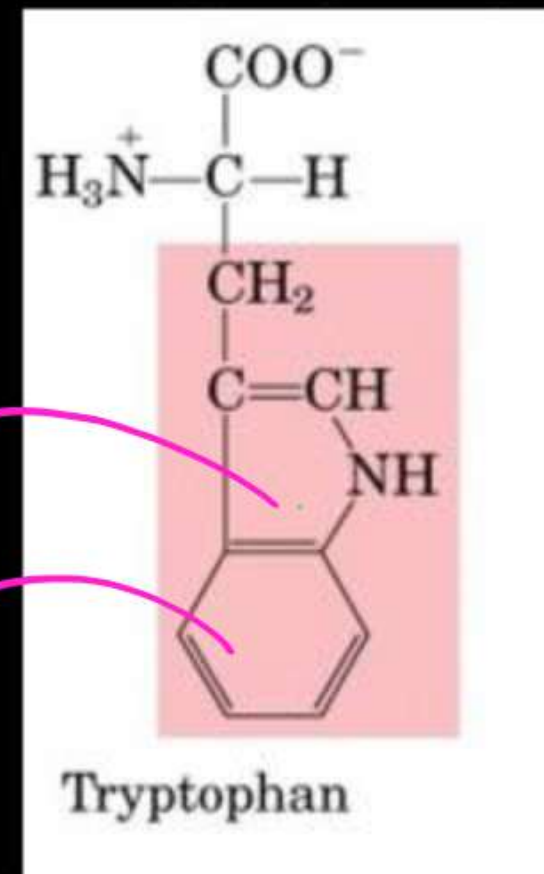
Proline,



Lysine, (+ve)



Tryptophan,



Protein

Structure

Bonding

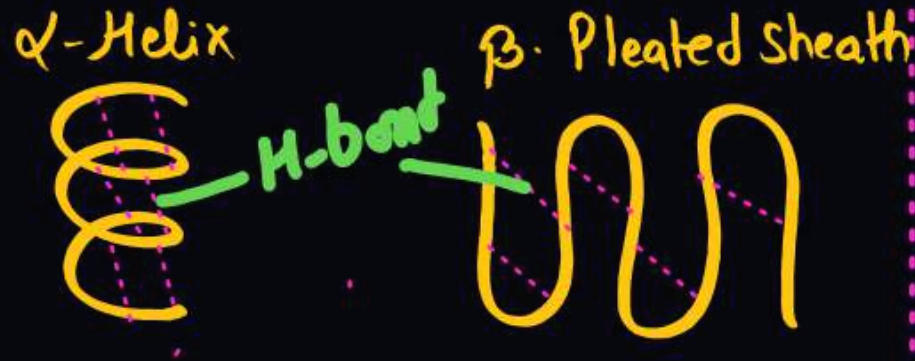
Example

Pri. Structure



Peptide bonds

Sec. Structure



Peptides + H-bond

^①
α-Helix = Keratin
β-Pleated Sheet = Silk -

Ter. Structure



Peptide, H-bond, Disulphide
Ionic, Hydrophobic,
Vanderwaals etc.

Enzymes + Myoglobin

Qua. Structure



Non Co-Valeant ✓
+
Disulphide ✓

^②
Insulin, Antibody - ^④
Haemoglobin - ^④

Function of Protein-

TABLE 9.5 Some Proteins and their Functions

Protein	Functions
Collagen	Intercellular ground substance
Trypsin	Enzyme
Insulin	Hormone
Antibody	Fights infectious agents
Receptor	Sensory reception (smell, taste, hormone, etc.)
GLUT-4	Enables glucose transport into cells

Collagen → Most abundant animal protein
→ +nt in Connective tissue. → Tendon, Ligament, Acellar ek Bones etc.

Tissue.

→ Triple Helix }
Structure

→ Vit. C play imp. role in Formation



Ru.BisCO → Most abundant Plant Protein. in whole of the biosphere

ENZYMES (Bio catalyst) = Stru. = 3-D

vvvimp

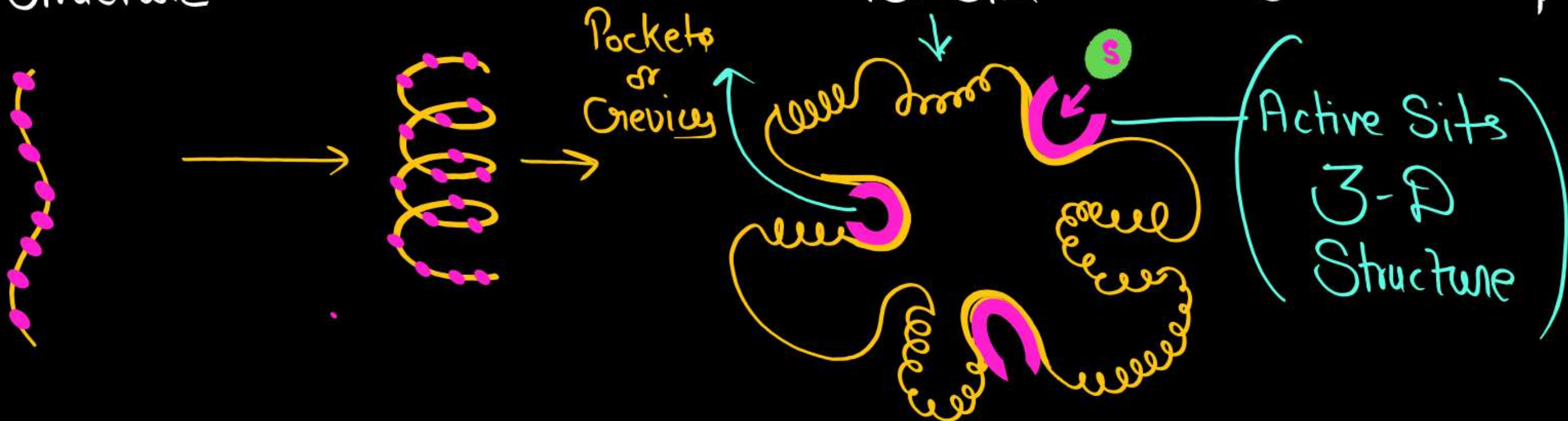
All Enzymes are protein.

Except $\left\{ \begin{array}{l} \text{Ribozyme} \\ \text{Ribonuclease} \end{array} \right\}$

Small Segments of RNA, which have Catalytic Activity

Structure of Enzyme.

Pri. Structure $\rightsquigarrow$ Sec. Stru. $\rightsquigarrow$ Ter. Stru.



- The reaction that occur in presence of catalyst proceed at very fast -



In absence of Enzyme
{ 200 mole. of H_2CO_3 }
formed / Hour

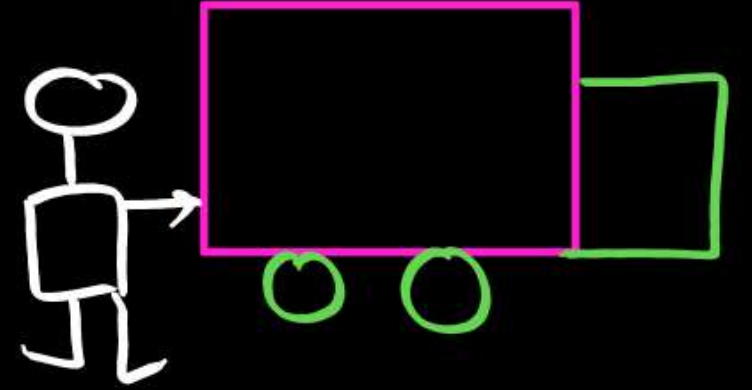
In presence of Enzyme.
eg - Carbonic anhydrase.

6,00,000 mole. of H_2CO_3
formed / Sec.

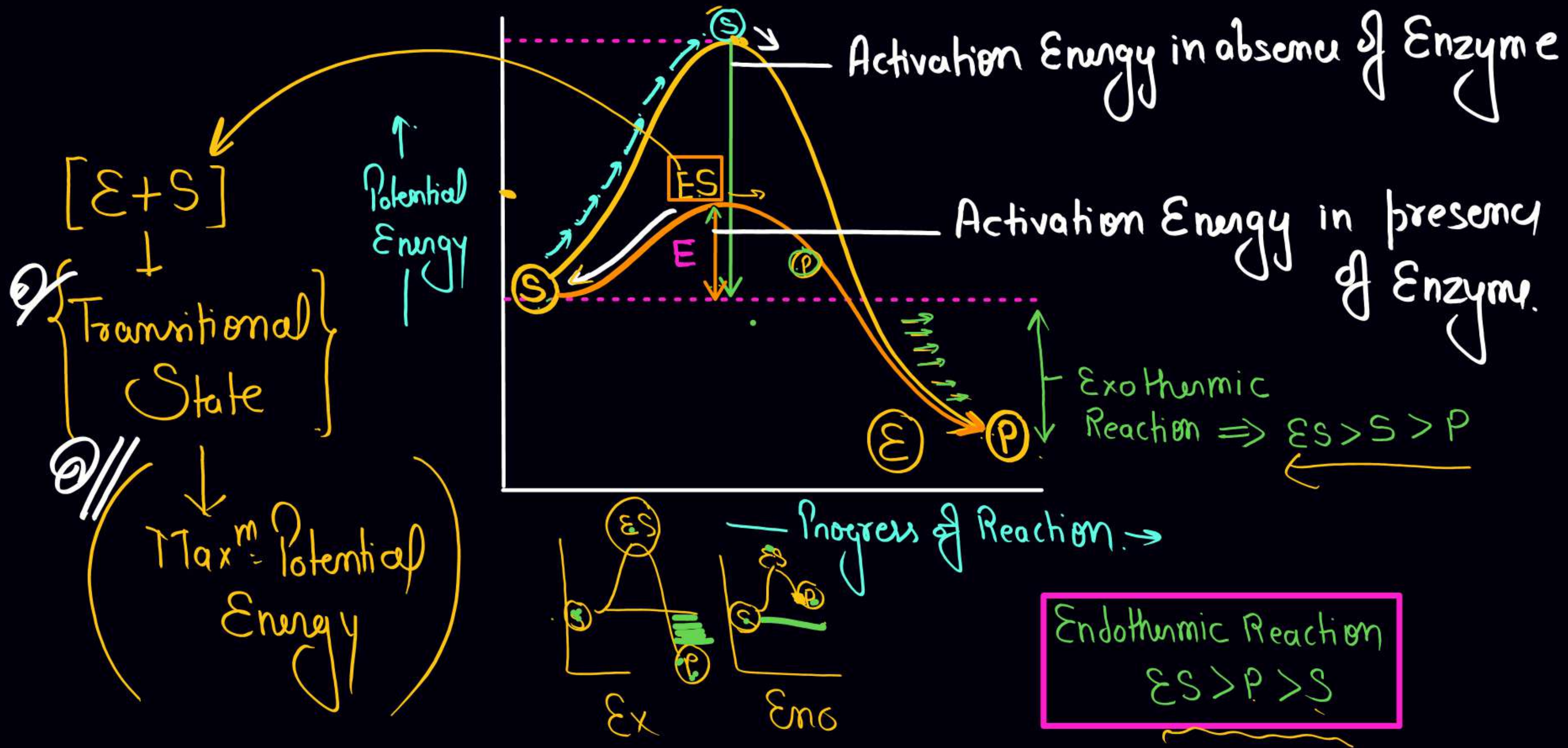
[Reaction 10 million Times]

How do Enzymes bring about such High Rates of Chemical Conversions?

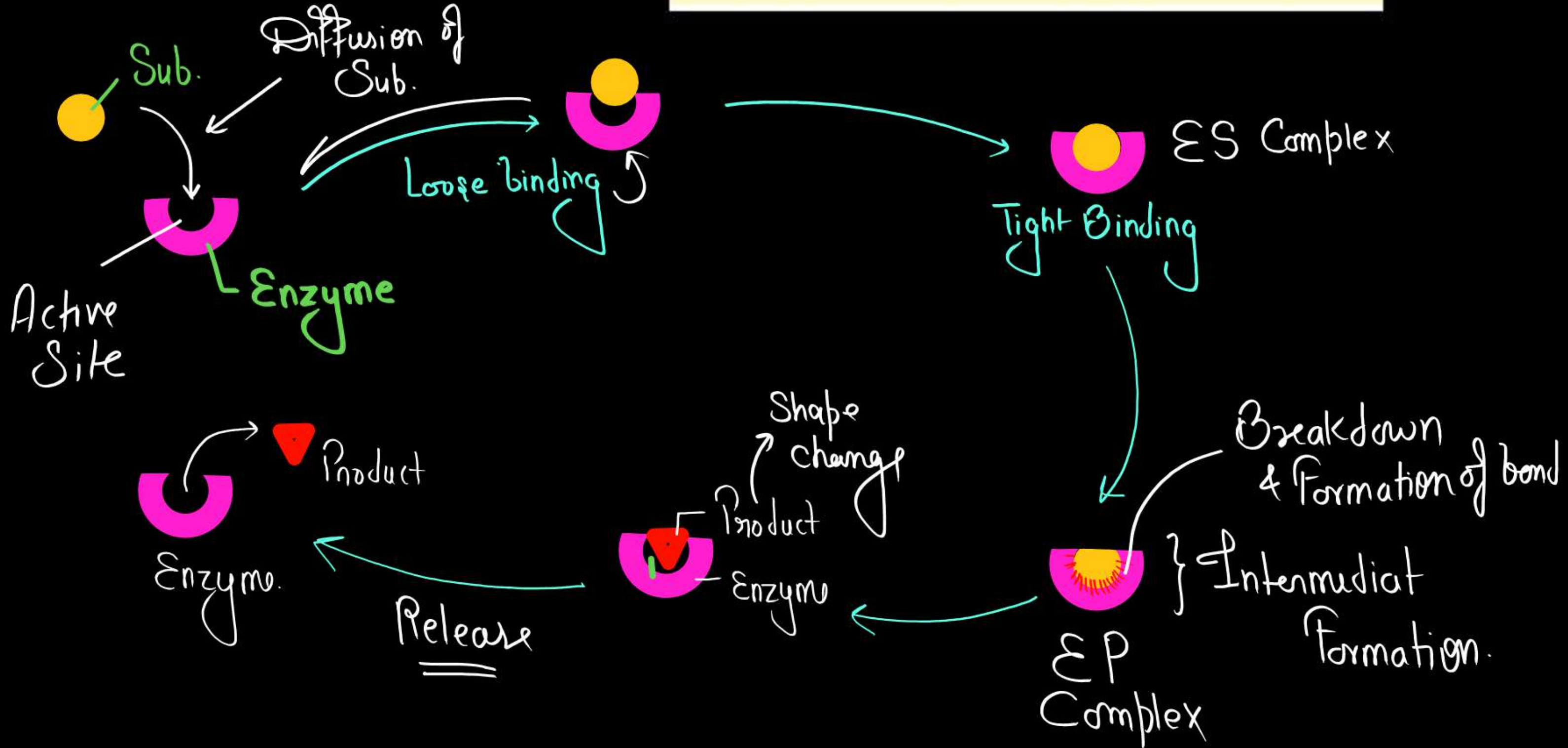
* Enzyme → Lower the activation Energy
Which require for reaction



Activation Energy } Energy require for
initiation of Chemical Reaction

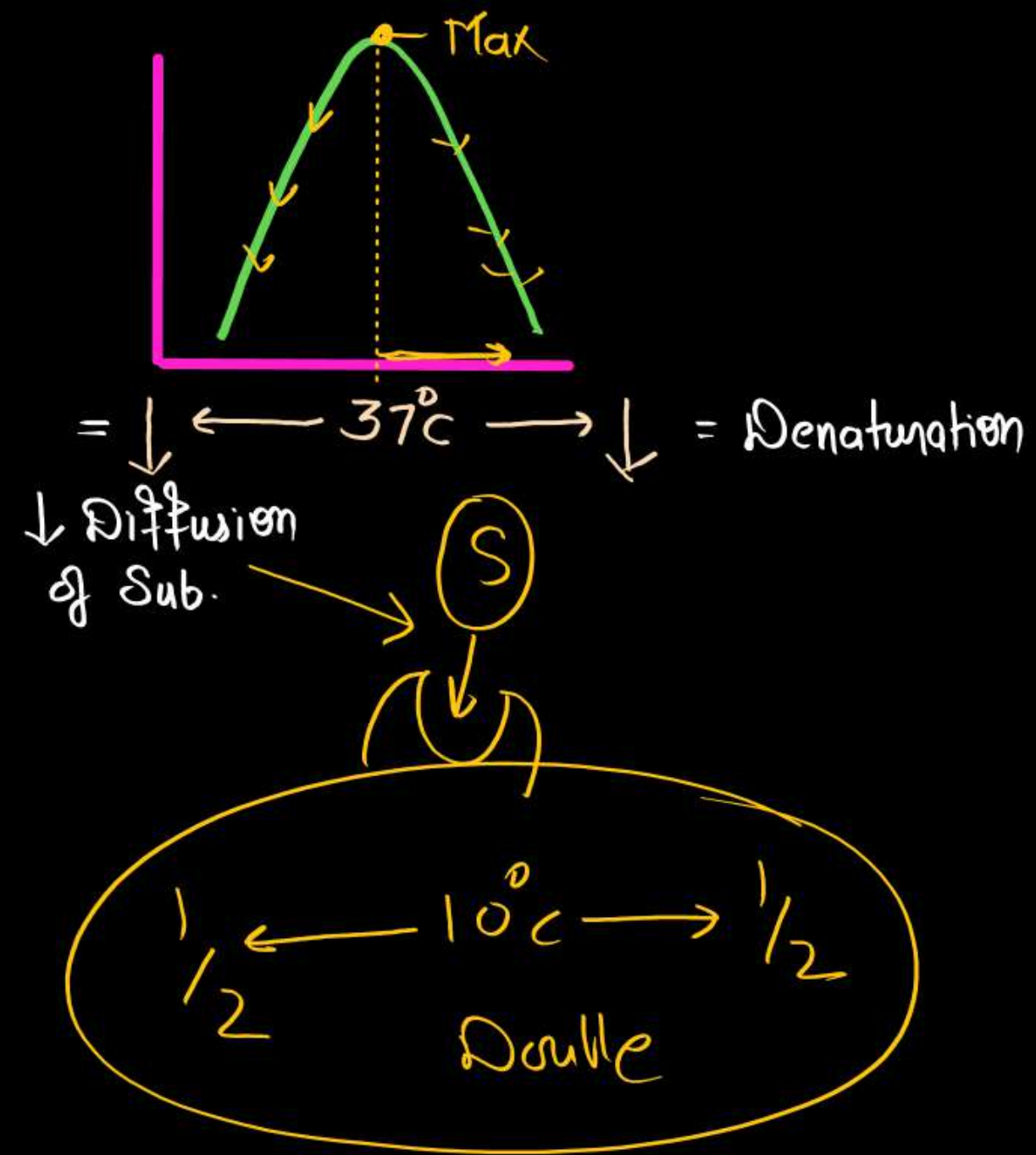
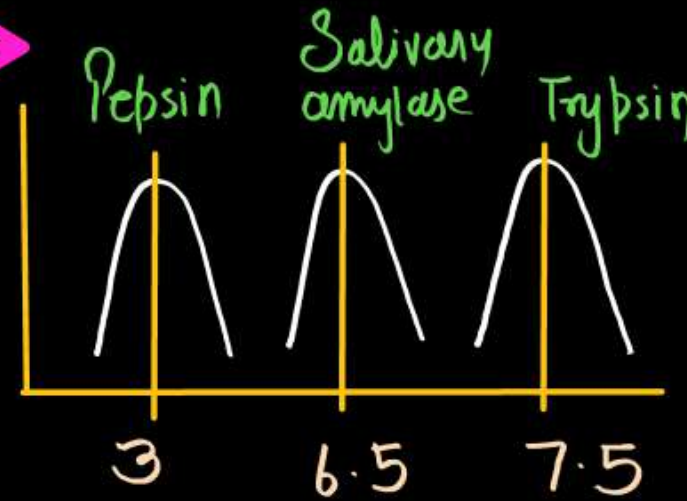


Nature of Enzyme Action -

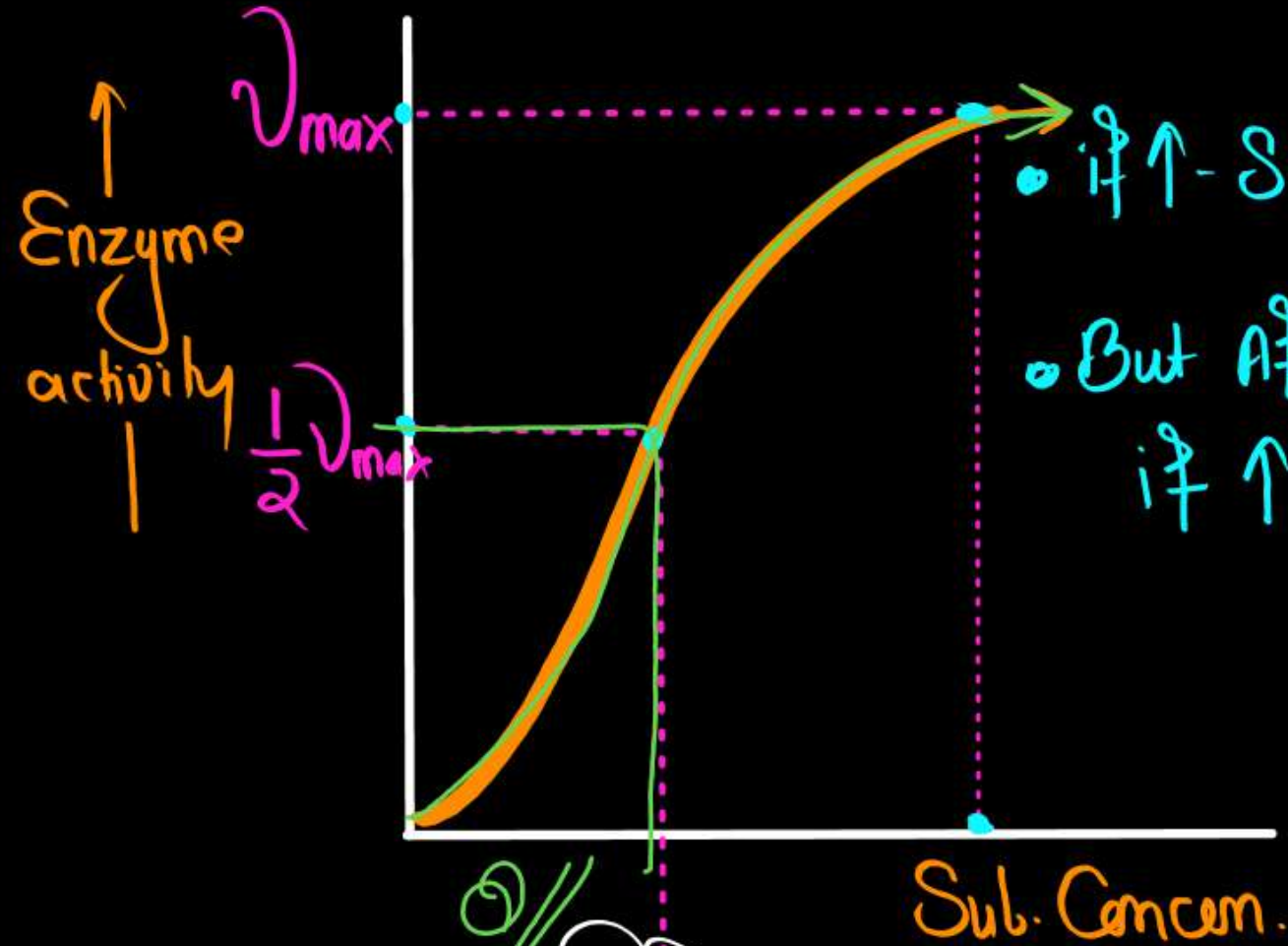


Factors Affecting Enzyme Activity

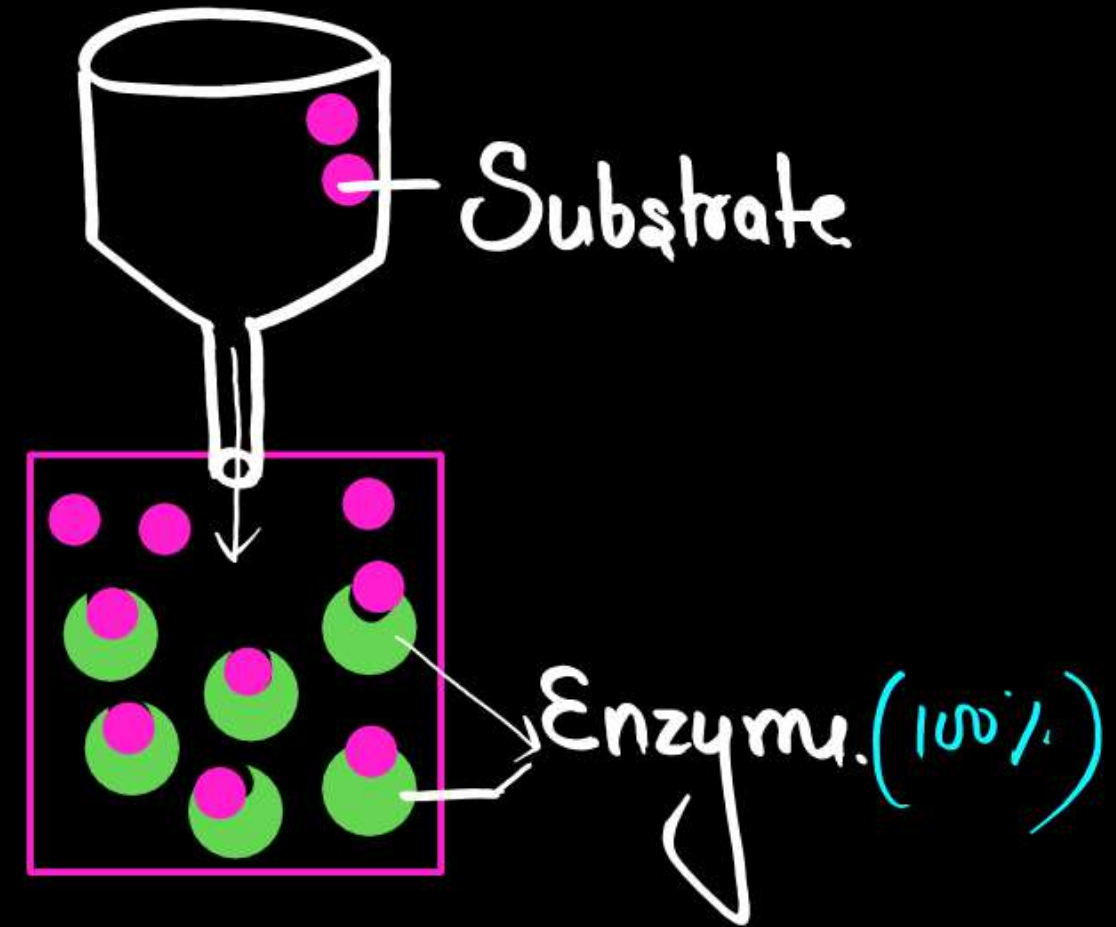
- Temperature →
- pH →
- Substrate concentration ✓
- Inhibitors



(3) Substrate concentration



- if $\uparrow$ Sub. Cons = $\uparrow$ Rate of Reaction.
- But After 100% Saturation
if $\uparrow$ Sub. Cons = Rate Cons.



K_m = Concentration of Substrate
at Velocity of Enzyme Reaction = $\frac{1}{2}$ of V_{max}

Competitive

Non competitive

Allosteric modulation

Reversible Binding

Common binding Site $\leftarrow \begin{matrix} S \\ I \end{matrix}$

In Presence of Inhibitor -

V_{max} = Same

Sub Cons. $= \uparrow = K_m = \uparrow$

eg - Methanol $\leftrightarrow$ Ethanol vvv

Succinic acid $\leftrightarrow$ Malonate

Irreversible Binding

Seperate binding Site $\leftarrow \begin{matrix} S \\ I \end{matrix}$

In Presence of Inhibitor

$V_{max} = \downarrow$

$K_m = \text{Cons.}$

eg - Cyanide.

Reversible binding

Seperate binding Site

K_m = Not follow

Product act as Inhibitor

eg - In Glycolysis Hexokinase.

Classification and Nomenclature of Enzymes

Enzymes are divided into 6 classes each with 4-13 subclasses and named accordingly by a four-digit number

- 1) - oxidoreductase
- 2) - Transferase.
- 3) - Hydrolase
- 4) - Lyase
- 5) - Isomerase
- 6) - Ligase

OTH-LIL

1st-digit
Class
(6)

6
1:1:1:1

E.C. No.

1st = Class

2nd = Sub class

3rd = Sub-Sub Class

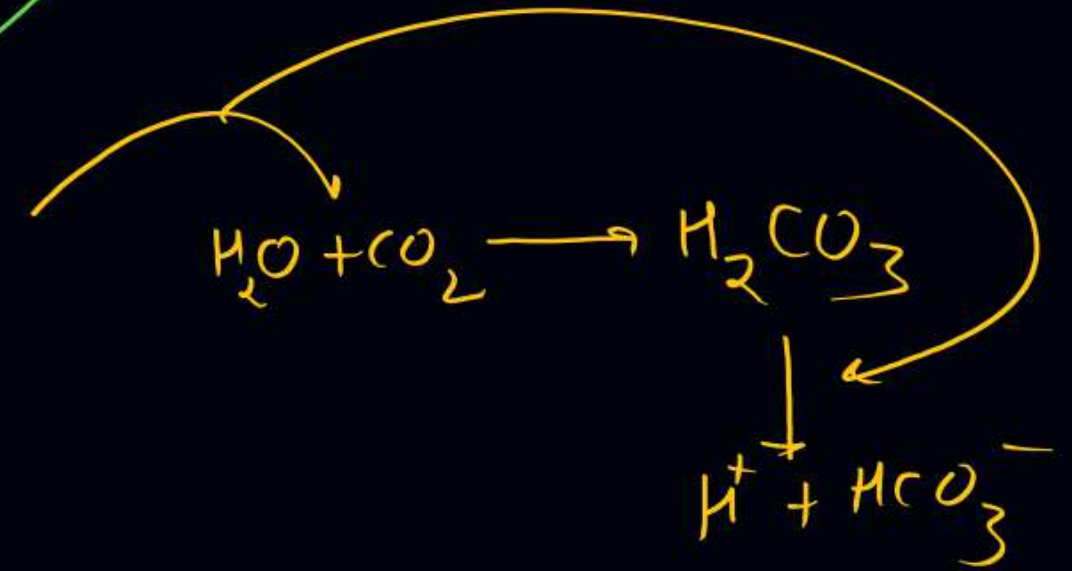
4th = Ser. No.

Class-1 Oxidoreductase/Dehydrogenase = Dehydrogenase, Reductase, Peroxydase

Class-2 Transferases = Transaminase, Hexokinase

Class-3 Hydrolases = All Digestive Enzymes ✓

Class-4 Lyases = Carbonic anhydrase



Class-5 Isomerases = Isomerase ✓

Class-6 Lygase = Ty-ligase

Types of Enzyme

Simple Enzyme

↓
Formed by only
Protein
[Active form]

② Conjugate Enzyme. Imp

Protein
[Apo enzyme]
[Inactive] + ③ Non Protein
[Co-factor]

(Holoenzyme)

Co-factors

Inorganic [Metal ions]
[Bind with - Coordination bond]

- Zn^{+2} → Carbonic anhydrase
→ Carboxy peptidase } Prob
- Cl^- → Salivary amylase
- Fe^{2+} → Cytochrome oxidase
→ Peroxydase
- Mg^{2+} → ATPase
→ Hexokinase
- Cu^{++} → Cytochrome oxidase

Organic

② Co-Enzyme [Non-Covalent] (Loose)
Prosthetic group [Covalent] (Strong)

Organic

Co-enzyme (Reversible binding)

[loosely bind $\rightarrow$ Non Covalent Interaction $\rightarrow$ Detach after Reaction]

* Derivative of Vit.

Riboflavin Vit.
(B₂)

F.A.D (Flavin adenine
dinucleotide)

Succinic Dehydrogenase

Thiamine Vit
B₁

T.P.P (Thiamine Pyro-
phosphate)

Pyruvic Dehydrogenase

Niacin Vit
B₃

NAD⁺ (Nicotinamide
Adenosine Dinucleotide)

Lactate Dehydrogenase

Co-Enzyme

Enzymes

~~v/vmp~~

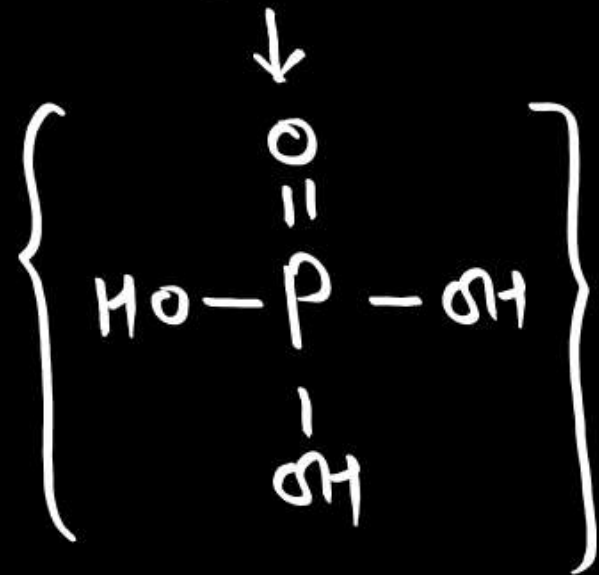
Prosthetic group

[Bind with - Covalent Interaction]

eg. Heme in Catalase & Peroxy-dase

Nucleic acid → Polymer of Nucleotides = unit of DNA/RNA

① Phosphate
(H_3PO_4)



② Pentose Sugar (5-C)

Ribose
↓
(RNA)

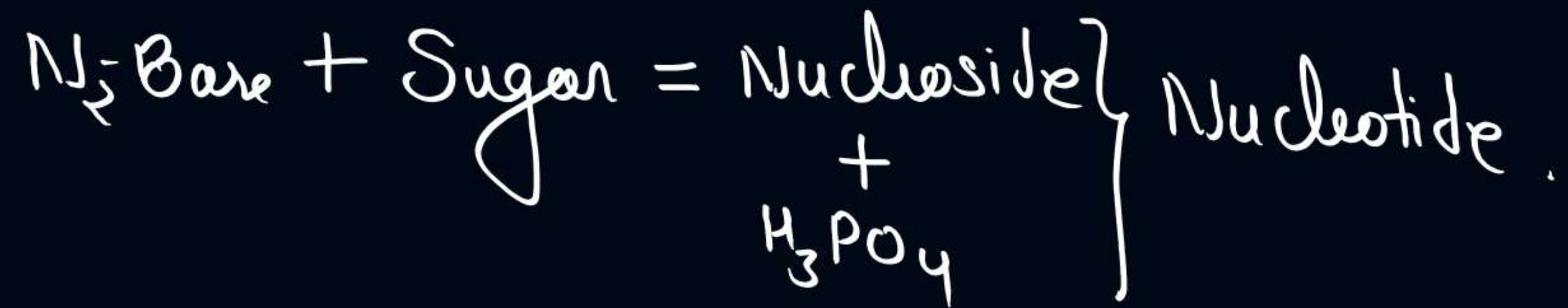
Deoxyribose
↓
(DNA)

③ N₂-Base

Purine
↓
(A+G)
Double Ring

Pyrimidine
↓
(C T U)
Single Ring

Difference b/w Nucleotide & Nucleoside.



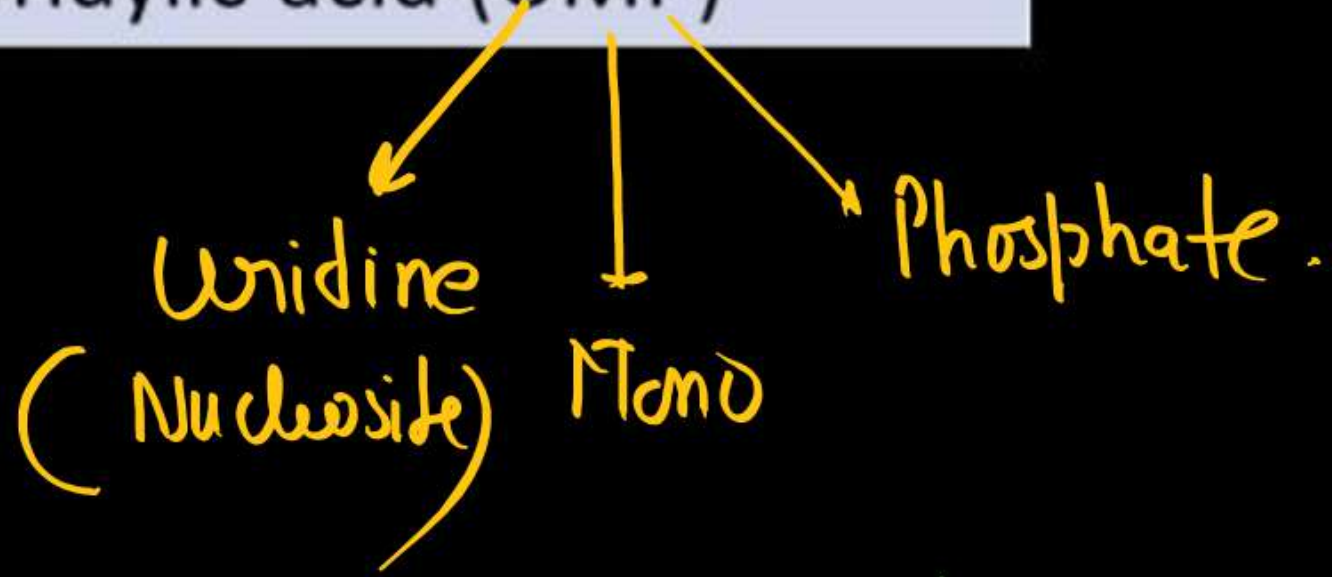
8/11/2022

Nitrogenous Base	Nucleoside	Nucleotide
Adenine (A) + S =	Adenosine + P =	Adenylic acid (AMP)
Guanine (G) + S =	Guanosine + P =	Guanylic acid (GMP)
Cytosine (C) + S =	Cytidine + P =	Cytidylic acid (CMP)
Thymine (T) + S =	Thymidine + P =	Thymidylic acid (TMP)
Uracil (U) + S =	Uridine + P =	Uridylic acid (UMP)

} Purines

} Pyrimidines

UMP



Types of Fatty Acids

Saturated

- ✓ Double bond -nt
- ✓ Solid at room temp.
- ✓ High M.P.

Mostly +nt in animal fat

eg - Palmitic acid
Stearic acid

Non-Essential

unsaturated

- ✓ = +nt
- ✓ = Liquid.
- ✓ = Low.

= Mostly +nt in Plant.

eg - oleic acid } MUFA

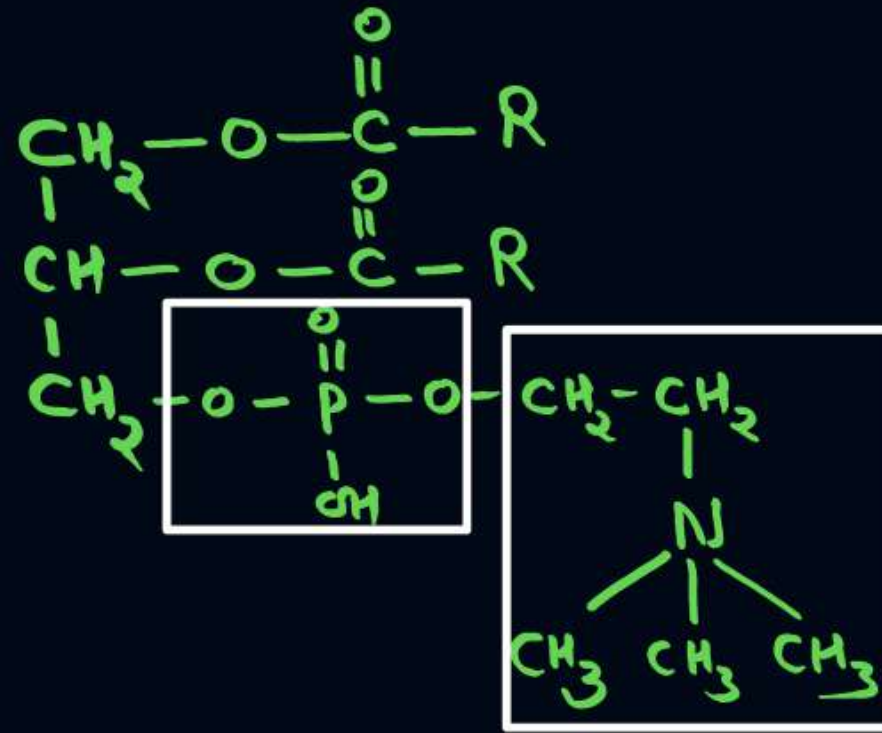
linoleic acid
linolenic acid
Arachidonic acid } PUFA (Essential)

~~high temp~~

Phospholipid

(Leathine)

vvvvfmb



Glycolipid

Carbohydrate

eg → Glycosphingo lipids
(+nt in Neurons)

- Fatty acids ✓
- Alcohol = Sphingosin
- Sugar = Galactose

Lipoprotein

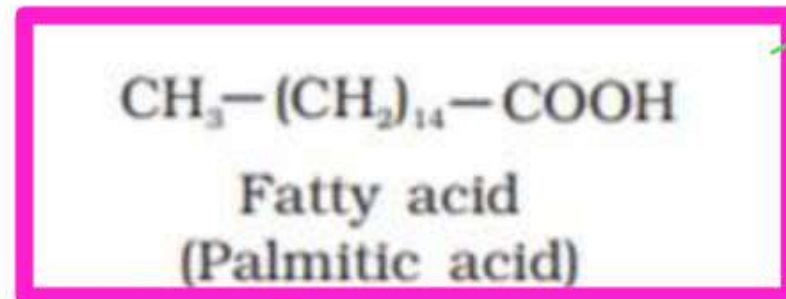
→ Chylomicron [Protein + lipid]

Chromolipid

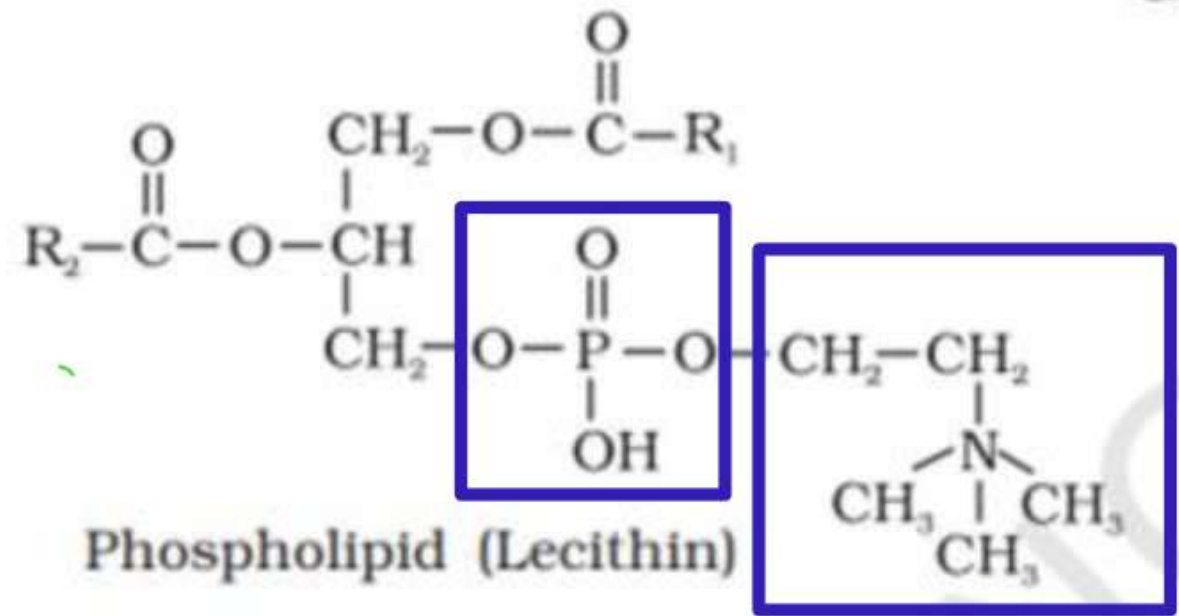
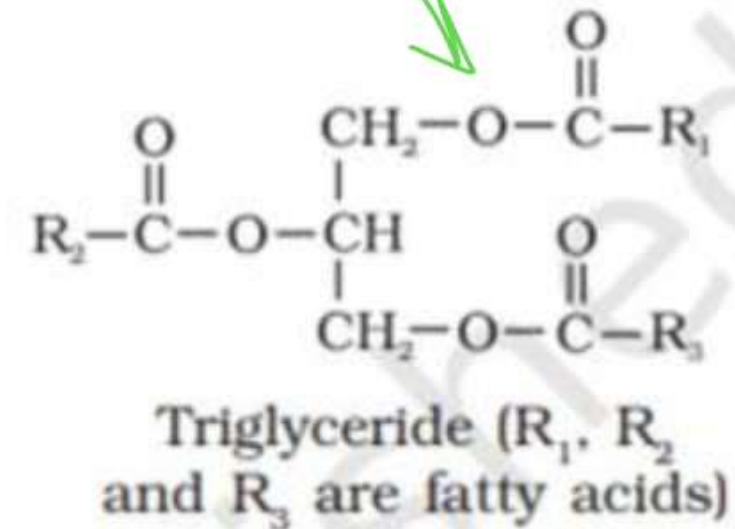
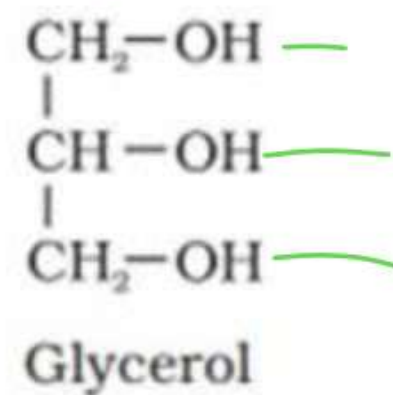
Pigment Containing Lipid

→ Carotene (Vit-A)

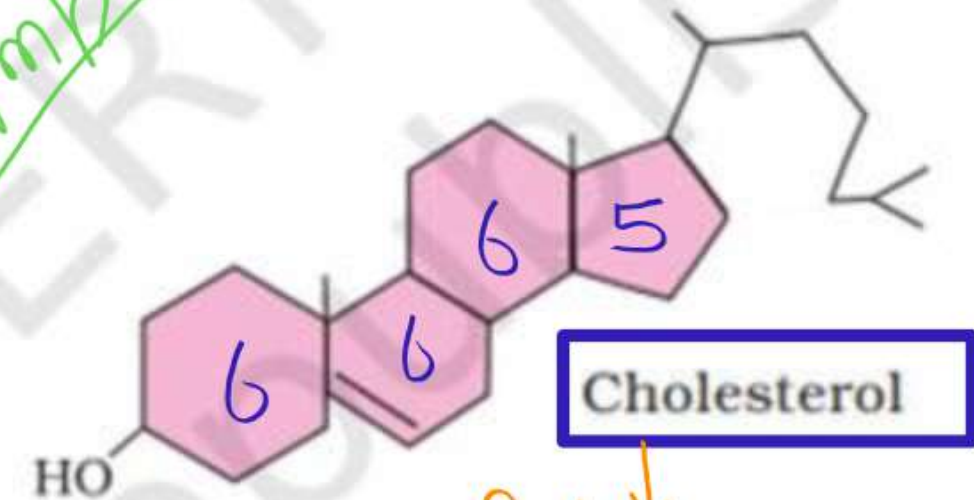
vvv imp



+



vvv imp



Fats and oils (lipids)

- Produce in Liver
- Most abundant **Stencids** in Animals
- Precursor of Hormones
- Fat like Property

Animal kingdom

Basis of Classification

Multicellular
Eukaryotic
Holozoic mode of Nutrition

⇒ Metazoa

Parazoa

eg Pontera

Eumetazoa

eg - Coelenterata - Chordata

1) - Level of Organisation

Cellular — Pontera
Tissue — Coelenterata & Ctenophora
Organ — Platyhelminthus
Organ System — Aschelminthus to Chordata.

2) - Pattern of Complexities

Digestive System

Incomplete



→ Coelenterata
Ctenophora
Platy.

Complete



→ Asch. → Chordata.

open

closed

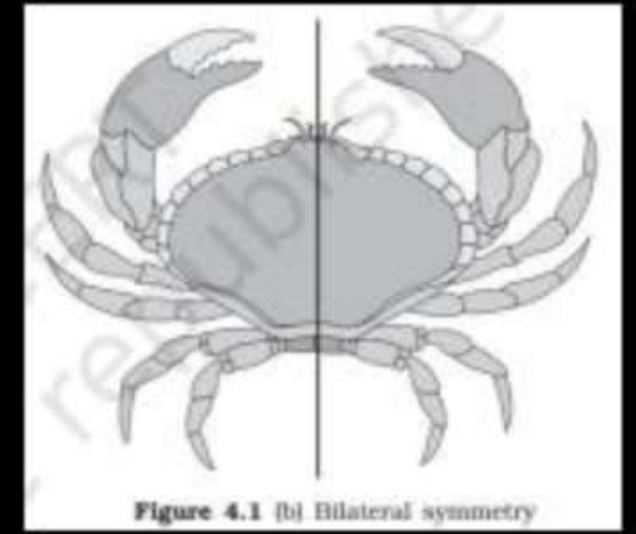
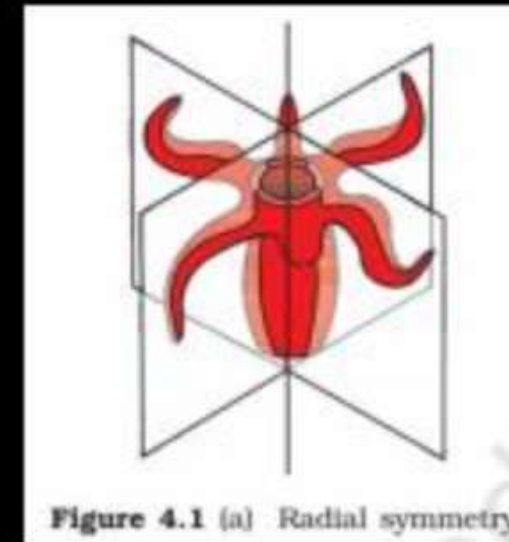
→ Arthropoda & Mollusca
Annelida & Chordata.

(3) Symmetry -

Radial → Some porifera, Coelenterata
Ctenophora, Adult Echino.

Bilateral → Platyhelminthes to Chordata

Asymmetry → Mostly Sponges eg- Spongilla

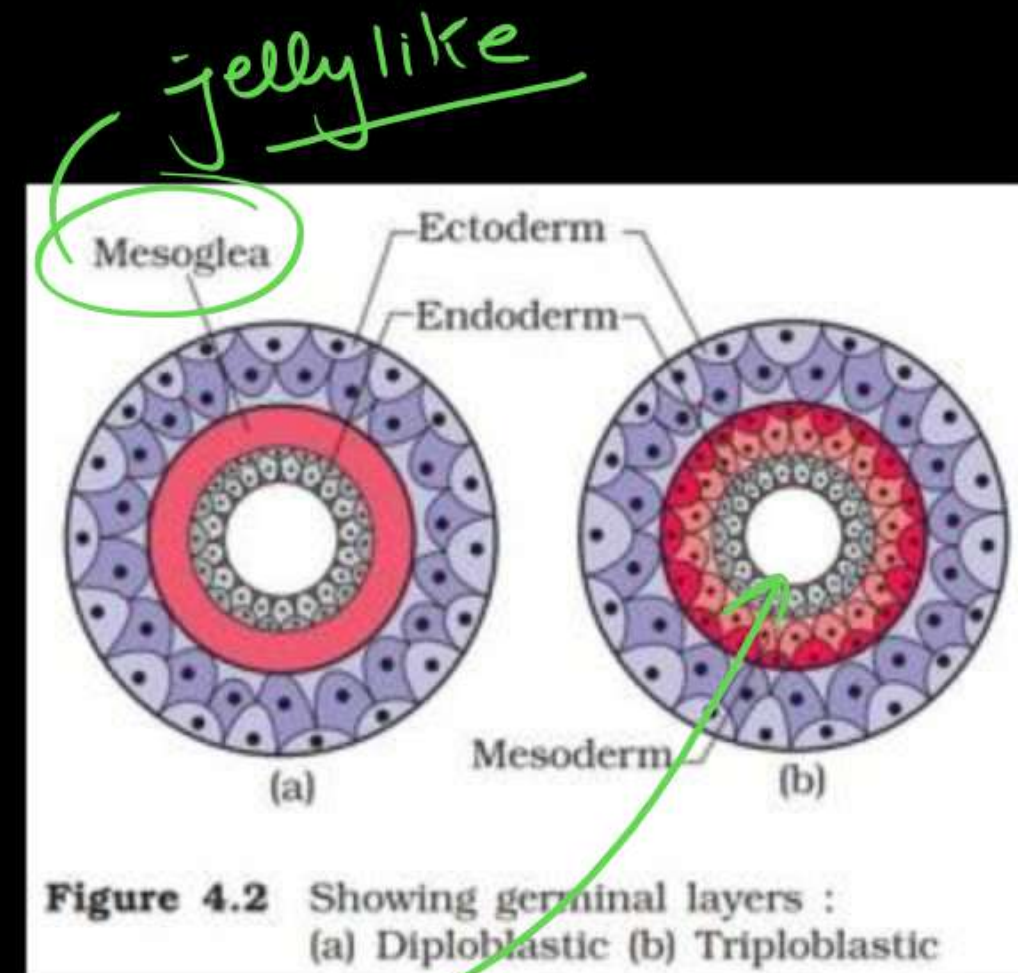


(4) Diploblastic and Triploblastic Organisation (Germ Layers)

Coelenterata
Ctenophora

Meso-
nt

Platyhelminthes
to
Chordata.



(5) Coelom - [Origin = Mesodermal → +nt b/w —
Outer body wall & Inner Gut Wall]

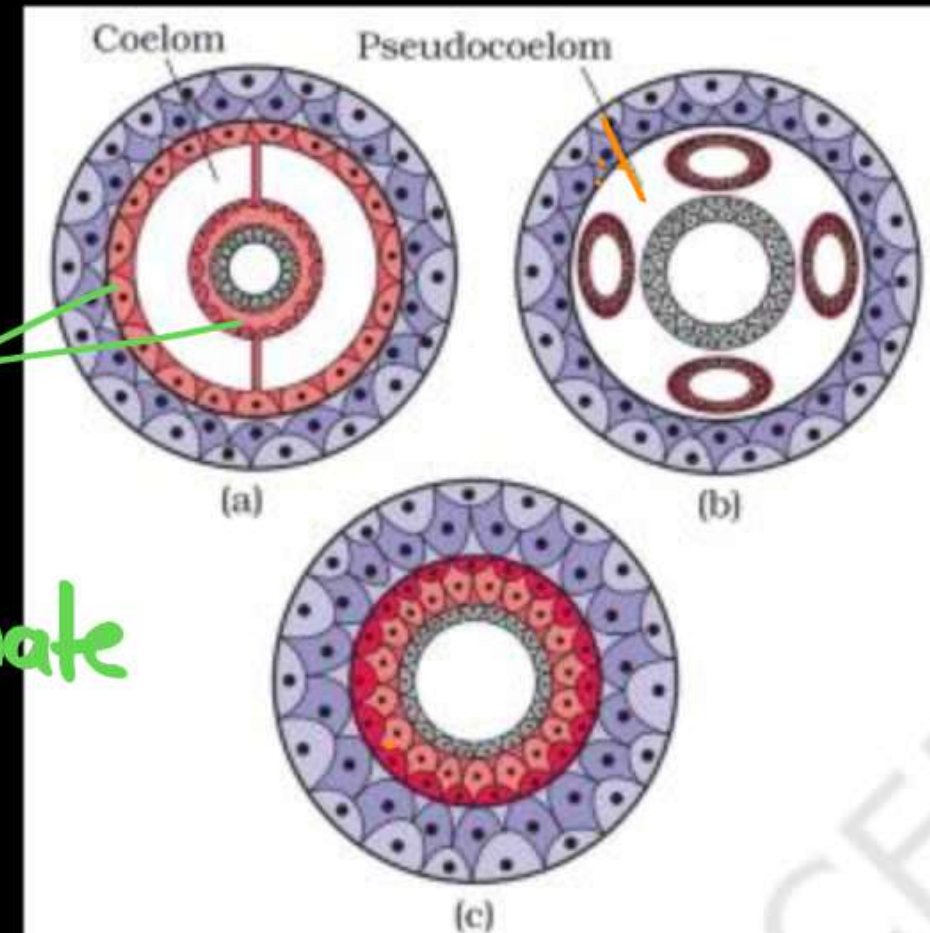


Figure 4.3 Diagrammatic sectional view of :
(a) Coelomate (b) Pseudocoelomate
(c) Acoelomate

vs

Acoelomate

eg → Coelenterata
Ctenophora
Platyhelminths
Solid Mesoderm

Pseudocoelomate

eg - Aschelminthes

Eucoelomate

Schizocoelom
[Splitting of Mesoderm]

eg → Annelida
Arthropoda
Mollusca

Entero.
[Outpocketing of Meso.]
Echinodermata
to
Chordata.

(7) Segmentation -



[Metameric Segmentation]

Position of Internal Organ according to External Segmentation

= eg → Annelida ✓
→ Arthropoda ✓
→ Chordata ✓

(8) Notochord -



Origin =

Mesodermal

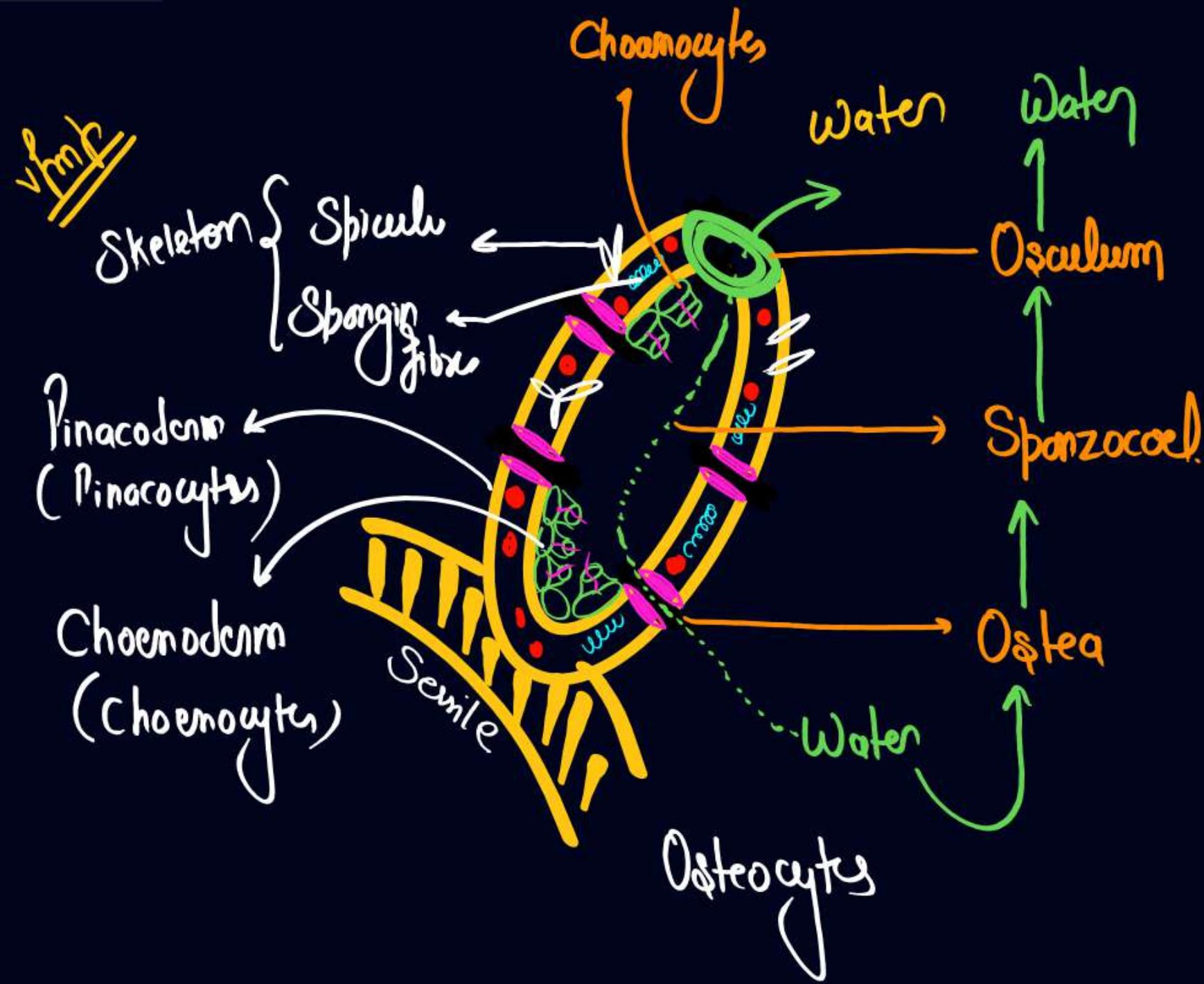
(Endoskeleton)

Nonchordata = Notochord - nt

Chordata = Notochord + nt

In some Chordata Notochord Replace by
V. Column Called = Vertebrata

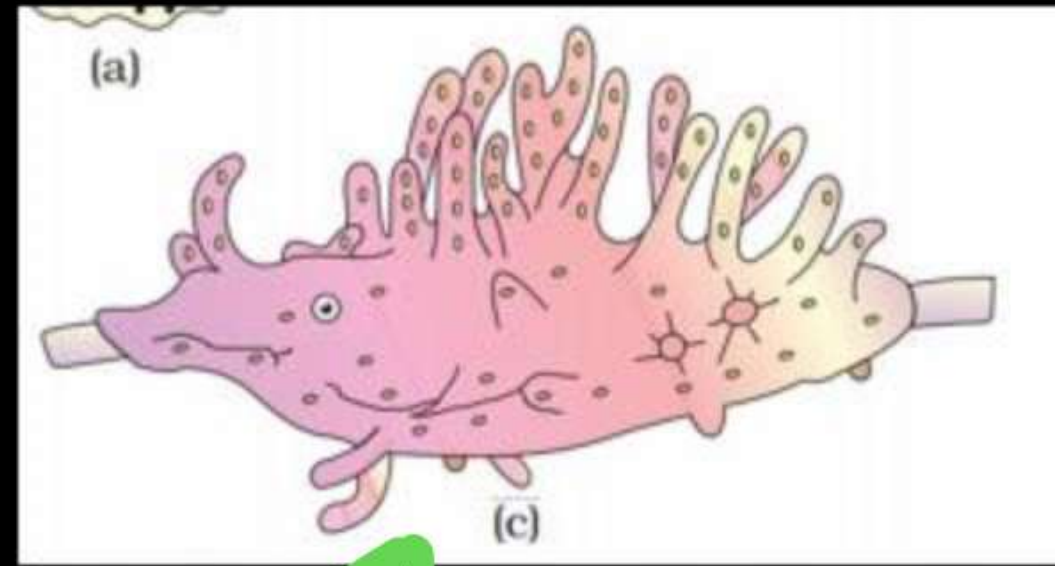
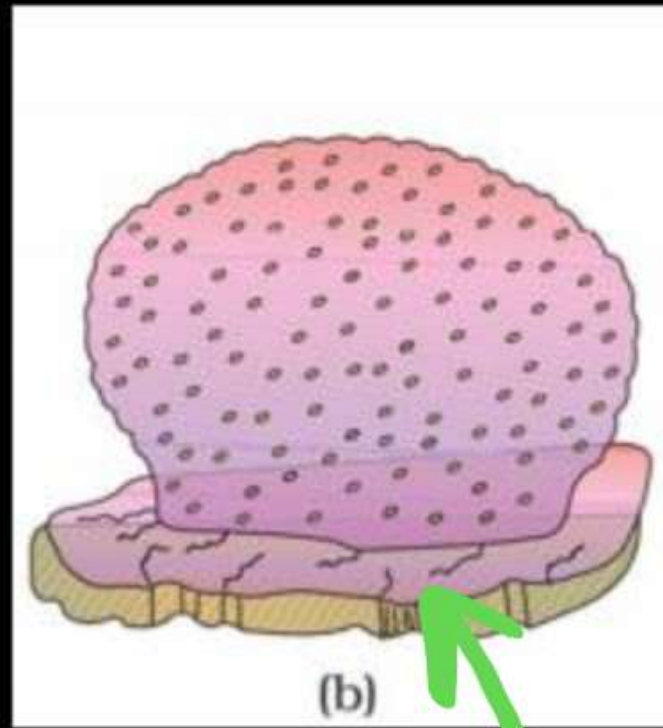
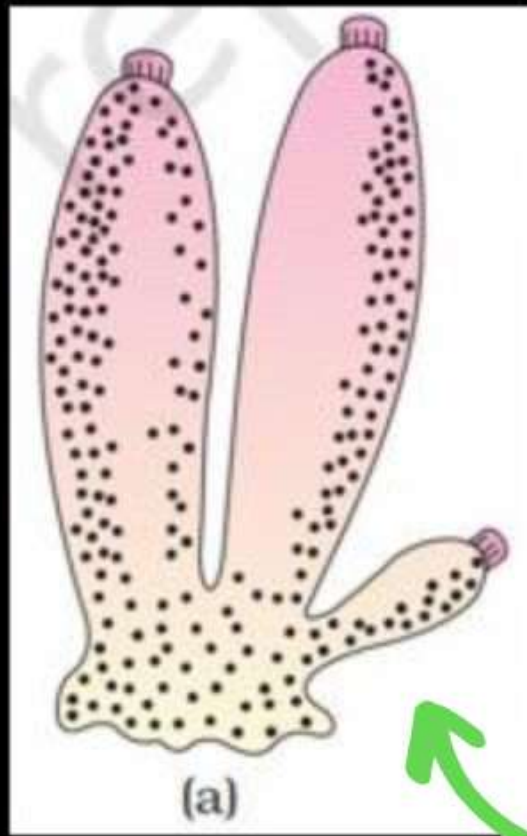
Porifera



~~Vamp~~
Water Canal System

Function

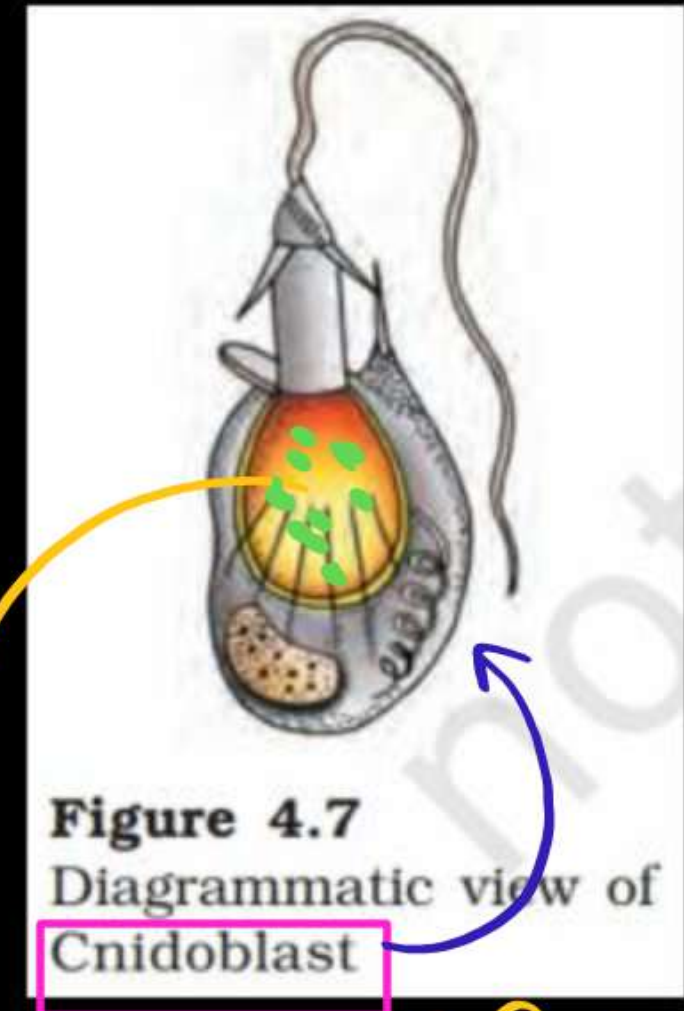
- Respiration ✓
- Excretion ✓
- Food Capture ✓
- Fertilization ✓
- etc.



Examples: Sycon (Scypha), Spongilla (Fresh water sponge) and Euspongia (Bath sponge).

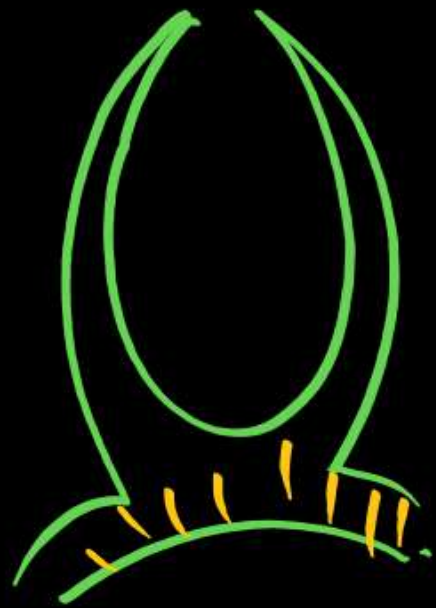
Phylum – Coelenterata (Cnidaria)

- The name cnidaria is derived from the **cnidoblasts** or **cnidocytes** (which contain the stinging capsules or **nematocysts**) present on the tentacles and the body.
- Cnidoblasts are used for **anchorage**, **defense** and for the **capture of prey**.
- Some of the cnidarians, e.g., corals have a skeleton composed of calcium carbonate.



Nematocyst
or Cnidocyst
v/v/m/b

vvv Members of this phylum show Bimorphic Condition



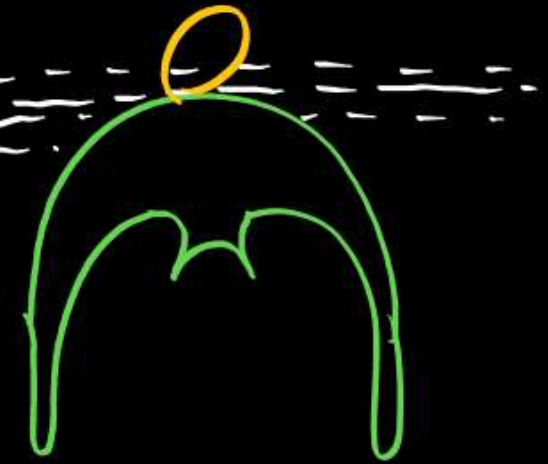
[Polyp]

- * Cylindrical
- * Sessile
- * Asexual Reproduction

eg = Hydra ✓
Obelia ✓
Adamsia ✓

[Medusa]

- * Umbrella like →
- * Free Swimming
- * Sexual Reproduction



eg = Aurelia
[jelly fish]



Examples: Physalia (Portuguese man-of-war), Adamsia (Sea anemone), Pennatula (Sea-pen), Gorgonia (Sea-fan) and Meandrina (Brain coral).



Phylum – Ctenophora

- Sea Walnut / Comb jelly — 8 Comb plates (Ciliated)
↓
Help in Locomotion

- Reproduction only Sexual

- Bioluminescence +nt

+nt only in Marine Water

Examples:

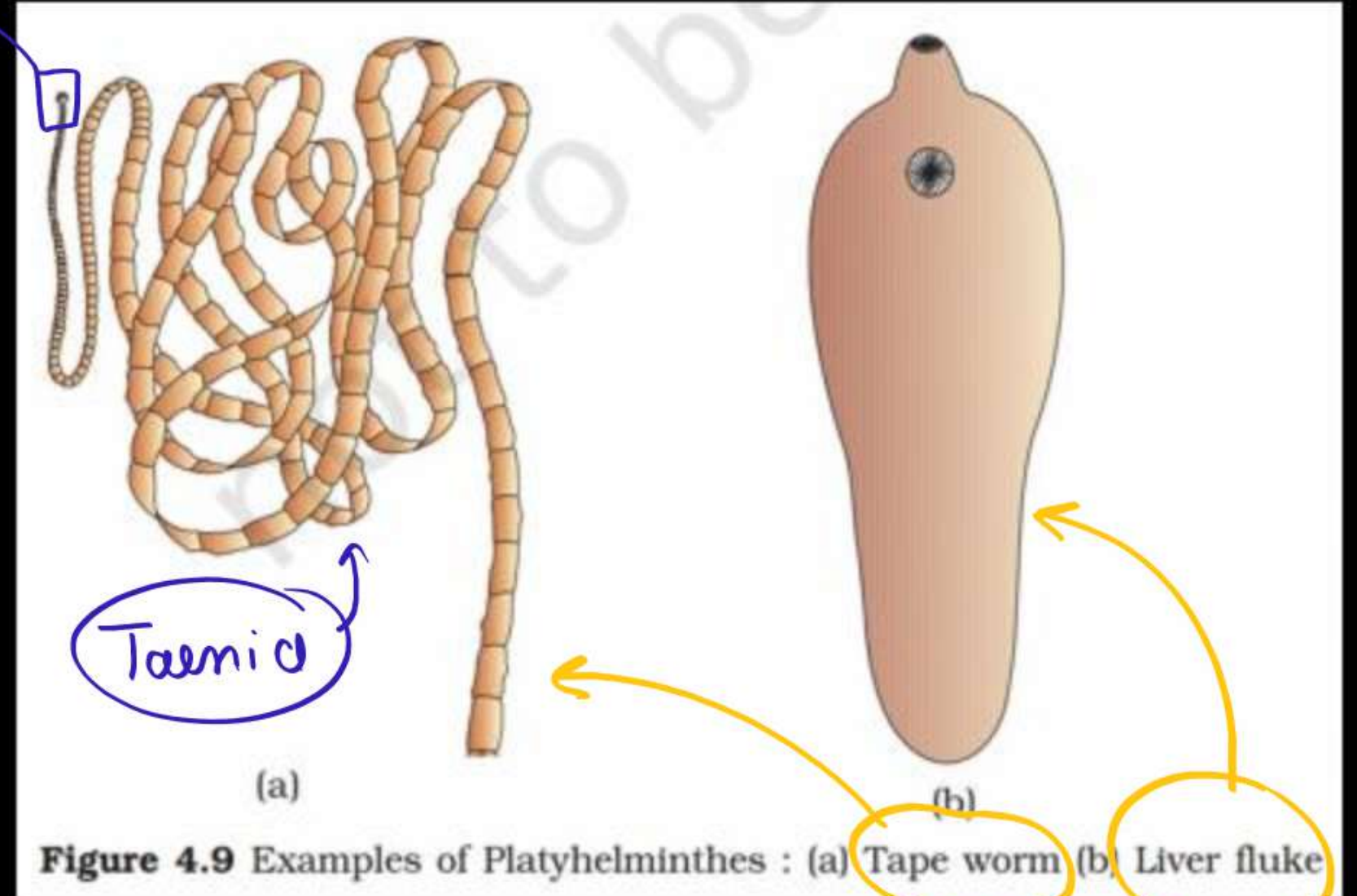
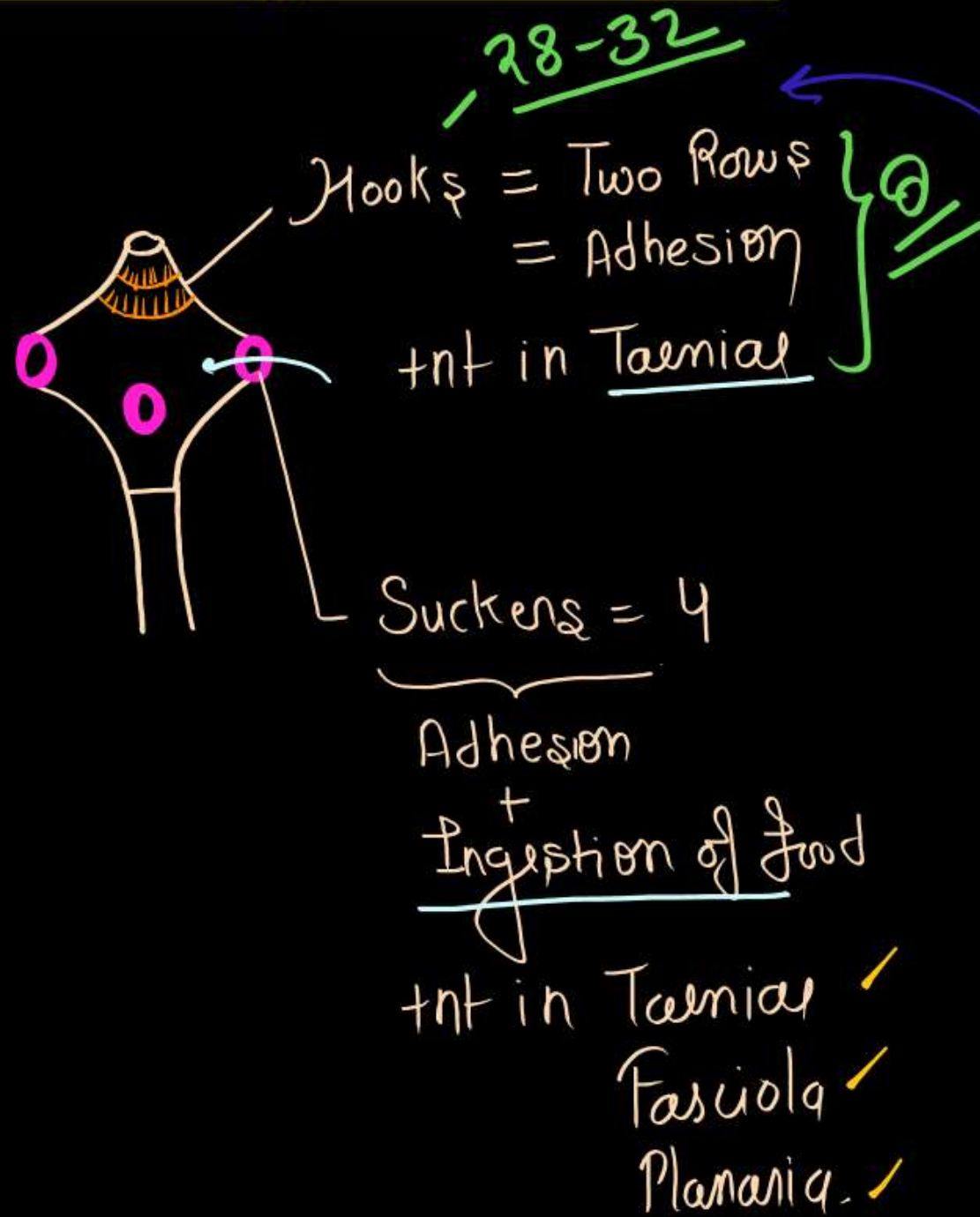
Ctenoplana.

Pleurobrachia

and



Phylum – Platyhelminthes



eg - Planaria = High Regeneration

Phylum – Aschelminthes

Sexes are separate (dioecious), i.e., males and females are distinct. Often females are longer than males.

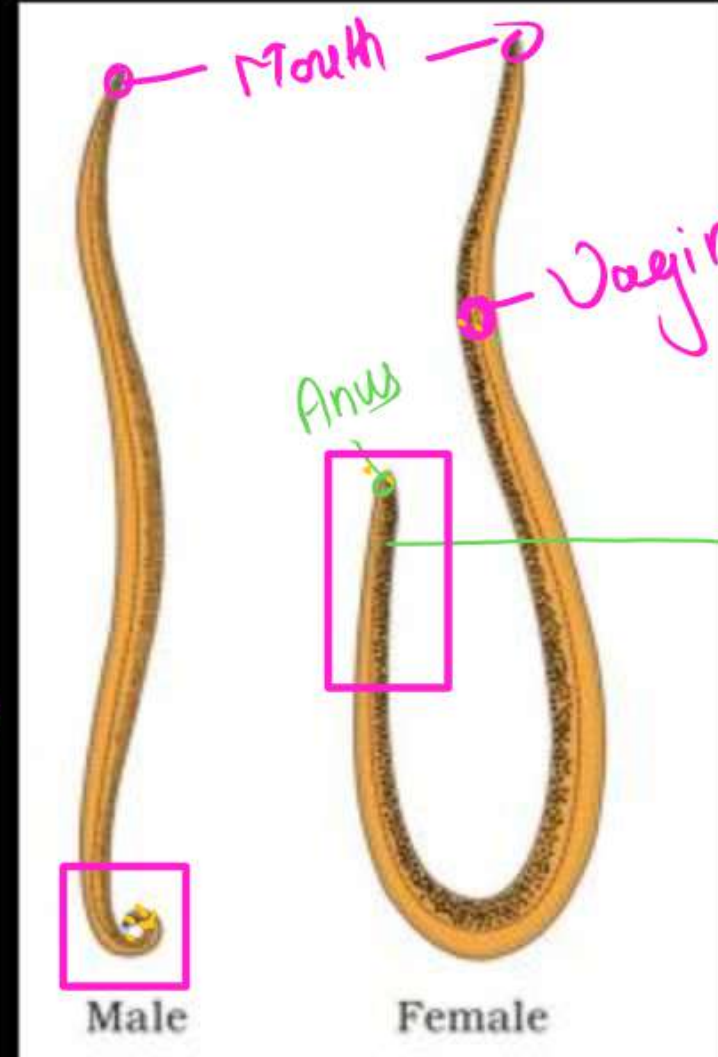
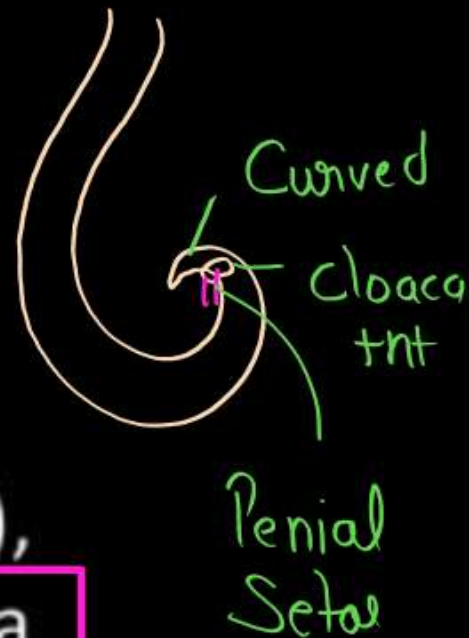
Examples

~~vvvvvvvvvv~~

Ascaris

(Roundworm),

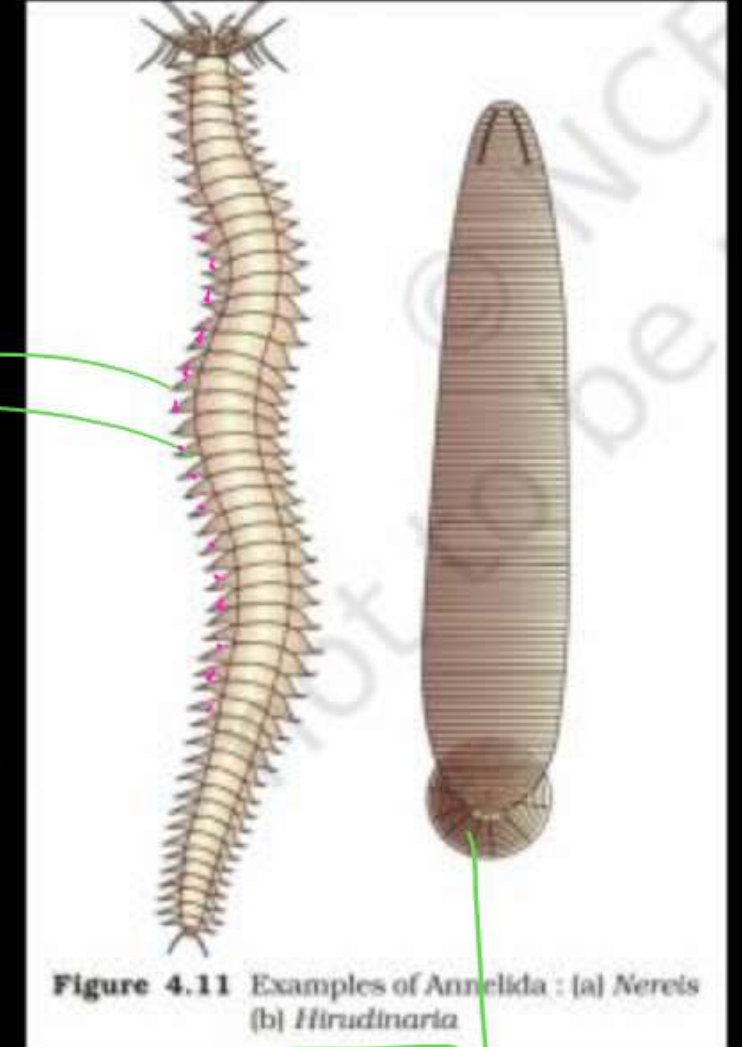
Wuchereria (Filaria worm), Ancylostoma (Hookworm).



Straight
Cloaca (nt)

Phylum – Annelida

② Parapodia
├── Locomotion
└── Respira.



✓
Examples : Nereis, Pheretima (Earthworm) and Hirudinaria (Blood sucking leech).

Phylum – Arthropoda (arthros-joint, poda-appendages)

1. This is the largest phylum of Animalia which includes insects.
2. Over two-thirds of all named species on earth are arthropods.

70%

Examples: Economically important insects –

Apis (Honey bee),

Bombyx (Silkworm),

Laccifer (Lac insect)

Vectors – Anopheles, Culex and Aedes
(Mosquitoes)

Gregarious pest – Locusta (Locust)

Limulus – living fossil

vvvvv fomp

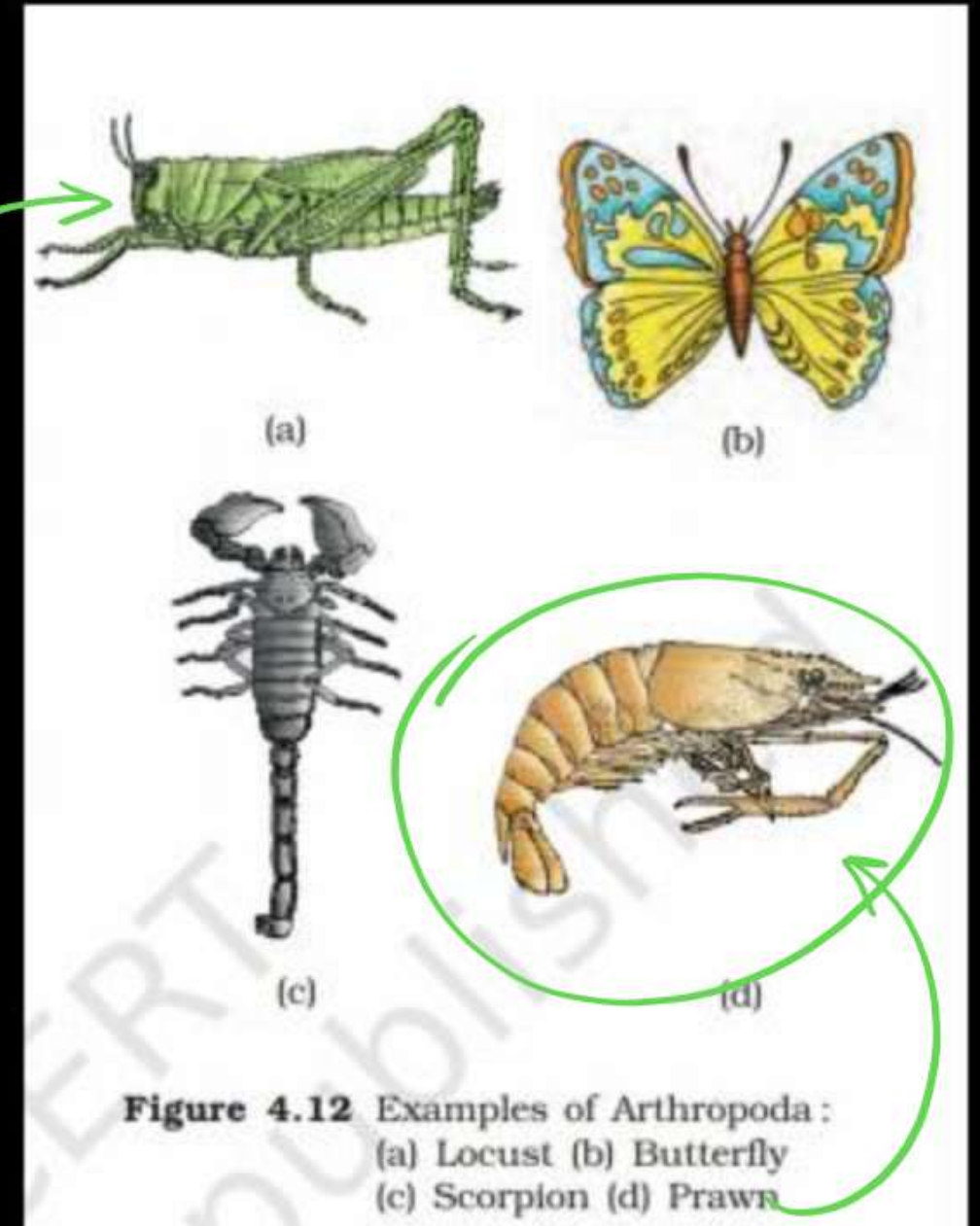
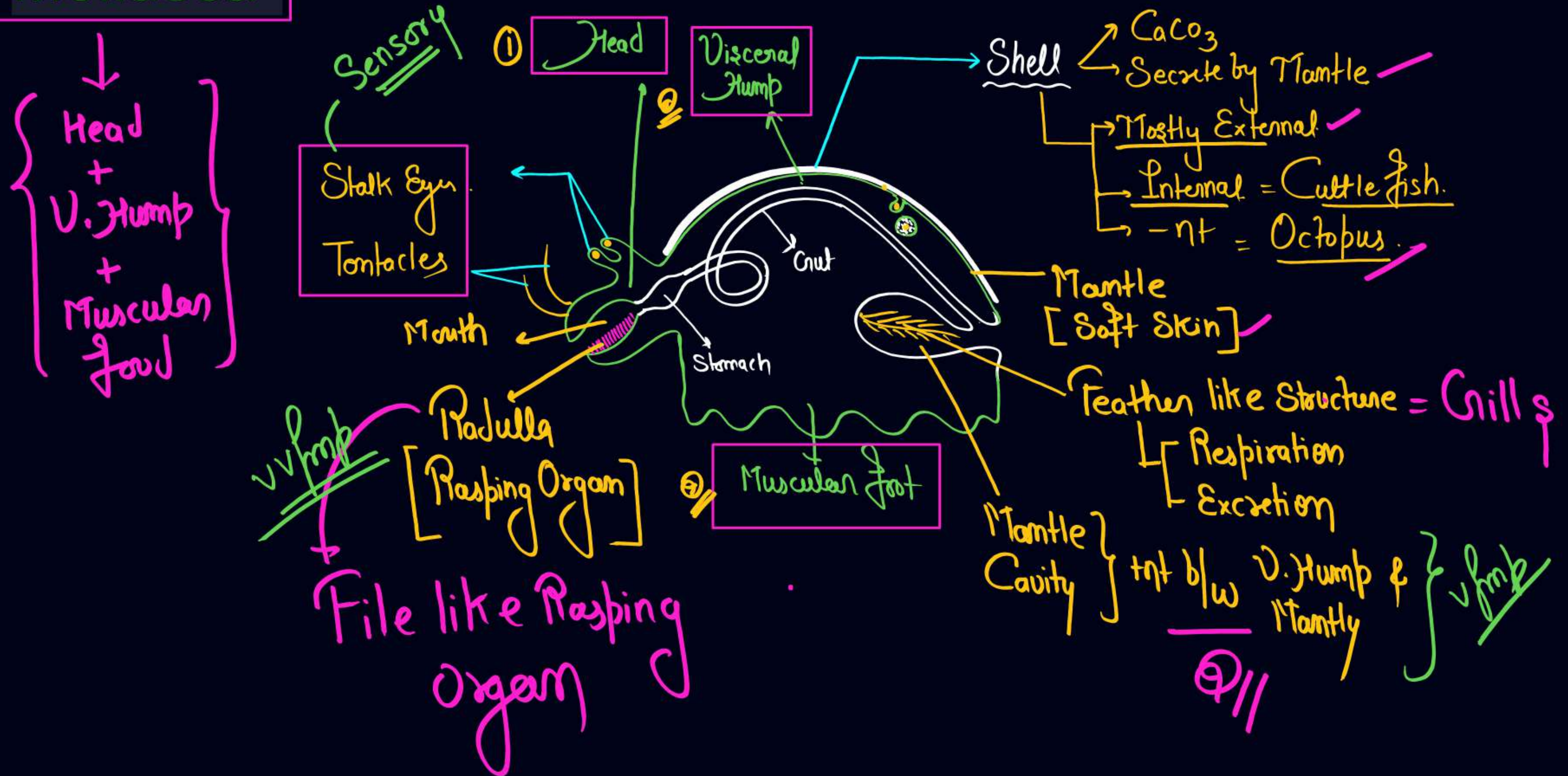
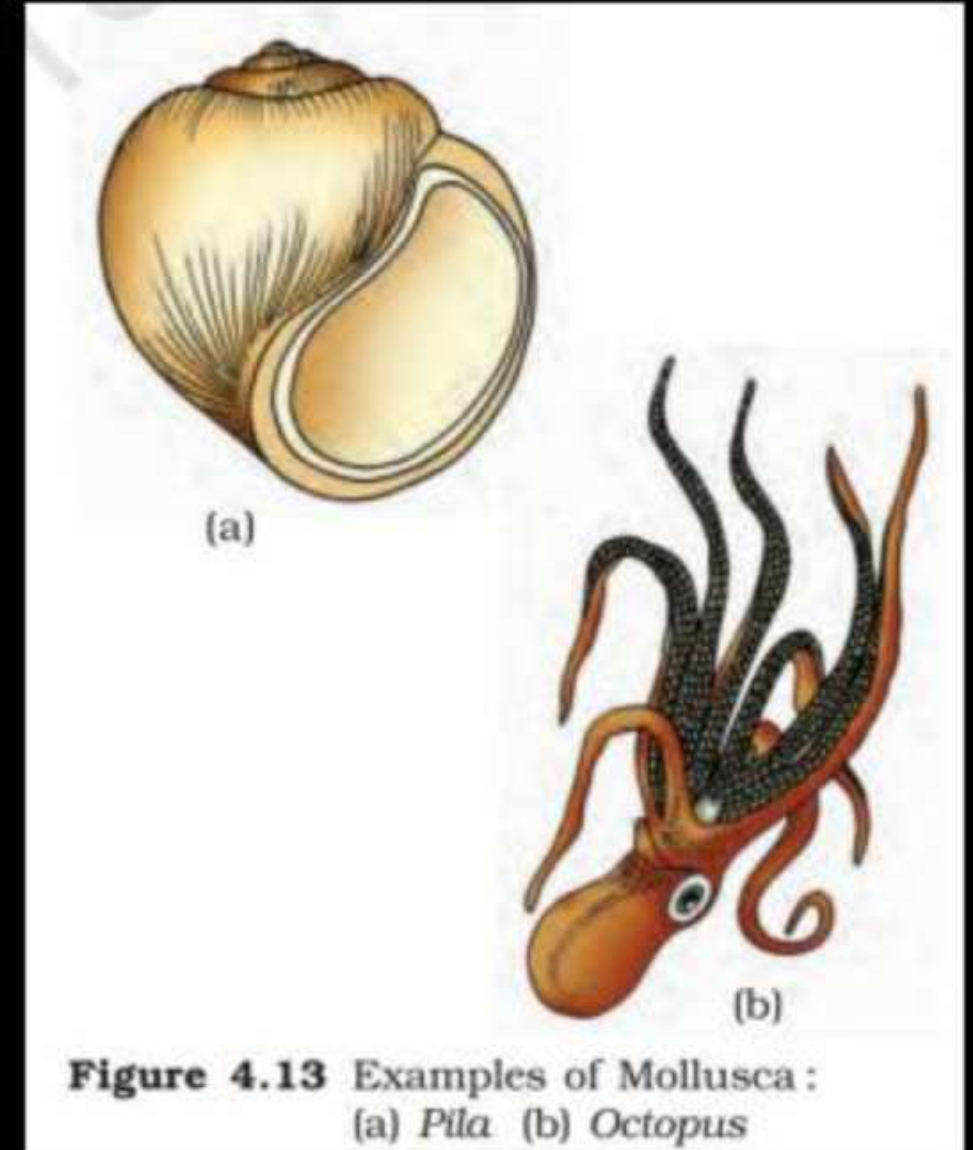


Figure 4.12 Examples of Arthropoda :
(a) Locust (b) Butterfly
(c) Scorpion (d) Prawn

Mollusca

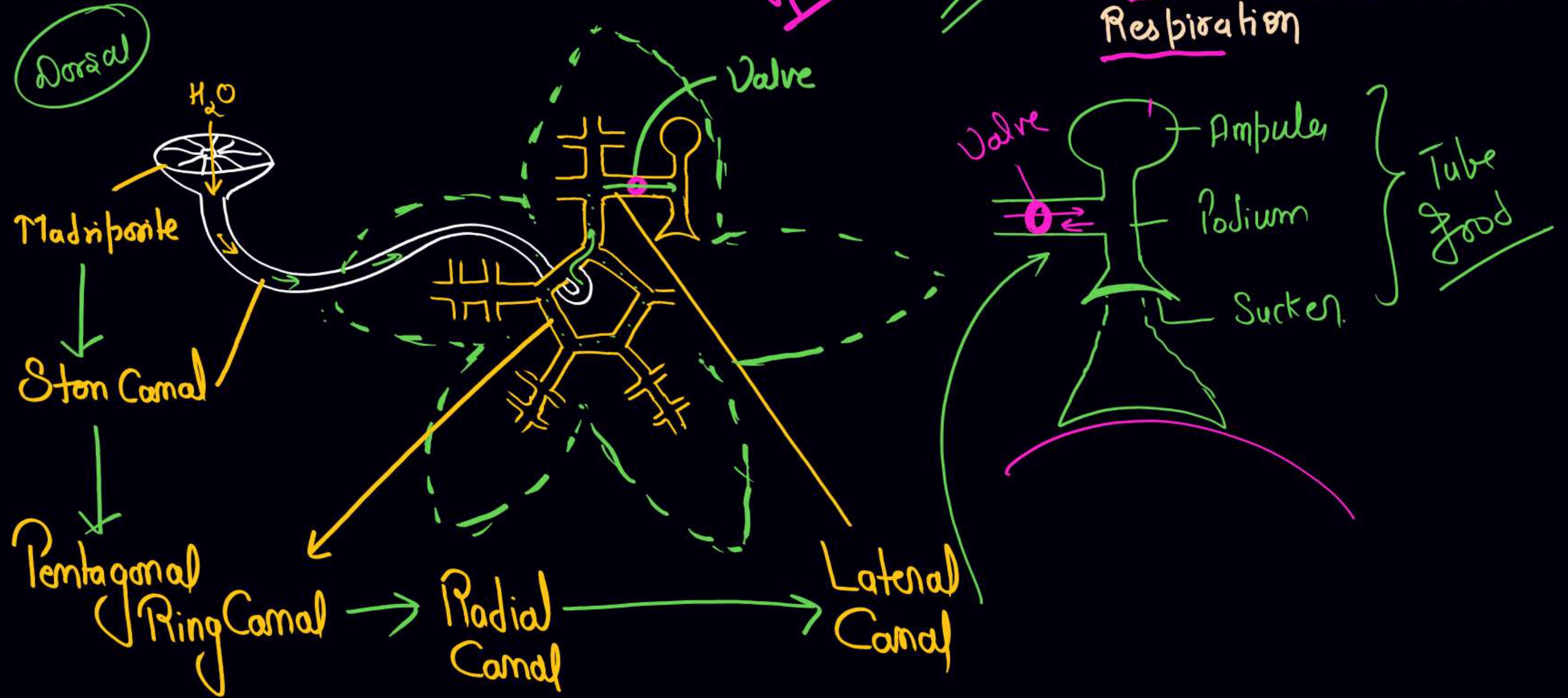


Examples: Pila (Apple snail), Pinctada (Pearl oyster), Sepia (Cuttlefish), Loligo (Squid), Octopus (Devil fish), Aplysia (Sea hare), Dentalium (Tusk shell) and Chaetopleura (Chiton).



Echinodermata

= Water Vascular System = Locomotion /
Capture & Transport of food
Respiration



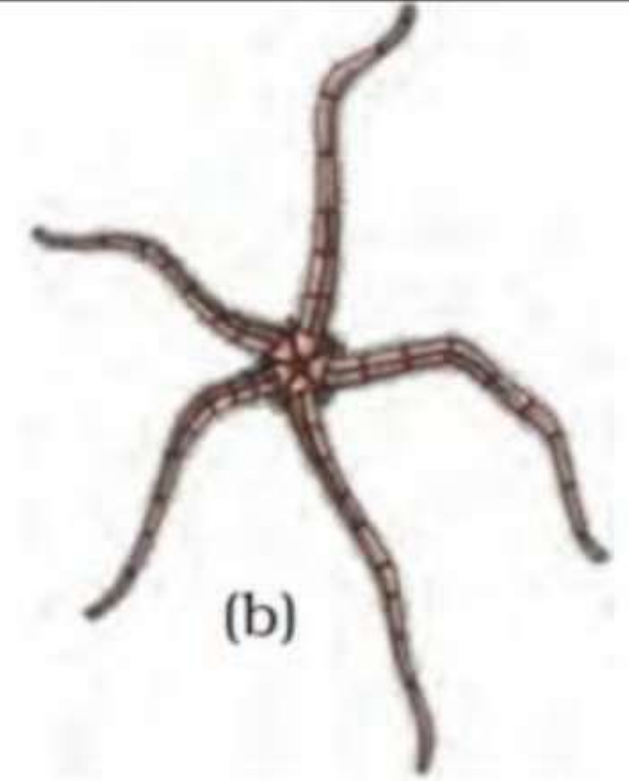
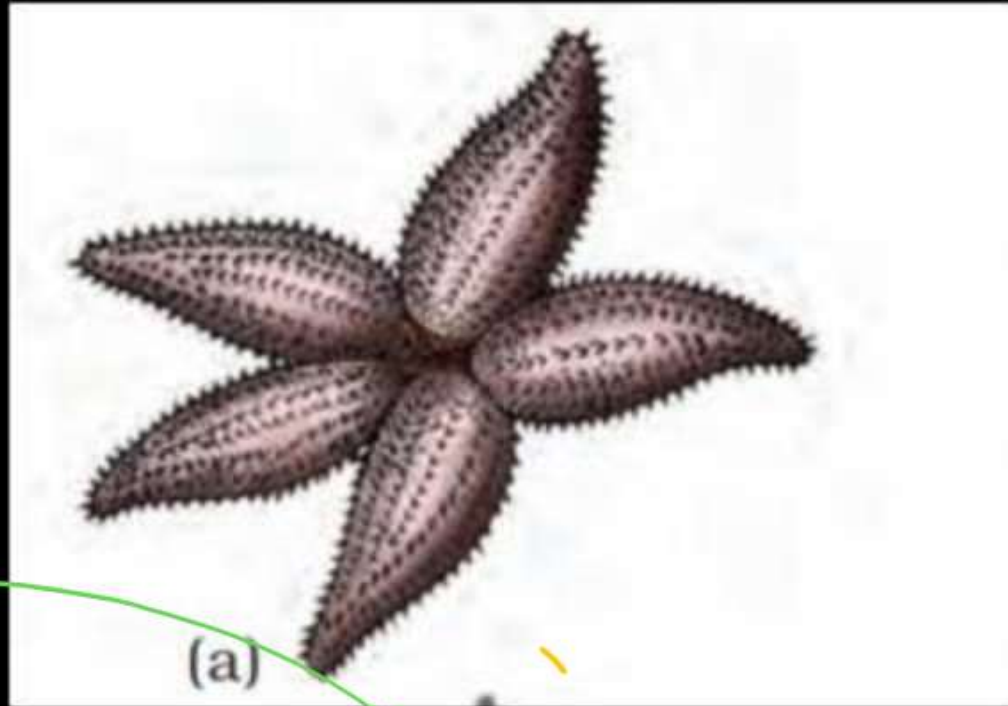
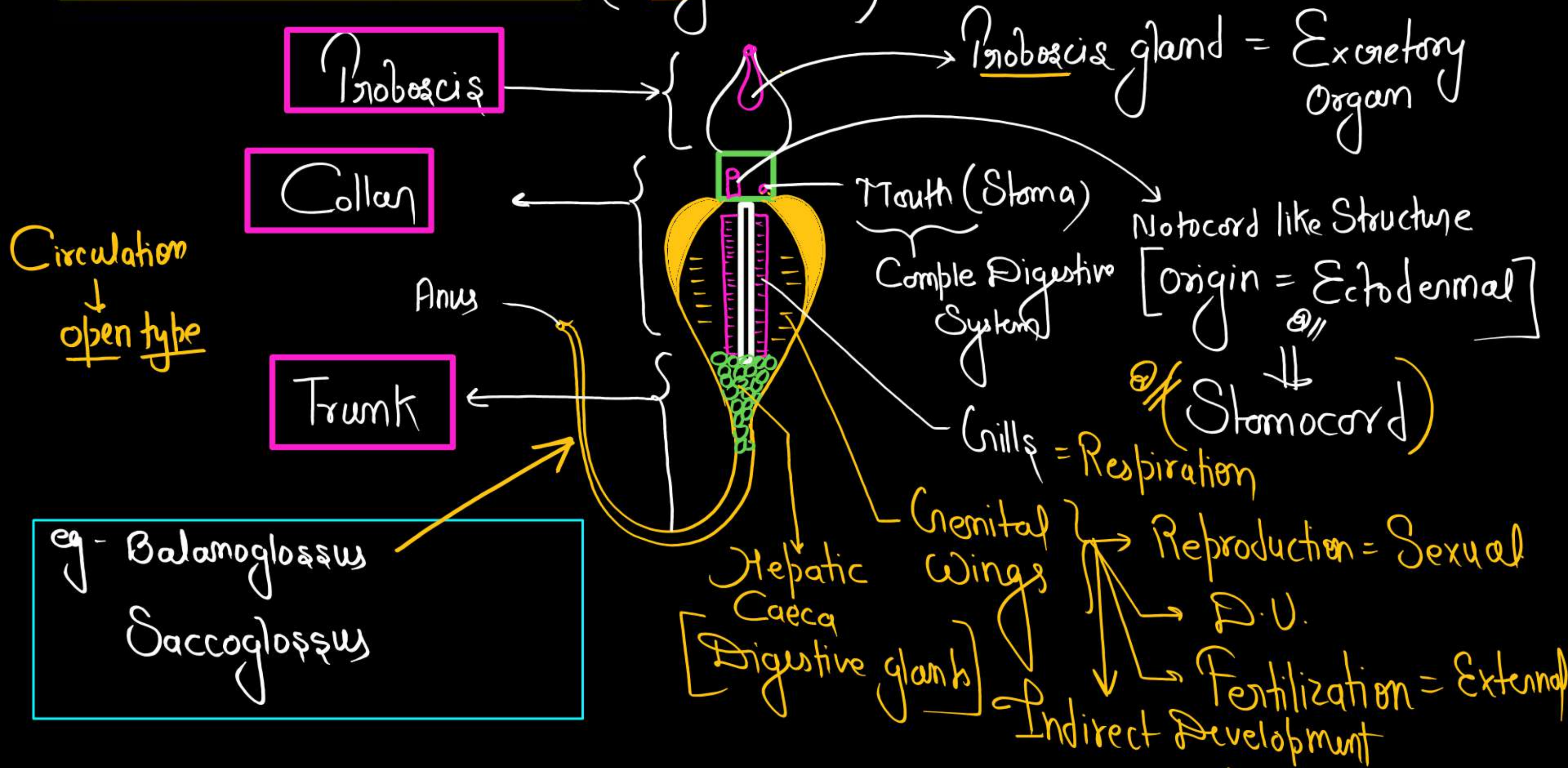


Figure 4.14 Examples of Echinodermata :
(a) *Asterias*
(b) *Ophiura*

Examples: Asterias (Star fish), Echinus (Sea urchin), Antedon (Sea lily), Cucumaria (Sea cucumber) and Ophiura (Brittle star).

Phylum – Hemichordate (Tongue Worm)



Chordata

Origin - Ectodermal
+ nt

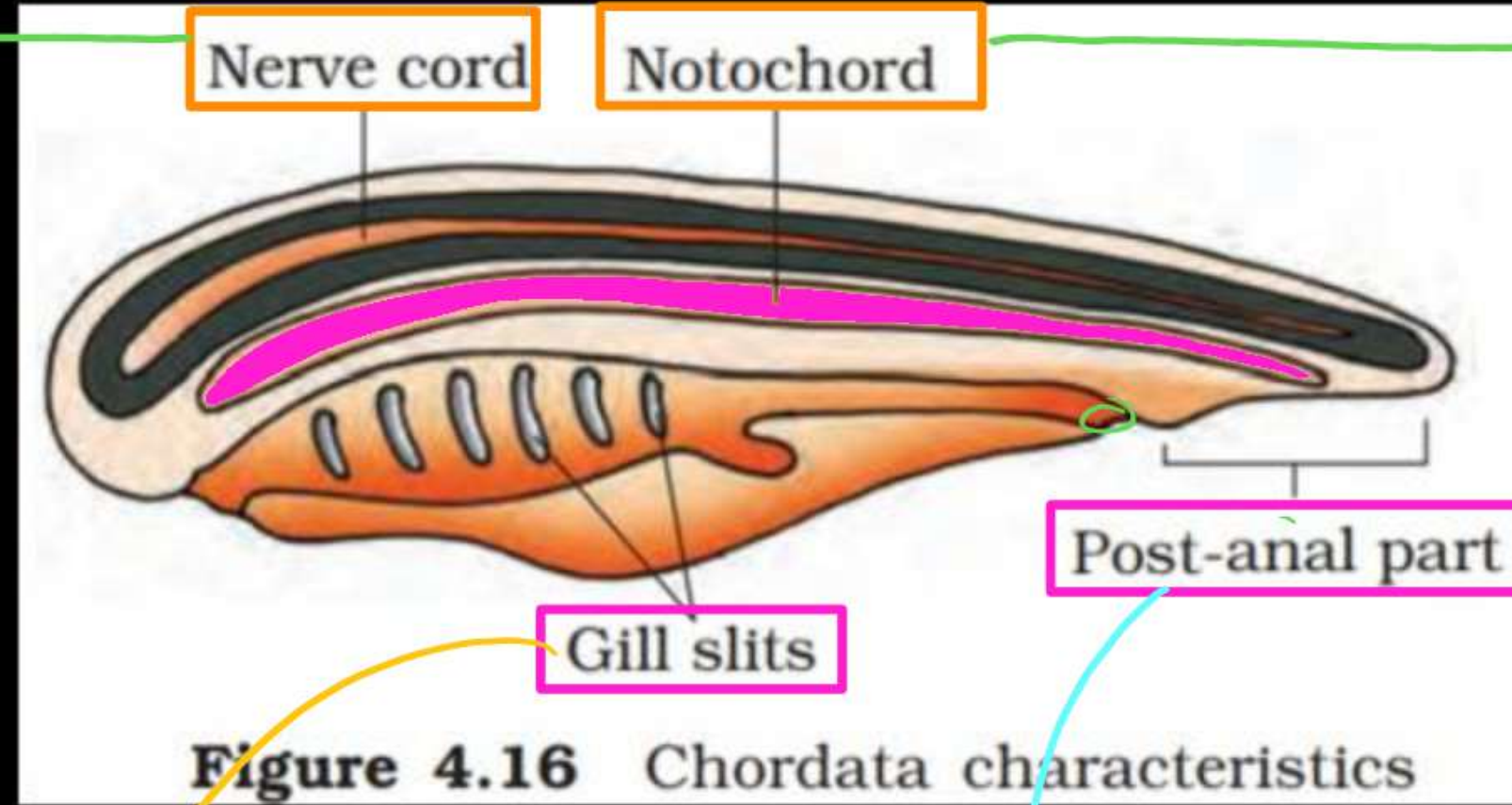
Nonchordates

Chordates

Ventral
Double
Solid

Dorsal
Single
Hollow

In some higher
Chordate Gill Slits
+ nt only in Embryonic
Stage.



Origin = Mesodermal
+ nt only in chordata
Supporting Structure
+ nt behind = Nerve Cord
above = Gut

for Balancing

In-Some chordate.
Notochord Replace by
V. Columns

Phylum Chordata

(Brain box)

Protochordata [Acraniata]

(Sub-Phylum)

Urochordata

Tail

* Notochord + nt only
in tail of Larva —
-nt in Adult

eg → Ascidia
Solpa
Doleolum

Cephalochordata

Head

* Notochord + nt —
Head to Tail in —
Larva & Adult both

eg → Branchiostoma
or
Amphioxus or lanulelet

Euchordata [Craniata]

Sub-Phylum

Vertebrata

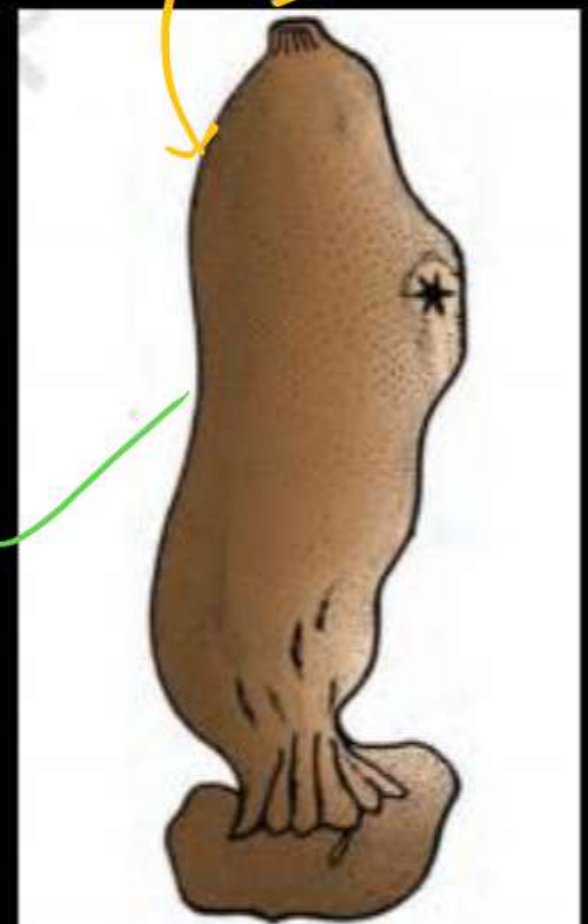
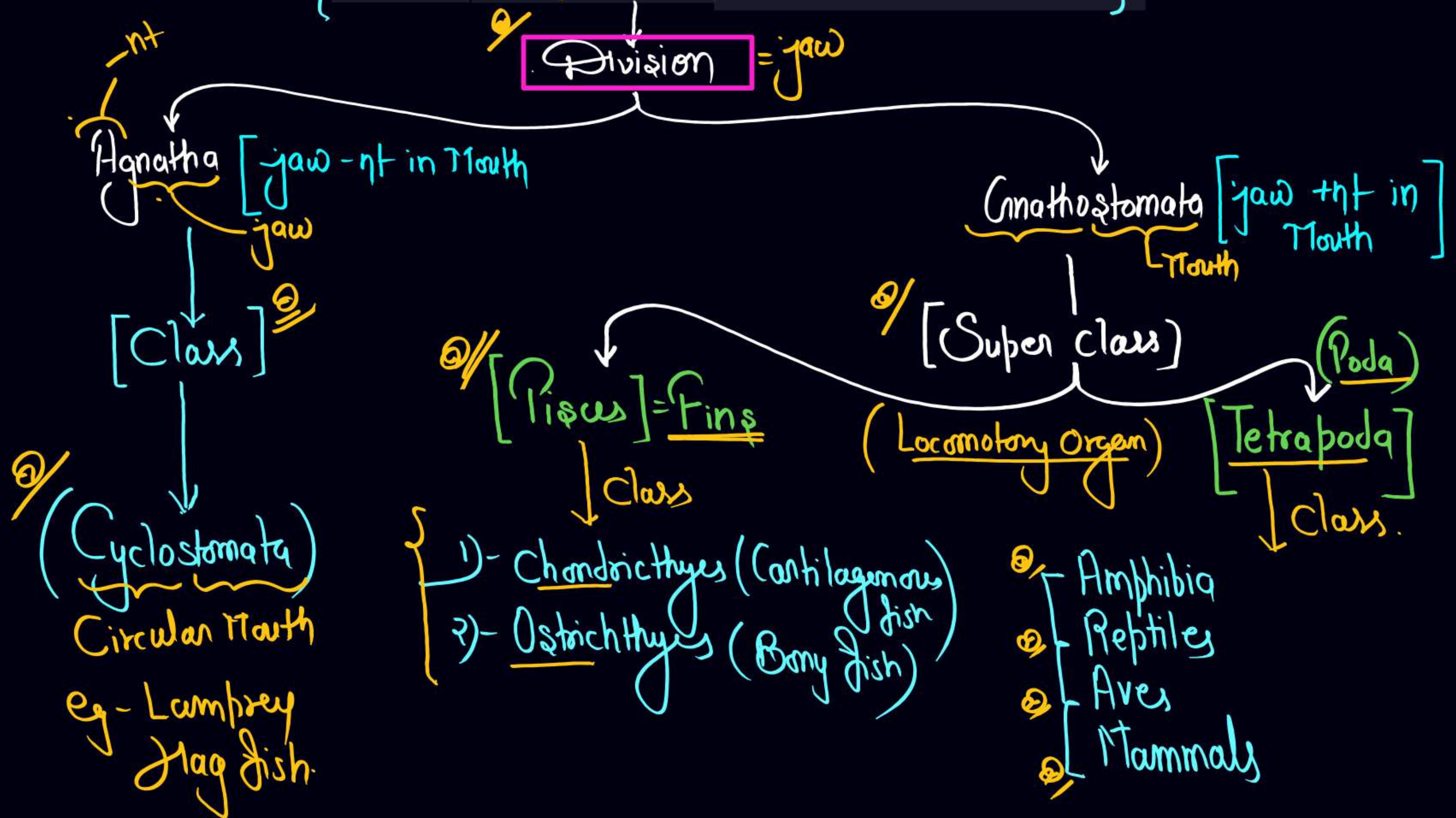


Figure 4.17 Ascidia

{ Sub-Phylum = Vertebrata }



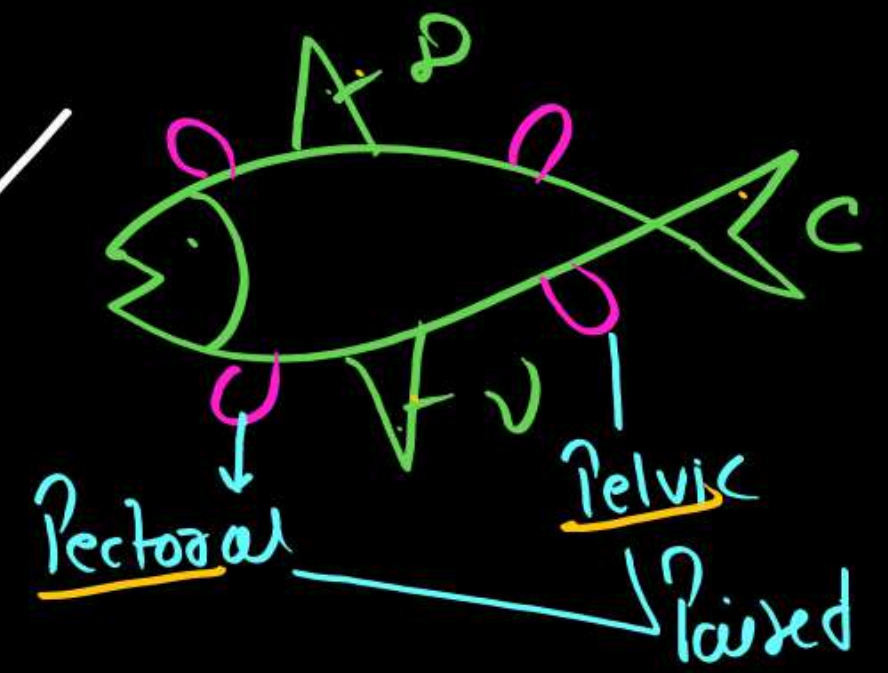
① Class – Cyclostomata

Circular Mouth

vvvmp

1. All living members of the class Cyclostomata are ectoparasites on some fishes.
2. They have an elongated body bearing 6-15 pairs of gill slits for respiration.
3. Cyclostomes have a sucking and circular mouth without jaws.
4. Their body is devoid of scales and paired fins. unpaired fins +nt
5. Cranium and vertebral column are cartilaginous. ✓
6. Circulation is of closed type.

vvvmp



Cartilaginous

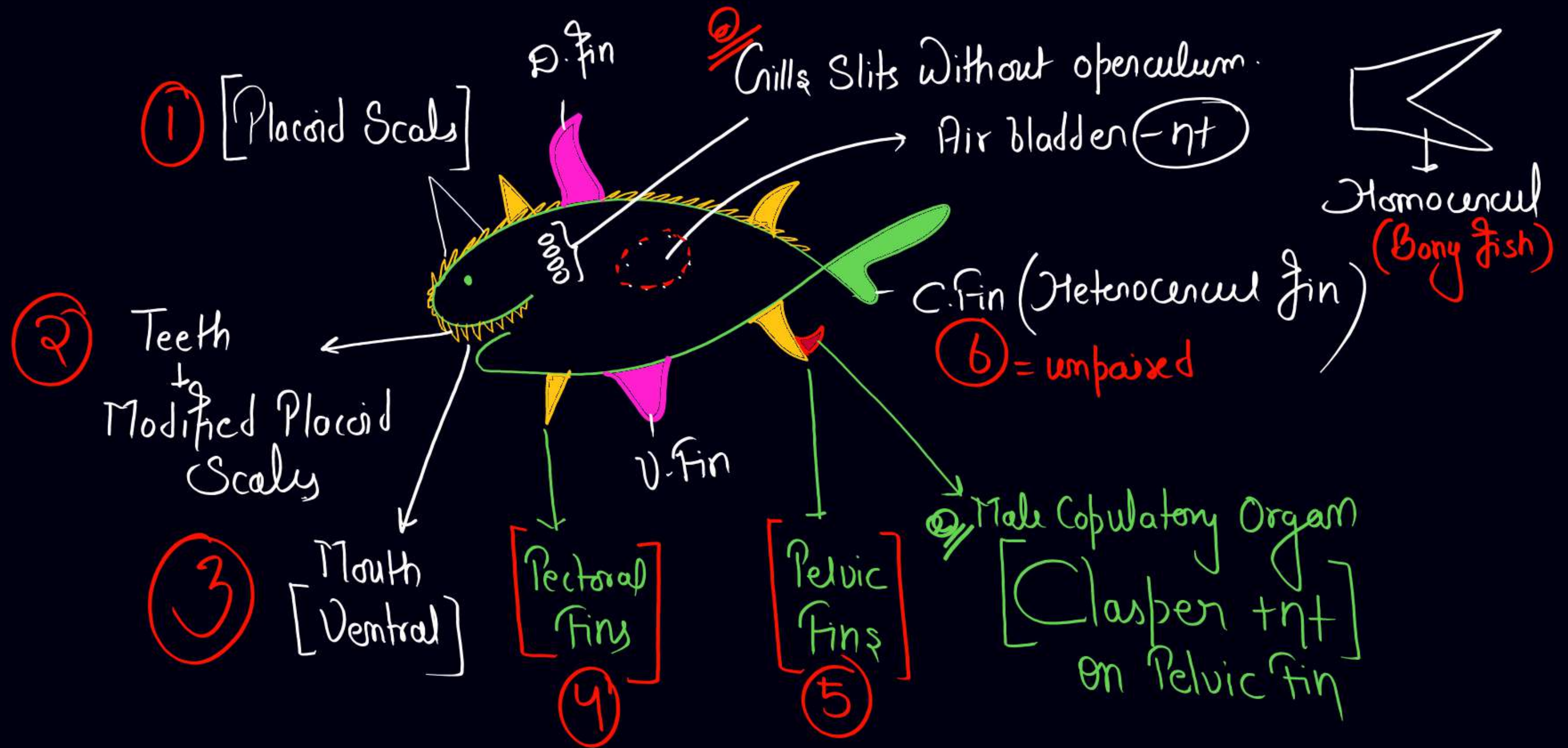
Class – Chondrichthyes

1. They are marine animals with streamlined body and have cartilaginous endoskeleton.
2. Mouth is located ventrally.
3. Notochord is persistent throughout life.
4. Gill slits are separate and without operculum (gill cover).
5. The skin is tough, containing minute placoid scales. Teeth are modified placoid scales which are backwardly directed.
6. Their jaws are very powerful.
7. These animals are predaceous.

Bony Fish.

Class – Osteichthyes

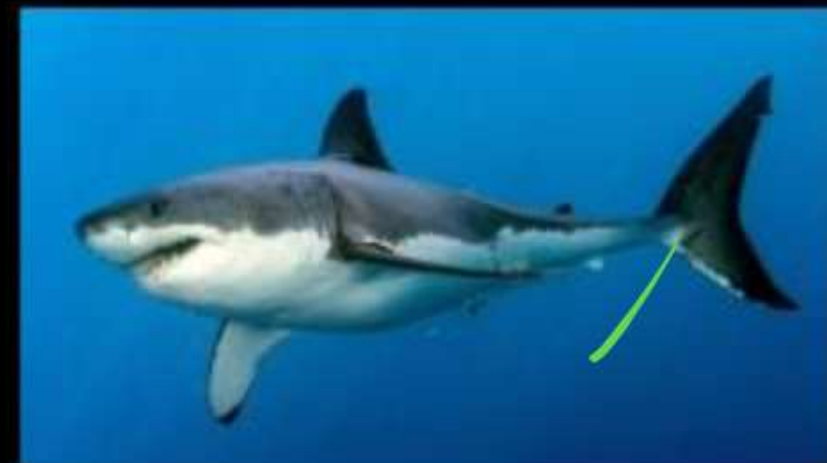
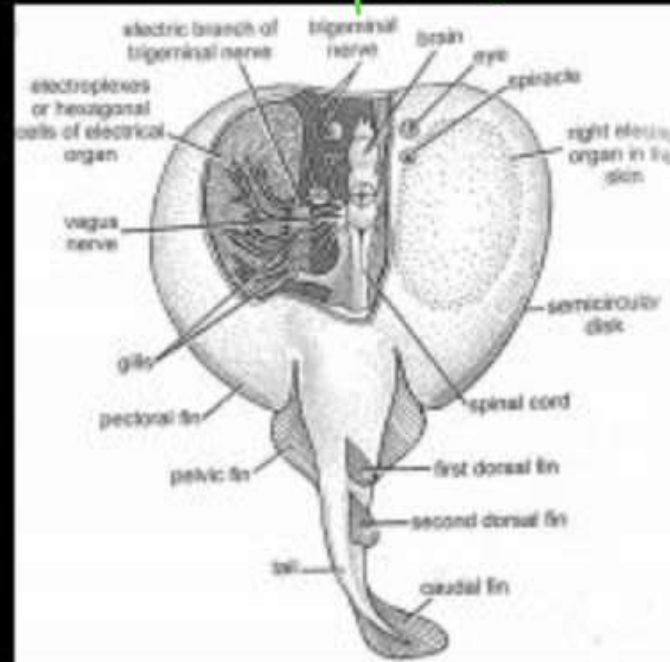
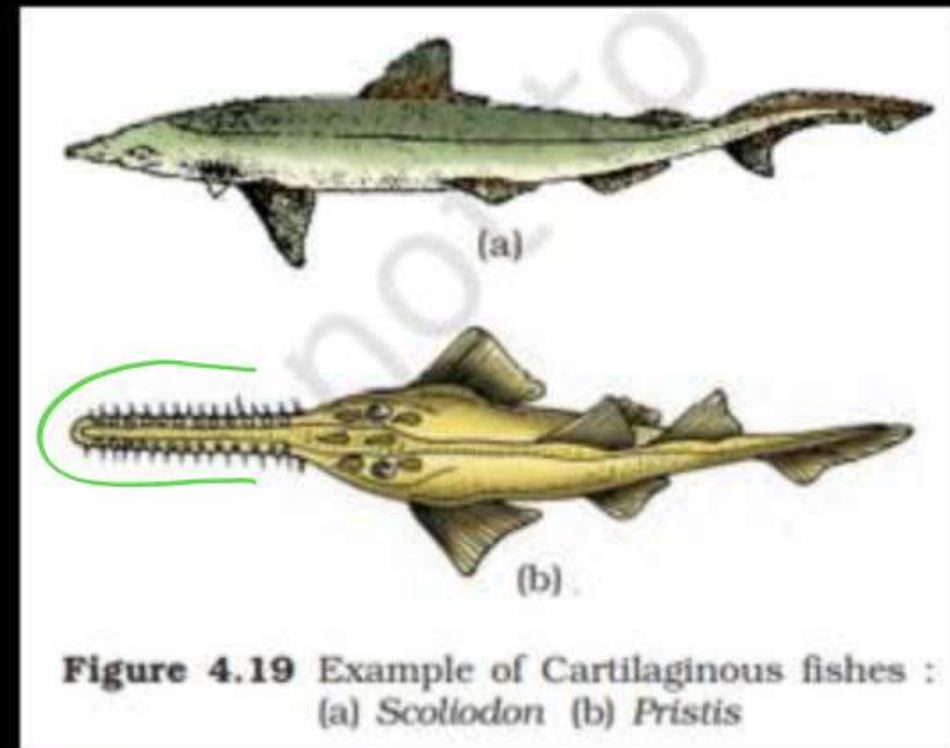
1. It includes both marine and fresh water fishes with bony endoskeleton. Their body is streamlined.
2. Mouth is mostly terminal.
3. They have four pairs of gills which are covered by an operculum on each side.
4. Skin is covered with cycloid/ctenoid scales.
5. Air bladder is present which regulates buoyancy.
6. Heart is two chambered (one auricle and one ventricle).
7. They are cold-blooded animals.



vvvvvmp

Examples: Scoliodon (Dog fish), Pristis (Saw fish), Carcharodon (Great white shark), Trygon (Sting ray).

Torpedo (Electric Ray)

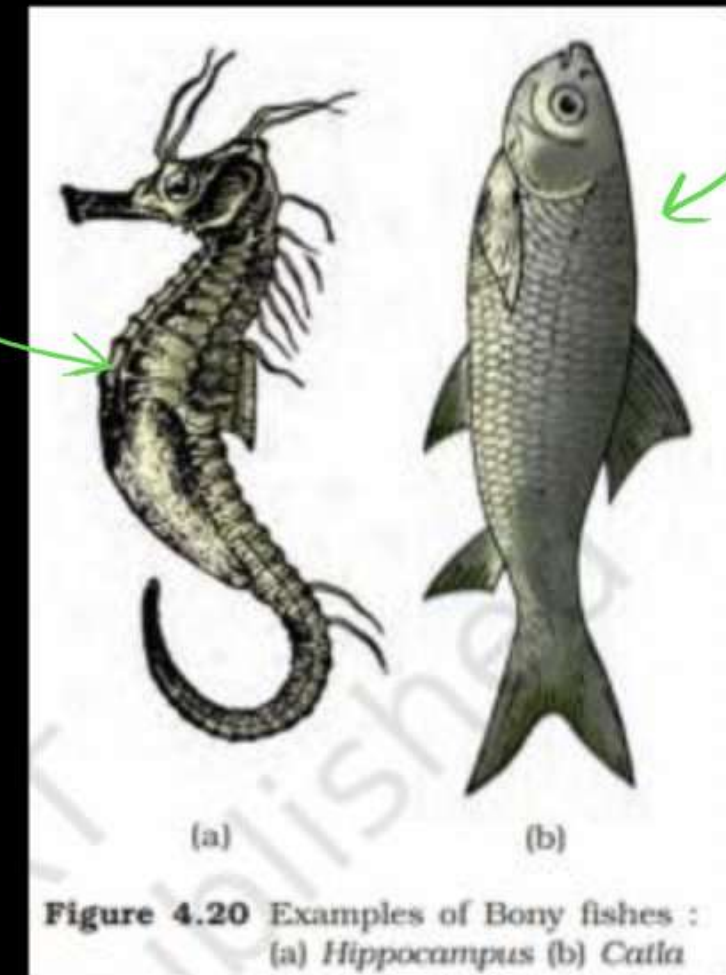


Trigem (Sting ray)

Examples:

• Marine – Exocoetus (Flying fish), Hippocampus (Sea horse);

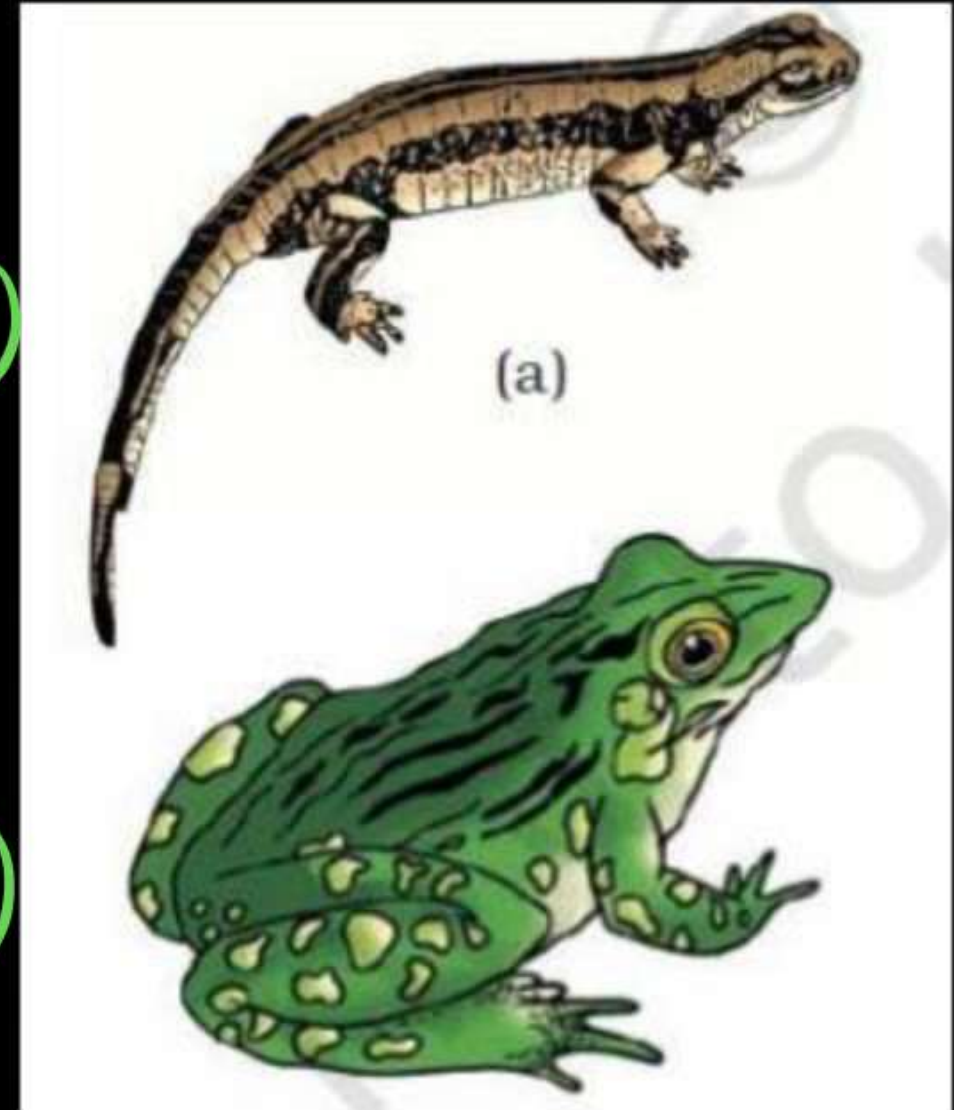
Freshwater – Labeo (Rohu), Catla (Katla), Clarias (Magur); Aquarium – Betta (Fighting fish), Pterophyllum (Angel fish)



Amphibia :

1. Bufo.....Toad.....,
2. Rana.....Frog.....
3. Hyla.....Tree frog.....
4. Salamandra.....Salamander.....
5. IchthyophisLimbless amphibian.....
Q// =

(a)



(b)

Reptiles:

1. Chelone.....Turtle — (b)
2. Testudo.....Tortoise
3. Chameleon.....Tree lizard — (c)
4. Calotes.....Garden lizard
5. Crocodilus.....Crocodile — (a)
6. Alligator.....Alligator Hemidactylus.....Wall lizard
7. Poisonous snakes – Naja.....Cobra — (d)
Bangarus (Krait), Vipera (Viper).

(a)



(b)



(c)



(d)



Aves

1. Corvus.....Crow
2. Pigeon.....Columba
3. Psittacula.....Parrot (d)
4. Struthio.....Ostrich — (a)
5. Peacock.....Pavo (b)
6. Aptenodytes.....Penguin
7. Neophron.....Vulture (c)

(a)



(b)



(c)



(d)



Mammals

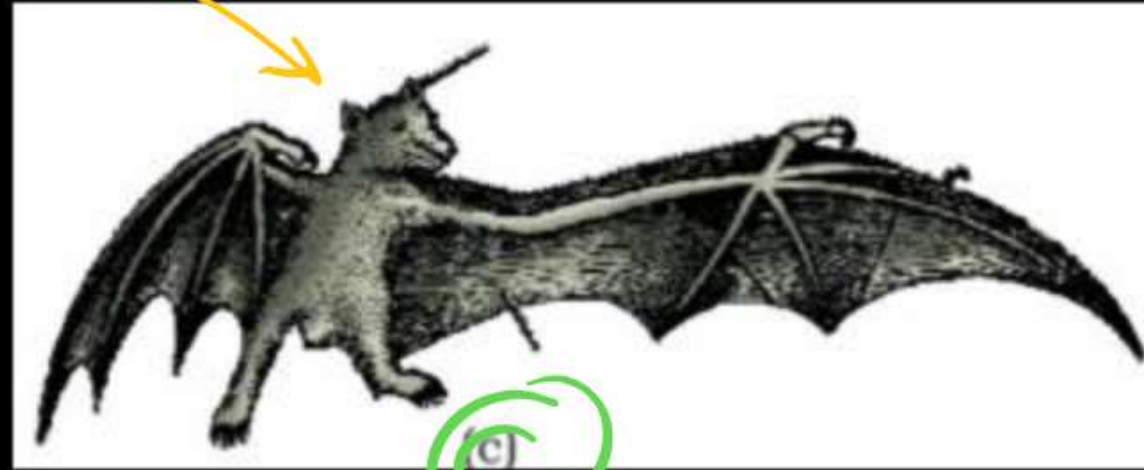
1. Ornithorhynchus..... *Platypus*
2. Macropus..... *Kangaroo*
3. Pteropus..... *Flying fox*
4. Balaenoptera..... *Whale*



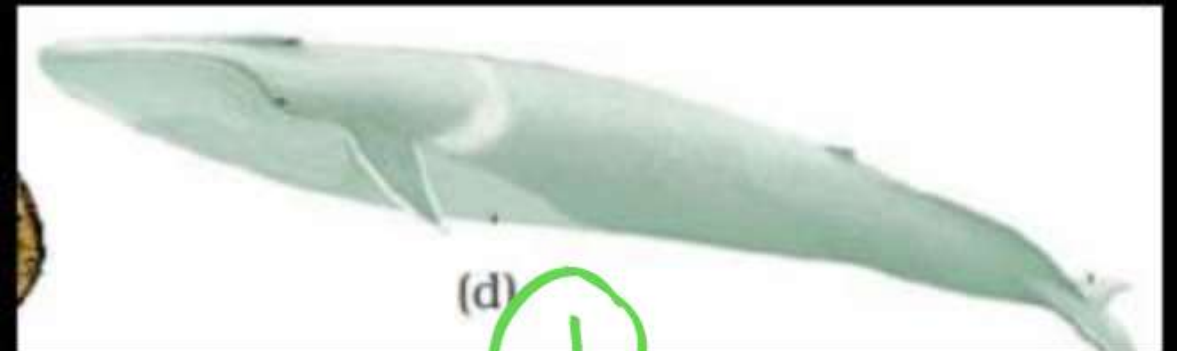
(a)



(b)



(c)



(d)

Frog

.

Body = Head + Trunk

Skin = Smooth & Slippery (Mucus + nt)

Cutaneous Respiration.

Nose - one pair

Ear = Internal Ear + Middle Ear + External Ear [Tympanum]

Dorsal = Olive Green

Ventral = Pale Yellow

4-Digits

Figure 7.1 External features of frog

3rd Eye
↓
(Nictitating Membrane)

Neck & Tail } - nt

Male Copulatory pad

Thorax + Abdomen

Trunk

Head

Eye

Fore limb

Hind limb

Frog

- **Rana tigrina** = The most common species of frog found in India.
- They do not have constant body temperature - **cold** blooded or poikilotherms. Q
- Summer sleep = Aestivation. H.W.
- Winter sleep = Hibernation. H.W.
- They have the ability to change the colour to hide them from their enemies = **camouflage**. This protective coloration is called = **mimicry**. Q Q

Morphology of Frog

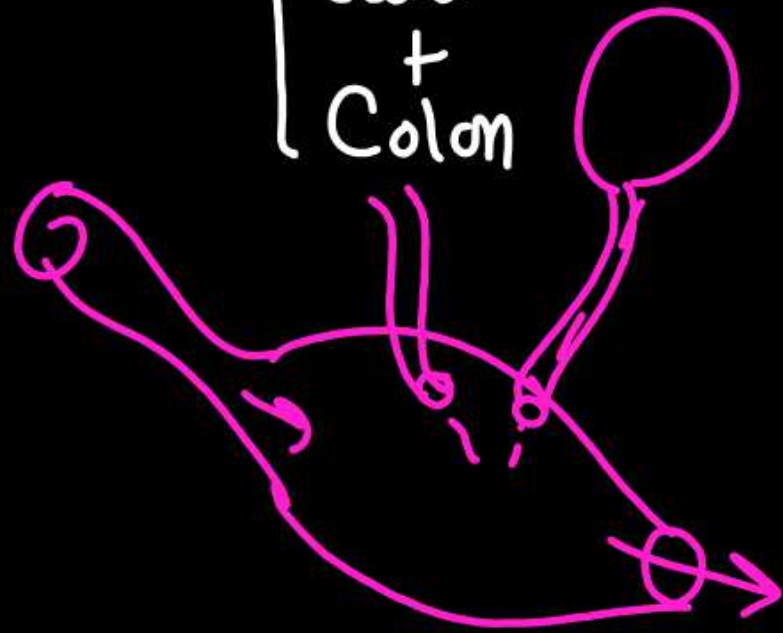
- The skin is smooth and slippery due to the presence of mucus. The skin is always maintained in a moist condition. = *Cutaneous Respiration*.
- Frog never drinks water but absorb it through the skin.
- Body of a frog is divisible into head and trunk =
- A neck and tail are absent
Above the mouth a pair of nostrils is present.
- Nictitating membrane
- Membranous tympanum (ear) =
- hind limbs end in = five digits ✓
- fore limbs that end in = four digits ✓
- Feet have webbed digits that help in swimming.
- Frogs exhibit sexual dimorphism, Male frogs - vocal sacs and copulatory pad on the first digit of the fore limbs.

Digestive System

1) Digestive Glands = Gastric, Pancreas, Liver

2) Digestive Tract = Mouth → Bilobed tongue 
[Carnivores = Small]

- nt {
 J
 J +
 Calcium
 +
 Colon



Stomach → HCl + Gastric juice } Chyme

Intestine → Duodenum
 ↓
 Illum
 [Villi & Microvilli] = for absorption

Bile = Fat

Pancreatic juice = Protein + Carbohydrate

Rectum

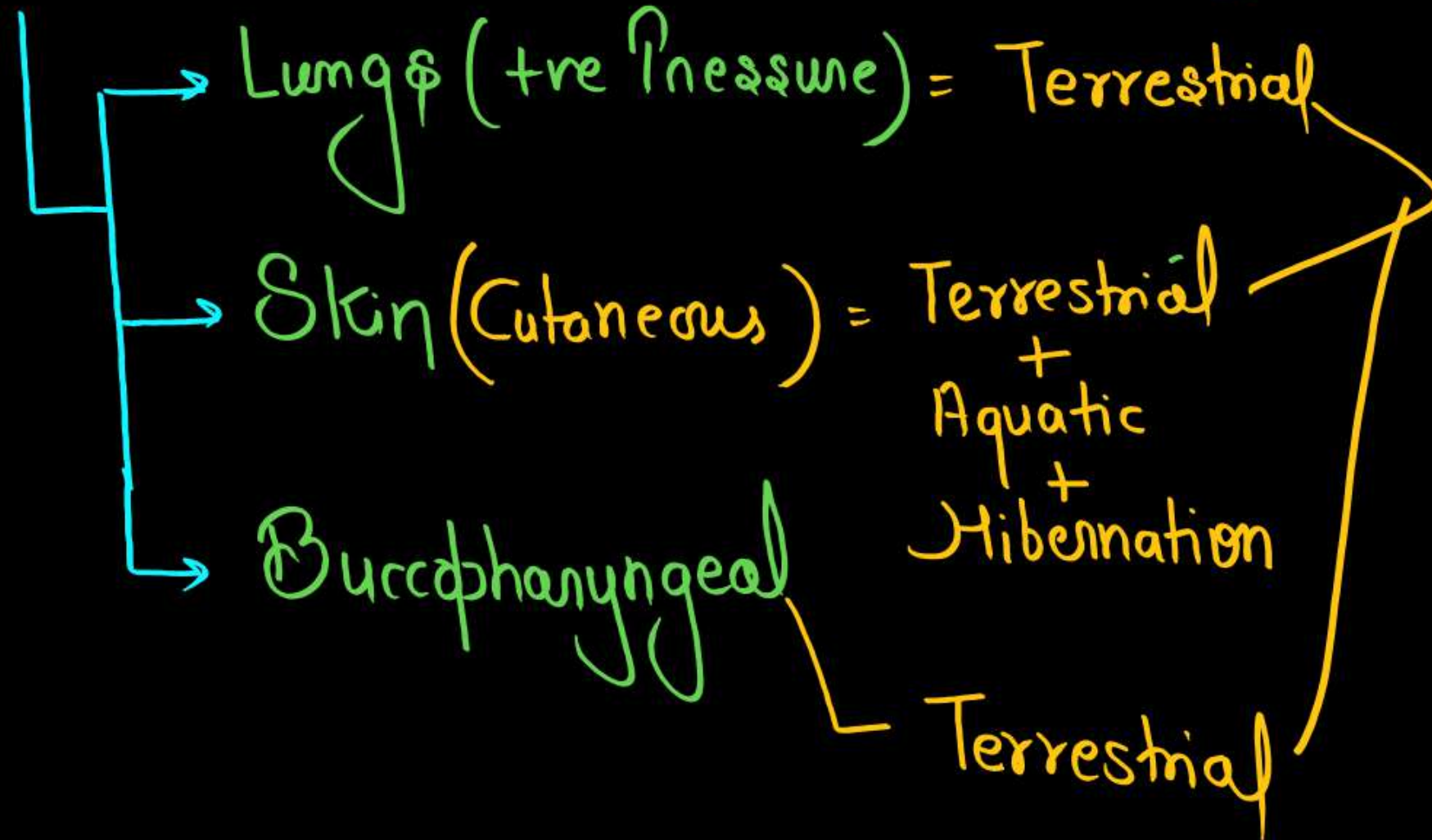
Cloaca [urine + sperm + undigested food]

Respiratory System

②

Respiratory Organs

(Adult)

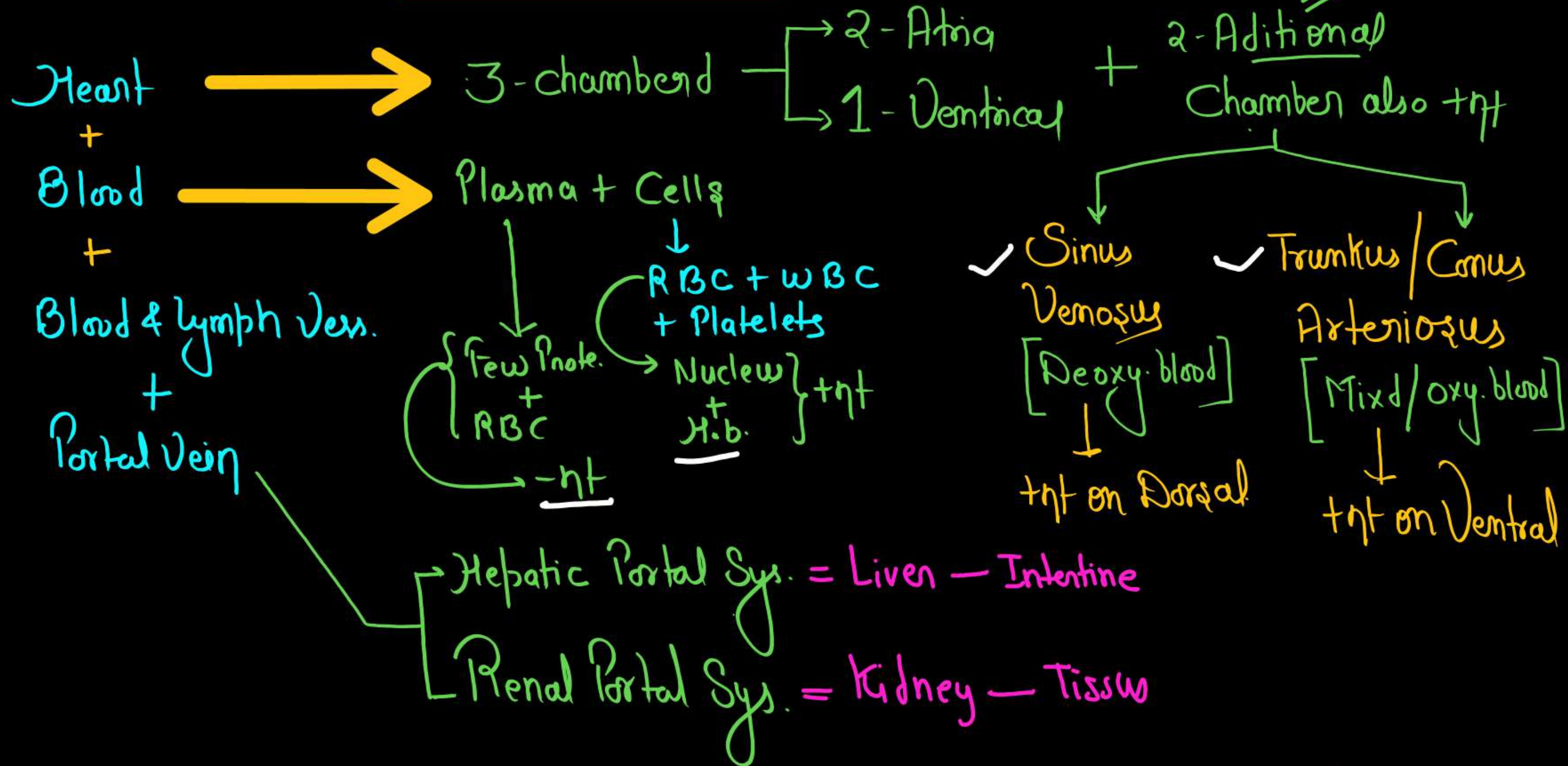


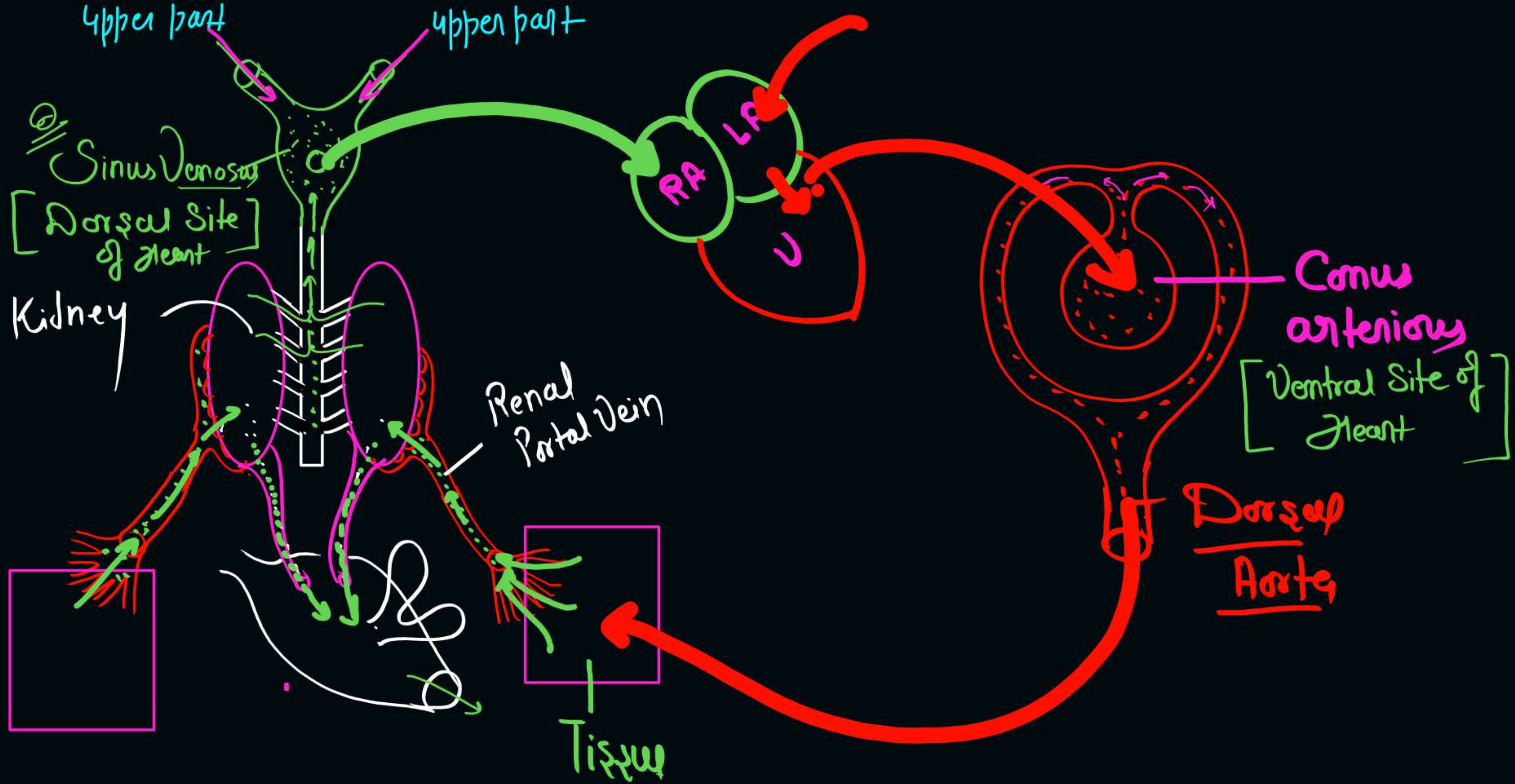
(Larva Tadpole)

Gills

Circulatory System

Closed Type





Excretory System

└─ Excretory Product = Urea in Adult (Ureotelic)
└─ Excretory Organ = Kidney (Mesonephric)

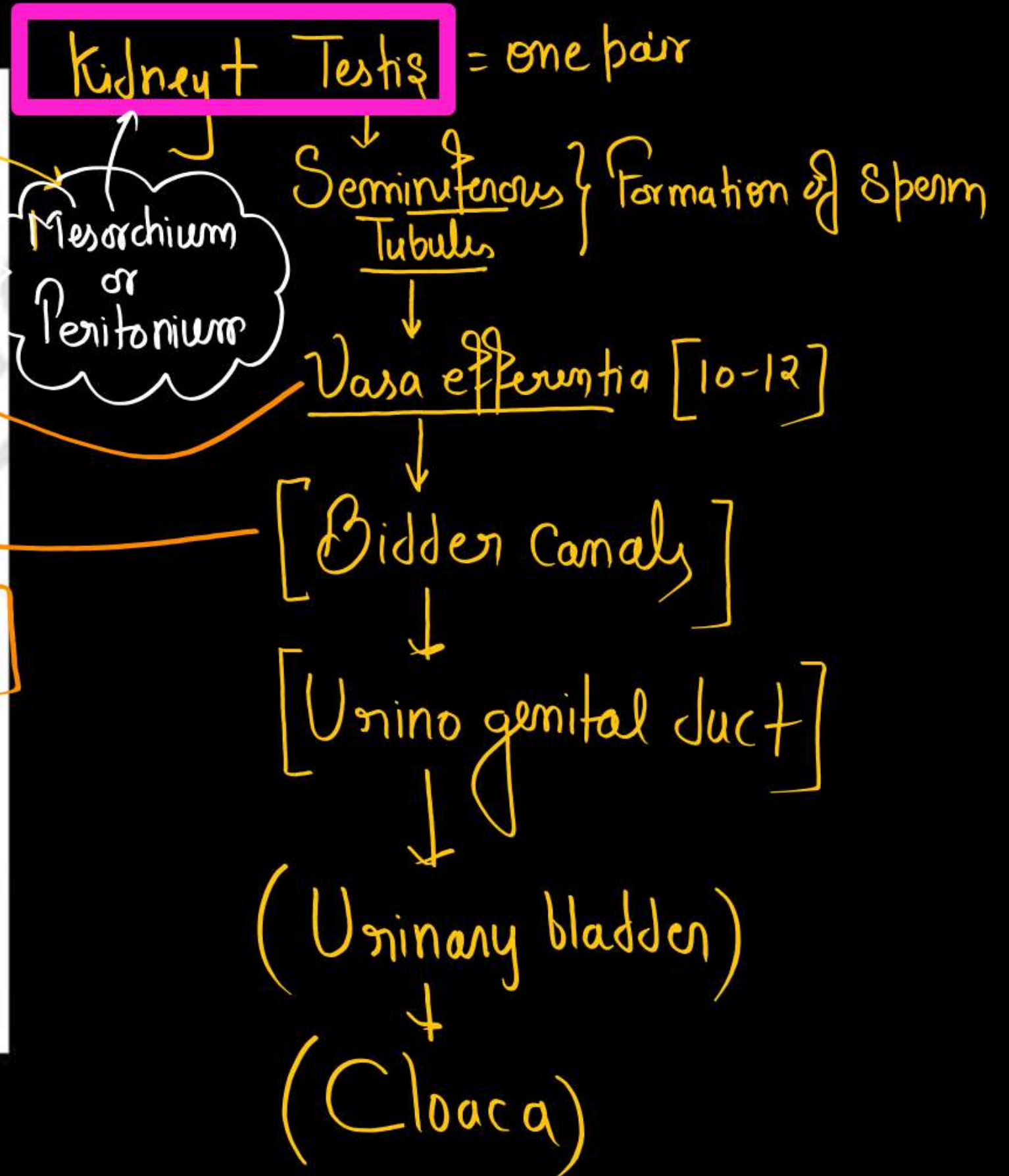
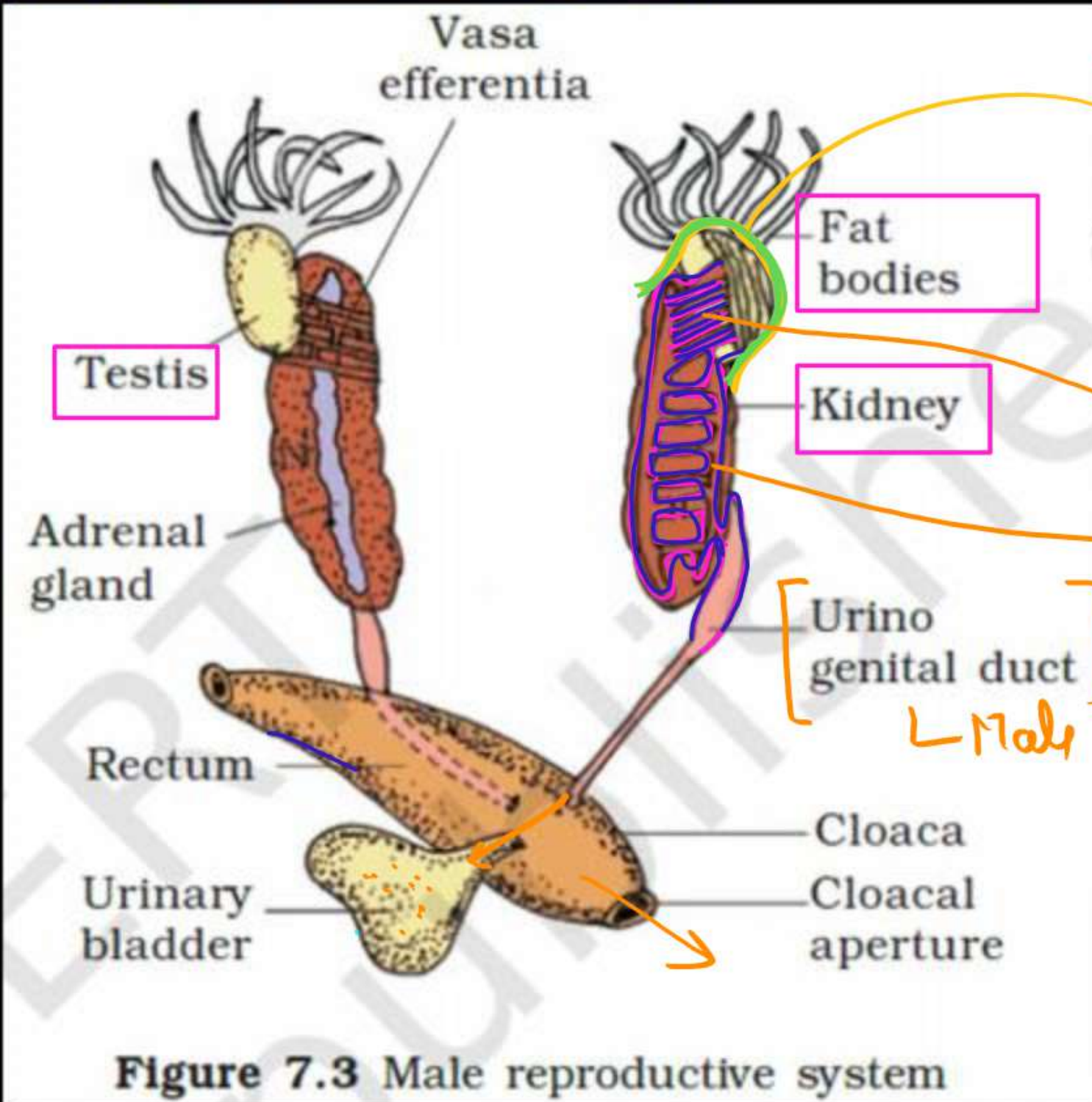
Nervous System →

- 1) - CNS = Brain + Spinal cord
2) - PNS = Cranial + Spinal Nerve
 (10-Pairs) (10-Pairs)

└─ ANS ─┬─ Parasympathetic
 └─ Sympathetic
└─ SNS

Male Reproductive System

lvvmb



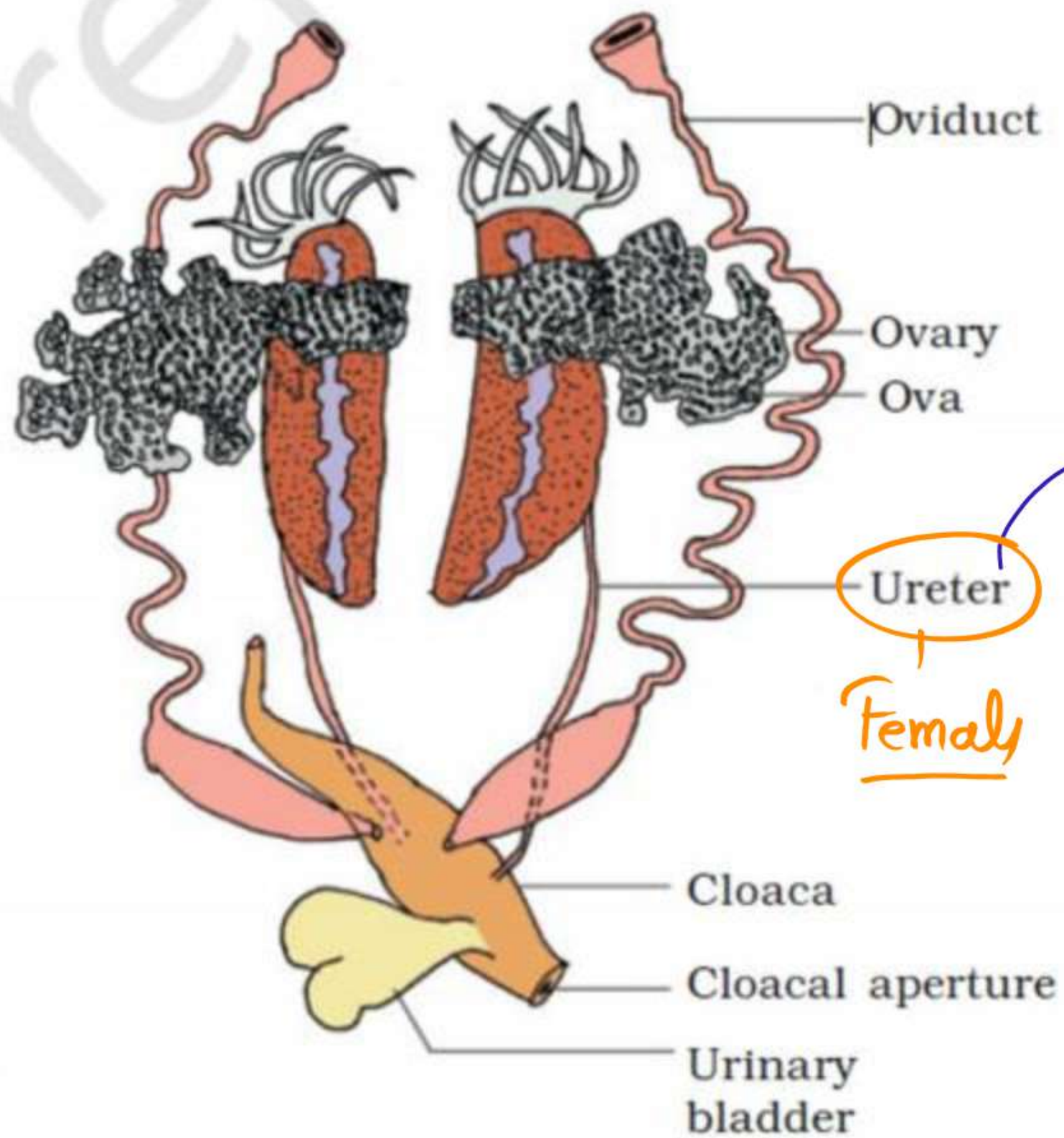
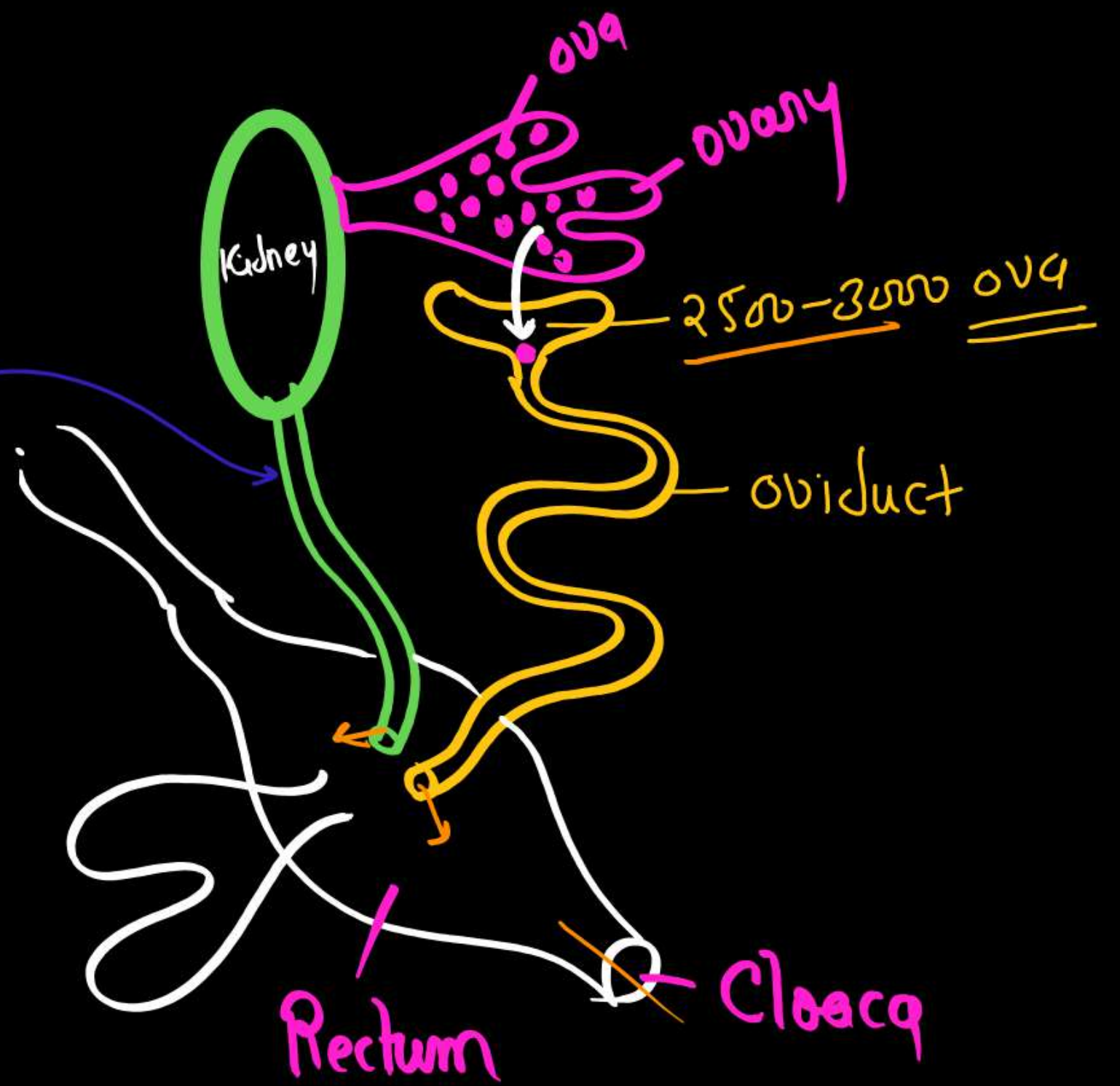


Figure 7.4 Female reproductive system



Read the following statements (1–3) and answer the question:

- (1) Tidal Volume (TV) is the amount of air inspired or expired during normal breathing, averaging about 500 mL.
- (2) A healthy adult breathes approximately 2,000–3,000 mL of air per minute during quiet breathing.
- (3) Residual Volume (RV) is the air left in the lungs after forced expiration, typically 1100–1200 mL.

How many of the above statements are correct?

- (1) Only 1 and 2
- (2) Only 2 and 3
- (3) Only 1 and 3
- (4) All correct

Under normal physical conditions in human beings every 100 ml oxygenated blood can deliver _____ ml of O_2 to the tissues.

- (a) 2 ml
- (c) 4 ml

- (b) 5 ml
- (d) 10 ml

(b)

(3)

Select the favorable conditions required for the formation of oxyhaemoglobin at the alveoli.

lungs

- (a) High pO_2 , low pCO_2 , less H^+ , lower temperature
- (b) Low pO_2 , high pCO_2 , more H^+ , higher temperature
- (c) High pO_2 , high pCO_2 , less H^+ , higher temperature
- (d) Low pO_2 , low pCO_2 , more H^+ , higher temperature

(a)

The partial pressures (in mmHg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are:

- (a) $pO_2=104$ and $pCO_2=40$
- (b) $pO_2=40$ and $pCO_2=45$
- (c) $pO_2=95$ and $pCO_2=40$
- (d) $pO_2=159$ and $pCO_2=0.3$

(a)

104
40

Match the items given in Column I with those in Column II and select the correct option given below:

Column I		Column II	
a.	Tidal volume	i.	2500-3000 mL
b.	Inspiratory Reserve volume	ii.	1100-1200 mL
c.	Expiratory Reserve volume	iii.	500-550 mL
d.	Residual volume	iv.	1000-1100 mL

- (a) a-i, b-iv, c-ii, d-iii
(b) a-iii, b-i, c-iv, d-ii
(c) a-iii, b-ii, c-i, d-iv
(d) a-iv, b-iii, c-ii, d-i

Match List I with List II.

vmp

List I	List II
A. P Wave	I. Heart muscles are electrically silent.
B. QRS complex	II. Depolarisation of ventricles.
C. T wave	III. Depolarisation of atria.
D. T-P gap	IV. Repolarisation of ventricles.

(joint Dia.)

Choose the correct answer from the options given below.

- (a) A-III, B-II, C-IV, D-I
- (b) A-II, B-III, C-I, D-IV
- (c) A-IV, B-II, C-I, D-III
- (d) A-I, B-III, C-IV, D-II

Match List I with List II:

	List I		List II
A.	P-wave	I.	Beginning of systole
B.	Q-wave	II.	Repolarisation of ventricles
C.	QRS	III.	Depolarisation of complex atria
D.	T-wave	IV.	Depolarisation of ventricles

Choose the correct answer from the options given below:

- (a) A-IV, B-III, C-II, D-I
- (b) A-II, B-IV, C-I, D-III
- (c) A-I, B-II, C-III, D-IV
- (d) A-III, B-I, C-IV, D-II

Which enzyme is responsible for conversion of inactive fibrinogen to fibrin?

- (a) Epinephrine (b) Thrombokinase
(c) Thrombin (d) Renin



Match the following columns and select the correct option.

	Column-I		Column-II
(1)	Eosinophils	(i)	Immune response
(2)	Basophils	(ii)	Phagocytosis
(3)	Neutrophils	(iii)	Release histamine, destructive enzymes
(4)	Lymphocytes	(iv)	Release granules containing histamine

- (a) (1)-(iv); (2)-(i); (3)-(ii); (4)-(iii)
(b) (1)-(i); (2)-(ii); (3)-(iv); (4)-(iii)
(c) (1)-(ii); (2)-(i); (3)-(iii); (4)-(iv)
(d) (1)-(iii); (2)-(iv); (3)-(ii); (4)-(i)

The QRS complex in a standard ECG represents:

- (a) Depolarization of ventricles
- (b) Repolarization of ventricles
- (c) Repolarization of auricles
- (d) Depolarization of auricles

a

Chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased.

- (1) Pneumonia
- (2) Tuberculosis
- (3) Emphysema
- (4) Asthma

3

All the components of the nodal tissue are autoexcitable. Why does the SA node acts as normal pacemaker?

- (a) SA node has the lowest rate of depolarization
- (b) SA node is the only component to generate Threshold
- (c) Only SA node convey the action potential to others
- (d) SA node has the highest rate of depolarization

Find the correct descending order of percentage proportion of leucocytes in human blood.

- (1) Neutrophils $\rightarrow$ Basophils $\rightarrow$ Lymphocytes $\rightarrow$ Acidophils (Eosinophils) $\rightarrow$ Monocytes
- (2) Monocytes $\rightarrow$ Neutrophils $\rightarrow$ Lymphocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils
- (3) Neutrophils $\rightarrow$ Lymphocytes $\rightarrow$ Monocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils
- (4) Lymphocytes $\rightarrow$ Acidophils $\rightarrow$ Basophils $\rightarrow$ Neutrophils $\rightarrow$ Monocytes

Given below are statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Nephrons are of two types: cortical and juxtamedullary, based on their relative position in cortex and medulla.

Reason R: Juxtamedullary nephrons have short loop of Henle, whereas cortical nephrons have longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

- (a) Both A and R are true, but R is NOT the correct explanation of A.
- (b) A is true, but R is false. b
- (c) A is false, but R is true.
- (d) Both A and R are true, and R is the correct explanation of A.

Erythropoietin hormone which stimulates R.B.C. formation is produced by

- (a) Alpha cells of pancreas
- (b) The cells of rostral adenohypophysis
- (c) The cells of bone marrow
- (d) Juxtaglomerular cells of the kidney

d

Which of the following statements are correct?

- A. An excessive loss of body fluid from the body switches off osmoreceptors. ~~X~~
- B. ADH facilitates water reabsorption to prevent diuresis. ~~/~~
- C. ANF causes vasodilation. ~~/~~
- D. ADH causes an increase in blood pressure. ~~/~~
- E. ADH is responsible for a decrease in GFR. ~~X~~

Choose the correct answer from the options given below:

- (a) ~~/~~ B, C and D only (b) A, B and E only
- (c) C, D and E only (d) A and B only

(a)

Which of the following would help in prevention of diuresis?

- (a) Atrial natriuretic factor causes vasoconstriction
- (b) Decrease in secretion of renin by JG cells
- (c) More water reabsorption due to under secretion of ADH C & D
- (d) Reabsorption of Na^+ and water from renal tubules due to aldosterone //

Presence of which of the following conditions in urine are indicative of Diabetes Mellitus?

- (a) Uremia and Ketonuria
- (b) Uremia and Renal Calculi
- (c) Ketonuria and Glycosuria C
- (d) Renal calculi and Hyperglycemia

13. Match the following parts of a nephron with their function:

Column I		Column II	
A	Descending limb of Henle's loop	i.	Reabsorption of salts only
B	Proximal Convoluted tubule	ii.	Reabsorption of water only
C	Ascending limb of Henle's loop	iii.	Conditional reabsorption of sodium ion and water
D	Distal convoluted tubule	iv.	Reabsorption of ion, water and organic nutrients

- (a) A-ii, B-iii, C-i, D-iv (b) A-i, B-ii, C-iii, D-iv
(c) A-ii, B-iv, C-i, D-iii (d) A-iv, B-i, C-ii, D-iii

15. Match the items given in column I with those in Column II and select the correct option given below:

Column I		Column II	
a.	Glycosuria	i.	Accumulation of uric acid in joints
b.	Gout	ii.	Mass of crystallized salts within the kidney
c.	Renal calculi	iii.	Inflammation in glomeruli
d.	Glomerular nephritis	iv.	Presence of glucose in urine

(a) a-ii, b-iii, c-i, d-iv

(b) a-i, b-ii, c-iii, d-iv

(c) a-iii, b-ii, c-iv, d-i

(d) a-iv, b-i, c-ii, d-iii

Which of the following statements is correct?

- (a) The ascending limb of loop of Henle is impermeable to water
- (b) The descending limb of loop of Henle is impermeable to water
- (c) The ascending limb of loop of Henle is permeable to water
- (d) The descending limb of loop of Henle is permeable to electrolytes

Match List I with List II:

List-I	List-II
A. Fibrous joints	I. Adjacent vertebrae, limited movement
B. Cartilaginous joints	II. Humerus and Pectoral girdle, rotational movement
C. Hinge	III. Skull, don't allow any movement
D. Ball and socket joints	IV. Knee, help in locomotion

Choose the correct answer from the options given below:

- (a) A-I, B-III, C-II, D-IV
- (b) A-II, B-III, C-I, D-IV
- (c) A-III, B-I, C-IV, D-II
- (d) A-IV, B-II, C-III, D-I

Match list 1 and list 2

Choose the correct answers from the options given below

	List - I		List - II
1.	Scapula	i.	Cartilaginous joints
2.	Cranium	ii.	Flat bone
3.	Sternum	iii.	Fibrous joints
4.	Vertebral column	iv.	Triangular flat bone

- (a) 1 – iv, 2 – ii, 3 – iii, 4 – i
(b) 1 – iv, 2 – iii, 3 – ii, 4 – i
(c) 1 – i, 2 – iii, 3 – ii, 4 – iv
(d) 1 – ii, 2 – iii, 3 – iv, 4 – i

During muscular contraction, which of the following events occur.

1. 'H' zone disappears
2. 'A' band widens
3. 'I' band reduces in width
4. Myosin hydrolysis ATP releasing ADP and P
5. Z line attaches to actin and pull inwards

- (a) 2,3,4 and 5 only
(b) 2,4,5 and 1 only
(c) 1,3,4 and 5 only
(d) 1,2,3 and 4 only
(e)

1. Match List I with List II:

List I	List II
A. Pons	I. Provides additional space for Neurons, regulates posture and balance
B. Hypothalamus	II. Controls respiration and gastric secretions
C. Medulla	III. Connects different regions of the brain
D. Cerebellum	IV. Neuro secretory cells

Choose the correct answer from the options given below:

- (a) A-III, B-IV, C-II, D-I
- (b) A-I, B-III, C-II, D-IV
- (c) A-II, B-I, C-III, D-IV
- (d) A-II, B-III, C-I, D-IV

The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear, etc., are

- (a) Corpora quadrigemina and hippocampus
- (b) Brain stem and epithalamus
- (c) Corpus callosum and thalamus
- (d) Limbic system and hypothalamus

Which of the following structures or regions is incorrectly paired with its functions?

- (a) Hypothalamus: production of releasing hormones and regulation of temperature, hunger and thirst
- (b) Limbic system: consists of fibre tracts that interconnect different regions of brain; controls movement
- (c) Medulla oblongata: controls respiration and cardiovascular reflexes
- (d) Corpus callosum: band of fibers connecting left and right cerebral hemispheres

Glucagon is a peptide hormone secreted from the alpha cells of the pancreatic islets of Langerhans
Match List I with List II.

List I	List II
A. Exophthalmic goiter	I. Excess secretion of cortisol, moon face & hyperglycemia
B. Acromegaly	II. Hypo-secretion of thyroid hormone and stunted growth.
C. Cushing's syndrome	III. Hyper secretion of thyroid hormone & protruding eye balls.
D. Cretinism	IV. Excessive secretion of growth hormone.

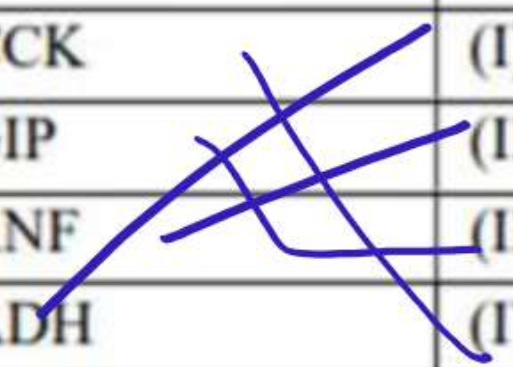
Choose the correct answer from the options given below:

- (a) A-IV, B-II, C-I, D-III
(b) A-III, B-IV, C-II, D-I
(c) A-III, B-IV, C-I, D-II
(d) A-I, B-III, C-II, D-IV

C

Match List I with List II.

List I	List II
(A) CCK	(I) Kidney
(B) GIP	(II) Heart
(C) ANF	(III) Gastric gland
(D) ADH	(IV) Pancreas



Choose the correct answer from the options given below:

- (a) A-III, B-II, C-IV, D-I
- (b) A-II, B-IV, C-I, D-III
- (c) A-IV, B-II, C-III, D-I
- (d) A-IV, B-III, C-II, D-I

Match the following columns and select the correct option.

Column I	Column II
(a) Pituitary gland	(i) Grave's disease
(b) Thyroid gland	(ii) Diabetes mellitus
(c) Adrenal gland	(iii) Diabetes insipidus
(d) Pancreas	(iv) Addison's disease

- (a) a – iii; b – i; c – iv; d – ii
(b) a – ii; b – i; c – iv; d – iii
(c) a – iv; b – iii; c – i; d – ii
(d) a – iii; b – ii; c – i; d – iv

Select the correct statement.

- (a) Insulin acts on pancreatic cells and adipocytes.
(b) Insulin is associated with hyperglycemia.
(c) Glucocorticoids stimulate gluconeogenesis.
(d) Glucagon is associated with hypoglycemia.

Match the column and select the correct option.

Column I	Column II
(a) Pituitary gland	(i) Steroid
(b) Epinephrine	(ii) Neuropeptide
(c) Endorphins	(iii) Peptide, protein
(d) Cortisol	(iv) Biogenic amines

(a) a-iv, b-i, c-ii, d-iv

(b) a-iii, b-iv, c-ii, d-i

(c) a-iv, b-iii, c-i, d-ii

(d) a-iii, b-iv, c-i, d-ii

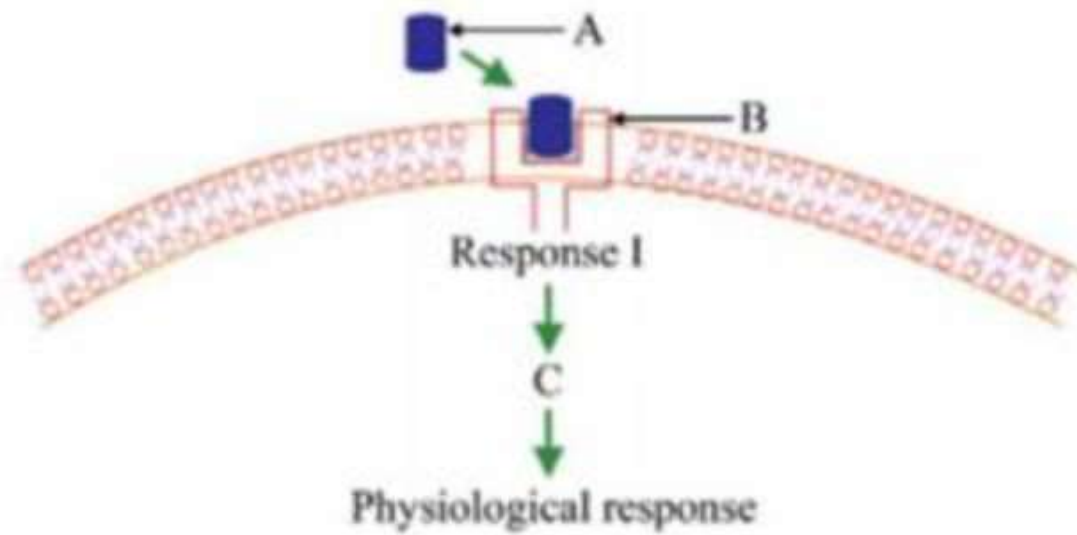
Match the following hormones with the respective disease.

- | | |
|---------------------|-------------------------|
| (p) Insulin | (i) Addison's disease |
| (q) Thyroxin | (ii) Diabetes insipidus |
| (r) Corticoids | (iii) Acromegaly |
| (s) Growth Hormones | (iv) Goitre |
| | (v) Diabetes mellitus |

Select the correct option.

- (a) $p \rightarrow (ii); q \rightarrow (iv); r \rightarrow (iii); s \rightarrow (i)$
- (b) $p \rightarrow (v); q \rightarrow (iv); r \rightarrow (i); s \rightarrow (iii)$
- (c) $p \rightarrow (ii); q \rightarrow (iv); r \rightarrow (i); s \rightarrow (iii)$
- (d) $p \rightarrow (v); q \rightarrow (i); r \rightarrow (ii); s \rightarrow (iii)$

Identify A, B and C in the diagrammatic representation of the mechanism of hormone action.



Select the correct option from the following.

- (a) A- steroid hormones, B- Hormone receptor complex, C- Protein
- (b) A- Protein hormone, B- Receptor, C- Cyclic AMP
- (c) A- Steroid hormone, B- Receptor, C- Second messenger
- (d) A- Protein hormone, B- Cyclic AMP, C- Hormone receptor complex

Which of the following statement is correct?

- (1) Essential oils like lemon grass oils are primary metabolites ~~X~~
- (2) Adenine and Guanine ~~X~~ are substituted pyrimidine
- (3) GLUT-4 enables glucose transport into cells
- (4) Proteins are linear chains of amino acids linked together by glycosidic ~~X~~ bonds

3

Match the following:

Column-I		Column-II	
1.	Primary structure	a.	Assembly of than one polypeptide.
2.	Secondary structure	b.	The sequence of amino acids.
3.	Tertiary structure	c.	Folded in the form of right handed helix.
4.	Quaternary structure	d.	Folded upon itself like a hollow woolen ball.

- (1) a-2, b-3, c-4, d-1
(2) a-1, b-4, c-3, d-2
(3) a-1, b-3, c-4, d-2
(4) a-2, b-4, c-1, d-3

Read the following statements (1 – 3) and answer the question which follows them:

1. Prosthetic groups are organic compounds and they are tightly bound to apoenzymes
2. In peroxidase and catalase, haem is the prosthetic group and it is a part of the active site of the enzyme
3. No enzymes require metal ions for their activity

How many of the above statements are correct?

- (1) Only 1 and 2
- (2) Only 2 and 3
- (3) Only 1, 2 and 3
- (4) All correct

1

Match List-I with List-II.

List-I

List-II

- | | |
|------------------|------------------|
| A. Adenosine | I. Nitrogen base |
| B. Adenylic acid | II. Nucleotide |
| C. Adenine | III. Nucleoside |
| D. Alanine | IV. Amino acid |

Choose the option with all correct matches.

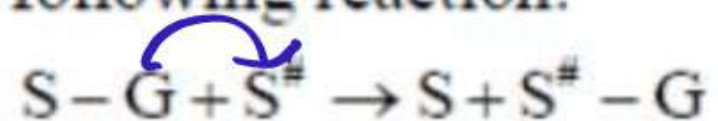
- (a) A-III, B-II, C-I, D-IV
- (b) A-II, B-III, C-I, D-IV
- (c) A-III, B-IV, C-II, D-I
- (d) A-III, B-II, C-IV, D-I

a

Which one of the following enzymes contains 'Haem' as the prosthetic group?

- (a) Succinate dehydrogenase
- ☒ (b) Catalase
- (c) RuBisCo
- (d) Carbonic anhydrase

Name the class of enzyme that usually catalyze the following reaction:



Where, $G \rightarrow$ a group other than hydrogen

$S \rightarrow$ a substrate

$S^{\#} \rightarrow$ another substrate

- | | |
|---|------------|
| ☒ (a) Transferase | (b) Ligase |
| (c) Hydrolase | (d) Lyase |

Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of

- (a) Feedback inhibition
- (b) Competitive inhibition
- (c) Enzyme activation
- (d) Cofactor inhibition

Choose the mismatch:

- (1) Dehydrogenase - catalyse oxidoreduction
- (2) Transferase - catalyses transfer of group
- (3) Hydrolase - enzyme catalyses hydrolysis of bonds like ester, ether etc.
- (4) Lyases - enzyme catalyses linking together of 2 compounds

4

Match List I with List II:

List I	List II
A. Lipase	I. Peptide bond
B. Nuclease	II. Ester bond
C. Protease	III. Glycosidic bond
D. Amylase	IV. Phosphodiester bond

Choose the correct answer from the options given below:

- (a) A-III, B-II, C-I, D-IV
(b) ~~A-II~~, ~~B-IV~~, C-I, D-III
(c) A-IV, B-I, C-III, D-II
(d) A-IV, B-II, C-III, D-I

(b)

Match the following columns:

Column I		Column II	
A.	Fatty acid	i.	Polymer of fructose
B.	Phospholipid	ii.	Modified sugar
C.	Glucosamine	iii.	Lecithin
D.	Inulin	iv.	Palmitic acid

- (1) A–(i), B–(ii), C–(iii), D–(iv)
(2) A–(iv), B–(iii), C–(ii), D–(i)
(3) A–(ii), B–(iii), C–(iv), D–(i)
(4) A–(iii), B–(iv), C–(i), D–(ii)

Malonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

- (a) Amylase
(b) Lipase
(c) Dinitrogenase
(d) Succinic dehydrogenase

Match list I with list II. Choose the correct answer from the options given below:

	List I		List II
(a)	Glycogen	(i)	Hormone
(b)	Globulin	(ii)	Biocatalyst
(c)	Steroids	(iii)	Antibody
(d)	Thrombin	(iv)	Storage product

(a) (a) - (iii); (b) - (ii); (c) - (iv); (d) - (i)

(b) (a) - (iv); (b) - (ii); (c) - (i); (d) - (iii)

(c) (a) - (ii); (b) - (iv); (c) - (iii); (d) - (i)

(d) (a) - (iv); (b) - (iii); (c) - (i); (d) - (ii)

Identify the incorrect pair.

(a) Lectins- Concanavalin A

(b) Drugs- Ricin

(c) Alkaloids- Codeine

(d) Toxin- Abrin

6

Match list-I with list-II.

List-I		List-II	
(a)	Protein	(i)	C=C bonds
(b)	Unsaturated fatty acids	(ii)	Phosphodiester bonds
(c)	Nucleic acids	(iii)	Glycosidic bonds
(d)	Polysaccharides	(iv)	Peptide bonds

(a) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(b) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(c) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(d) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

Secondary metabolites such as nicotine, strychnine, and caffeine are produced by plants for their

- | | |
|---------------------|----------------------------|
| (a) Defense action | (b) Effect on reproduction |
| (c) Nutritive value | (d) Growth response |

Match the following.

- | | |
|-------------------------------------|---------------|
| (a) Inhibitor of catalytic activity | (i) Ricin |
| (b) Possess peptide bonds | (ii) Malonate |
| (c) Cell wall material in fungi | (iii) Chitin |
| (d) Secondary metabolite | (iv) Collagen |

Choose the correct option from the following.

- (a) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(b) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(c) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
(d) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Role of the water vascular system in Echinoderms is:

- A. Respiration and Locomotion
- B. Excretion and Locomotion
- C. Capture and transport of food
- D. Digestion and Respiration
- E. Digestion and Excretion

Choose the correct answer from the options given below:

- | | |
|------------------|---------------------|
| (a) B and C Only | (b) B, D and E Only |
| (c) A and B Only | (d) A and C Only |

All living members of the class Cyclostomata are:

- | | |
|-----------------|------------------|
| (a) Symbiotic | (b) Ectoparasite |
| (c) Free living | (d) Endoparasite |

The unique mammalian characteristics are

- (a) Hairs, pinna and mammary glands
- (b) Hairs, pinna and indirect development
- (c) Pinna, monocondylic skull and mammary glands
- (d) Hairs, tympanic membrane and mammary glands

Match List I with List II:

List I	List II
A. Pleurobrachia	I. Mollusca
B. Radula	II. Ctenophora
C. Stomochord	III. Osteichthyes
D. Air bladder	IV. Hemichordata

Choose the correct answer from the options given below:

- (a) A-II, B-I, C-IV, D-III
- (b) A-II, B-IV, C-I, D-III
- (c) A-IV, B-III, C-II, D-I
- (d) A-IV, B-II, C-III, D-I

Match List I with List II:

List I	List II
A. <i>Pterophyllum</i>	I. Hag fish
B. <i>Myxine</i>	II. Saw fish
C. <i>Pristis</i>	III. Angel fish
D. <i>Exocoetus</i>	IV. Flying fish

Choose the correct answer from the options given below:

- (a) A-III, B-I, C-II, D-IV
- (b) A-IV, B-I, C-II, D-III
- (c) A-III, B-II, C-I, D-IV
- (d) A-II, B-I, C-III, D-IV

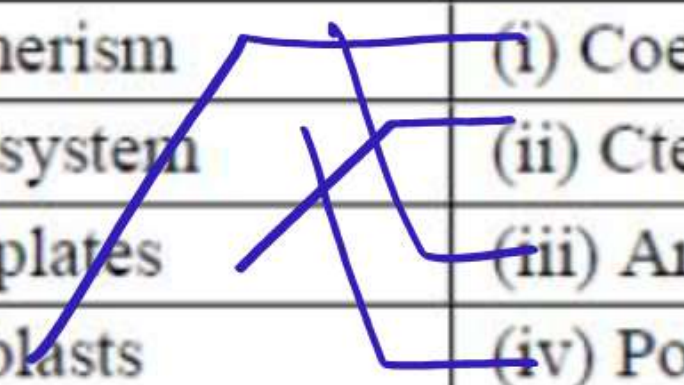
Match the following:

Column I	Column II
(p) <i>Physalia</i>	(i) Pearl oyster
(q) <i>Limulus</i>	(ii) Portuguese man of war
(r) <i>Ancylostoma</i>	(iii) Living fossil
(s) <i>Pinctada</i>	(iv) Hookworm

- (a) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(iv)}$; $r \rightarrow \text{(ii)}$; $s \rightarrow \text{(i)}$
(b) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(iv)}$; $s \rightarrow \text{(i)}$
(c) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(i)}$; $s \rightarrow \text{(iv)}$
(d) $p \rightarrow \text{(ii)}$; $q \rightarrow \text{(iii)}$; $r \rightarrow \text{(iv)}$; $s \rightarrow \text{(i)}$

Match the following columns and select the correct option.

Column I	Column II
(p) Metamerism	(i) Coelenterata
(q) Canal system	(ii) Ctenophora
(r) Comb plates	(iii) Annelida
(s) Cnidoblasts	(iv) Porifera



- (a) $p \rightarrow (i)$; $q \rightarrow (ii)$; $r \rightarrow (iii)$; $s \rightarrow (iv)$
(b) $p \rightarrow (iii)$; $q \rightarrow (iv)$; $r \rightarrow (i)$; $s \rightarrow (ii)$
(c) $p \rightarrow (iii)$; $q \rightarrow (iv)$; $r \rightarrow (ii)$; $s \rightarrow (i)$
(d) $p \rightarrow (iv)$; $q \rightarrow (iii)$; $r \rightarrow (i)$; $s \rightarrow (ii)$

Match the following columns and select the correct option.

Column I	Column II
(a) 6-15 pairs of gill slits	(i) <i>Trygon</i>
(b) Heterocercal caudal fin	(ii) Cyclostomes
(c) Air bladder	(iii) Chondrichthyes
(d) Poison sting	(iv) Osteichthyes

- (a) a – iv; b – ii; c – iii; d – i
 (b) a – i; b – iv; c – iii; d – ii
 (c) a – ii; b – iii; c – iv; d – i
 (d) a – iii; b – iv; c – i; d – ii

Match the following columns and select the correct option.

Column I	Column II
(a) Gregarious polyphagous pest	(i) <i>Asterias</i>
(b) Adult with radial symmetry and larva with bilateral symmetry	(ii) Scorpion
(c) Book lungs	(iii) <i>Ctenoplana</i>
(d) Bioluminescence	(iv) <i>Locusta</i>

- (a) a – iii; b – ii; c – i; d – iv
 (b) a – ii; b – i; c – iii; d – iv
 (c) a – i; b – iii; c – ii; d – iv
 (d) a – iv; b – i; c – ii; d – iii

Which of the following statements are true for the Phylum Chordata?

- (i) In Urochordata, notochord extends from head to tail, and it is present throughout their life.
- (ii) In Vertebrata, notochord is present during the embryonic period only.
- (iii) Central nervous system is dorsal and hollow.
- (iv) Chordata is divided into 3 subphyla:
Hemichordata, Tunicata and Cephalochordata.

- | | |
|--------------------|--------------------|
| (a) (i) and (ii) | (b) (ii) and (iii) |
| (c) (iv) and (iii) | (d) (iii) and (i) |

Match the following organisms with their respective characteristics.

- | | |
|--------------------------|-------------------------|
| (p) <i>Pila</i> | (i) Flame cells |
| (q) <i>Bombyx</i> | (ii) Comb plates |
| (r) <i>Pleurobrachia</i> | (iii) Radula |
| (s) <i>Taenia</i> | (iv) Malpighian tubules |

- (a) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(iv)}$; $r \rightarrow \text{(ii)}$; $s \rightarrow \text{(i)}$
- (b) $p \rightarrow \text{(ii)}$; $q \rightarrow \text{(iv)}$; $r \rightarrow \text{(iii)}$; $s \rightarrow \text{(i)}$
- (c) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(iv)}$; $s \rightarrow \text{(i)}$
- (d) $p \rightarrow \text{(iii)}$; $q \rightarrow \text{(ii)}$; $r \rightarrow \text{(i)}$; $s \rightarrow \text{(iv)}$

